Tight Black-Box Privacy Preservation and Adversarially-Adaptive Composition for Entropic Secrets

Hanshen Xiao[*,1,2] and Zhouyi Zhang[1]

[1]Department of Computer Science, Purdue University
[2]NVIDIA

June 15, 2026

Abstract

We address, in an asymptotic sense, two open questions concerning *tight* privacy preservation for entropic secrets sampled from an *arbitrary, specified* prior distribution. First, given an arbitrary *black-box* leakage procedure, where the user has no structural knowledge and only limited end-to-end simulation access, how can one certify and calibrate a randomized release mechanism with *minimal* utility loss and exhausting a prescribed Bayesian inference-hardness privacy budget against a computationally unbounded adversary? Despite a rich body of Bayesian leakage-analysis tools, there is currently no known unbiased and operationally-efficient approach for tightly bounding posterior success risk under arbitrary priors and arbitrary inference tasks. Second, in an interactive setting, how can one tightly account for cumulative privacy risk when a reused entropic secret is exposed through a sequence of black-box leakage procedures adaptively selected by an adversary?

In contrast to the regime of prior-free leakage measured by Input-Independent Indistinguishability (III), where strong impossibility results show that privacy guarantees are impossible to validate even for white-box leakage in the worst case, we show that the landscape changes fundamentally for specified entropic secrets. We introduce Prior–Reference-Weighted (PRW) divergence, a conservative finite-order posterior-risk certificate whose limit equals the exact Bayesian posterior success rate. This yields an asymptotically tight learning-based framework for black-box privacy–utility calibration. We further show through a *bounded-ratio* trick, we can generally break the exponential sample complexity in validating high-order PRW moments with instead an logarithmic order. We further study adaptive composition. We give a transcript-dependent local calibration rule whose branch-local equalities compose into a globally tight finite-order PRW accounting identity for the entire adaptive transcript mechanism. We also provide a mechanism-agnostic variant, deriving general composition upper bounds from marginal divergence measurements alone. Our results have direct relevance to side-channel mitigation, especially for physical leakage channels that lack tractable analytical models. We evaluate our framework on power-consumption leakage in Advanced Encryption Standard (AES) secret-key generation and timing leakage in RSA encryption as illustrative validations of the theory [1].

1 Introduction

Any processing of a secret can potentially leak information. As one of the most extensively-studied security problems, privacy essentially asks the following: *given the information revealed through*

[*]The authors contribute equally.
[1]The codes are available at `https://github.com/Hanshen-Xiao/Optimal_Blackbox_Privacy/tree/main`

leakage, how much of the underlying secret can an adversary recover? Despite its intuitive interpretation, rigorous formalization and provable protection are highly non-trivial for many applications in practice. At a high level, modern privacy research can be organized along three main directions: ❶ *definition-level*—how to characterize privacy risk and privacy guarantees, both *semantically* and *mathematically*; ❷ *measurement/accounting-level*—how to *tightly* quantify the privacy risk induced by a given leakage; and ❸ *operation-level*—how to construct *optimal* mitigations for a required privacy guarantee.

1.1 Privacy Definition

On ❶, existing privacy definitions can be largely categorized by two types of secrets to protect: **(A)** entropic secrets sampled from an *accessible distribution*[2], and **(B)** secrets for which such entropy is not tractable. A canonical example of a Type-(A) secret is a uniformly-generated secret key in a cryptographic protocol. For Type-(A) secrets, a natural notion of privacy is Bayesian: the adversary's optimal posterior success probability for recovering the secret [1]. In particular, when the adversary's objective is *identification* (i.e., the success criterion is exact recovery of the secret), this risk coincides with the classic notion of (conditional) *min-entropy* studied in [2–4]. Related notions include *guessing entropy* [5] and g-leakage [6], which capture the number of guesses required for a successful inference.

A key prerequisite of Bayesian viewpoint is the ability to determine the prior distribution $\mathcal{D}_{\mathsf{x}}$ over the secret x, which is often inappropriate for Type-(B) secrets where we may not know, or may not have access to, the secret generation[3]. To obtain a rigorous notion in a *prior-free* setting, Shannon [7], in 1949, initiated the concept of *input-independent indistinguishability* (III), which essentially considers the worst case over *all* priors. Rather than directly measuring the adversary's posterior knowledge, III evaluates the worst-case *posterior advantage*: the maximal change from prior to posterior when distinguishing between two secret hypotheses under an *arbitrary* prior distribution. Goldwasser and Micali [8] further introduced a computational form of III against computationally-bounded adversaries, laying the foundation of modern cryptography. They also showed that negligible III for binary hypotheses in the worst case implies negligible advantage for *any* efficient inference task. As a prior-free worst-case notion, III admits a clear semantic interpretation: regardless of the adversary's (subjective) prior belief about the secret, observing the leakage should not significantly change their posterior belief. Beyond cryptography, many information-theoretic privacy definitions for Type-(B) secrets, including *Differential Privacy* (DP) [9, 10], *Pufferfish Privacy* [11], and *Maximal Leakage* [12], are also rooted in the III philosophy.

1.2 Universal Risk Measurement and Mitigation

The ❷ measurement problem and the ❸ mitigation problem require a deeper study of the leakage itself. Broadly speaking, leakage can be any quantity statistically correlated with a secret x that contains sensitive information, and it can be generally represented as the output of a function $\mathcal{F}(\mathsf{x})$. In practice, leakage $\mathcal{F}(\mathsf{x})$ may arise from the release of any algorithmic processing of sensitive data x, for instance, average salary or a machine learning model trained on health data [13]. Leakage can also arise as a physical signal related to the processing of x: a large body of work has identified

[2] An *accessible distribution* refers to a secret-generation procedure from which we can repeatedly sample instances of the secret.

[3] Even if one could posit that all possible data is generated by some complicated stochastic process, in many real scenarios the observation is effectively unique and cannot be resampled at the relevant moment.

many side channels, such as timing [14,15], power [16], memory [17] consumption, or network traffic patterns [18], that may enable successful attacks.

To mitigate leakage (❸), most existing approaches introduce additional (algorithmic) randomization into $\mathcal{F}(\mathsf{x})$. A widely used strategy is *perturbation*, releasing a noisy version $\mathcal{F}(\mathsf{x}) + \mathsf{e}$. When $\mathcal{F}(\mathsf{x})$ is a statistic (e.g., a mean salary estimate or neural-network weights), e may be independent Gaussian [10] or Laplace [9] noise, as is standard in the DP literature. For physical signals $\mathcal{F}(\mathsf{x})$, e may also be introduced physically, e.g., by sending dummy queries in anonymous communication systems [18]. In any case, privacy generally entails a fundamental tradeoff between information propagation (utility) and preservation and the utility loss depends on the application: in anonymous communication, for example, the loss may be the delay introduced by dummy messages, with the relevant metric determined by latency sensitivity or protocol constraints.

Given the complexity of practical leakage functions $\mathcal{F}(\cdot)$ and the diversity of utility metrics, it is desirable to have a *universal* framework that addresses ❷ (tight measurement) and ❸ (optimal mitigation) simultaneously. Unfortunately, in the Type-(B) secret regime, neither tight measurement nor optimal mitigation can be efficiently determined under III-style definitions in general. Indeed, Xiao and Tao [19] show that computing sensitivity, the maximal outcome difference

$$\sup_{\bar{\mathsf{x}},\bar{\mathsf{x}}'} \|\mathcal{F}(\bar{\mathsf{x}}) - \mathcal{F}(\bar{\mathsf{x}}')\|$$

over arbitrary secret hypotheses $(\bar{\mathsf{x}}, \bar{\mathsf{x}}')$, is NP-hard in the worst case. This suggests that even *verifying* whether a leakage function $\mathcal{F}(\cdot)$ satisfies a given III-style guarantee can be computationally intractable.

As a consequence, known III-style privacy analyses for Type-(B) secrets typically apply only in specific *white-box* settings, and one of the most representative methodologies is *decompose-then-compose*. In particular, over the last two decades the DP literature has developed a rich collection of such tools, including *subsample-then-aggregate* [20], the *Sparse Vector Technique* [21], *Differentially Private SGD* (DP-SGD) [22–24], PATE [25], and *private evolution* [26]. In these frameworks, a complex processing pipeline is decomposed into simpler components[4], each of which is separately privatized and then composed. However, this methodology can lead to overly conservative (loose) noise requirements, especially for high-dimensional tasks with intricate dependencies among sensitive inputs, such as clustering [27] and machine learning model inference [28]. These limitations can be even more severe for settings without tractable algorithmic structure, including many of the physical side channels [14,16,17] mentioned earlier.

In contrast, for Type-(A) entropic secrets, the landscape can be starkly different. Some special Bayesian decision problems, most notably exact identification under a specified prior, have sharp information-theoretic characterizations through Arimoto–Rényi conditional entropy [29] and related hypothesis-testing quantities [30]. Recent work PAC Privacy [1,31] demonstrates that universal posterior-risk analysis is possible even in a restricted *black-box* setting. Here, *black box* means that we assume no particular algorithmic structure for $\mathcal{F}(\cdot)$; instead, the user only has oracle access to end-to-end evaluations of $\mathcal{F}(\cdot)$. In the context of ❷ and ❸, this means that both risk measurement and mechanism design must be derived from a finite number m of evaluations of $\mathcal{F}(\cdot)$ on selected inputs $\mathsf{X} = \{\bar{\mathsf{x}}_i : i \in [m]\}$. Follow-up works [32–36] show that the provable Bayesian upper bounds derived from PAC Privacy can be significantly sharper than the prior-free worst case in many applications as the secret's inherent entropy makes inference intrinsically harder.

[4]Ideally, each component in a *decompose-then-compose* framework ensures bounded sensitivity $\sup_{\bar{\mathsf{x}},\bar{\mathsf{x}}'} \|\mathcal{F}(\bar{\mathsf{x}}) - \mathcal{F}(\bar{\mathsf{x}}')\| \leq c$ for some threshold c under an appropriate metric $\|\cdot\|$. In the DP context, a common structure is *aggregation*: each data point can be clipped to a bounded range and contributes *independently* to the released output [22].

Fundamentally, this difference in addressing ❷ and ❸ with respect to Type-(A) and Type-(B) secrets arises because an III-style guarantee in a Type-(B) scenario demands divergence control between leakages from *arbitrary* pairs of secret hypotheses.[5] This strict "arbitrary" requirement is the root cause of the computational impossibility [19]. For Type-(A) secrets, however, the Bayesian posterior risk is dominated by the case of *specified* ground-truth distribution $\mathcal{D}_{\mathsf{x}}$ [37], as an adversary using a mis-specified or biased prior *cannot* outperform the optimal Bayesian adversary using $\mathcal{D}_{\mathsf{x}}$. Additionally, even if extreme worst cases exist and are hard to identify, they may be negligible under $\mathcal{D}_{\mathsf{x}}$, enabling near-optimal Bayesian analysis. Moreover, higher entropy in x (more uncertainty) should allow substantially cheaper privacy mechanisms than conservative prior-free worst-case approaches.

However, two tightness questions remained open for (black-box) posterior-risk privacy in Type-(A) regime: how to calibrate randomization so that the certified risk is conservative at finite samples yet asymptotically exhausts the requested privacy budget, and how to compose such guarantees under adversarially adaptive reuse of the same entropic secret. Existing f-divergence/Fano-style certificates can be loose because they summarize the Bayesian decision problem through coarser Bernoulli-kind divergence statistics. We formalize the first tightness problem for static leakage in the black-box setting as follows.

Problem 1 (Tight black-box privacy analysis with entropy)**.** *Given a prior distribution $\mathcal{D}_{\mathsf{x}}$ of an entropic secret x and an arbitrary leakage function $\mathcal{F}(\cdot) : \mathcal{X} \to \mathcal{O}$, we want to design a randomized mechanism-selection algorithm $\mathsf{Alg} : (\mathcal{X} \times \mathcal{O})^m \to \mathcal{R}$ that outputs a release rule $R \in \mathcal{R}$ with small overhead (measured by some metric $\kappa(R)$), ideally approaching the best feasible utility–privacy tradeoff within the admissible class $\mathcal{R}$, such that for an arbitrary inference task captured by a report space $\mathcal{A}_\rho$, criterion $\rho : \mathcal{A}_\rho \times \mathcal{X} \to \{0, 1\}$, and target posterior success rate δ_ρ, the following experiment is* impossible *for an informed adversary:*

1. *The user selects m strings $\mathsf{X} = \{\bar{\mathsf{x}}_i \in \mathcal{X} : i \in [m]\}$ [6], evaluates $\mathcal{F}(\bar{\mathsf{x}}_i)$, and applies Alg to the simulation data $\mathsf{S} = \{(\bar{\mathsf{x}}_i, \mathcal{F}(\bar{\mathsf{x}}_i)) : i \in [m]\}$. The algorithm outputs a randomized release rule $\mathsf{Alg}(\mathsf{S}) = R$.*

2. *The user samples an $\mathsf{x} \sim \mathcal{D}_{\mathsf{x}}$, samples a released value $Y \sim R(\cdot \mid \mathcal{F}(\mathsf{x}))$ (e.g., $Y = \mathcal{F}(\mathsf{x}) + \mathsf{e}$ for additive perturbation e), and sends Y to the adversary.*

3. *The adversary, who has full knowledge of $\mathcal{F}(\cdot)$ and the user's mechanism Alg, outputs a report $\hat{a} \in \mathcal{A}_\rho$ such that $\rho(\hat{a}, \mathsf{x}) = 1$ with probability greater than δ_ρ.*

Throughout, we always assume the secret's prior distribution $\mathcal{D}_{\mathsf{x}}$ is specified and the user can sample from it and evaluate its mass. We follow the PAC Privacy formalization [1] and model an arbitrary adversarial inference task via a criterion function $\rho(\cdot, \cdot)$, measuring privacy risk by the Bayesian posterior success probability δ_ρ. The criterion ρ can be chosen so that the preimage of $\rho(\cdot, \mathsf{x}) = 1$ captures all *unacceptable* reports about x to the user. For instance, given an l-bit secret string x, exact identification is the special case $\mathcal{A}_\rho = \mathcal{X}$ and $\rho(\hat{a}, \mathsf{x}) = 1$ iff $\hat{a} = \mathsf{x}$; approximate reconstruction with at most c-bit errors can be captured by reports $\hat{a} \in \mathcal{X}$ satisfying $\rho(\hat{a}, \mathsf{x}) = 1$ iff the Hamming distance between $\hat{a}$ and x is at most c. As for the utility-loss metric κ, a common choice for additive mechanisms is the variance of the injected noise e; in general, κ may be any application-specific expected quality loss of randomized rule R.

[5] This is necessary if one wants to upper bound the posterior advantage under *arbitrary* priors [8].

[6] **Problem** 1 allows any finite black-box evaluation design; our concrete calibration algorithms later use i.i.d. samples from $\mathcal{D}_{\mathsf{x}}$.

It is worth noting that the protection strategy need *not* be restricted to input-independent additive noise. In full generality, R can be a randomized channel whose conditional law depends on the raw leakage instance, public context, or realized transcript available to the user. Independent perturbation as used in many existing works [10,22], is only one straightforward instantiation, which can be suboptimal as shown in experiments Fig. 2, 3 (f). For notation simplicity, later algorithms still write the decision variable as an additive perturbation $\mathsf{e} \sim \mathcal{D}_e$ while $\mathcal{D}_e$ can be input-dependent. We also emphasize the adversarial model in the following remark.

Remark 1 (White-Box Adversarial View)**.** *Throughout the paper, the* black-box *assumption is from the user's perspective: the user lacks structural knowledge of how leakage is incurred. The adversary, in contrast, is fully informed of the leakage function $\mathcal{F}$, the prior $\mathcal{D}_{\mathsf{x}}$, the inference task, the admissible mechanism class, and the user's calibration algorithm* Alg*. This follows the standard Kerckhoffs' principle [38]: algorithms and distributions are public, while only realized private randomness is hidden. In particular, when* Alg *samples private training and validation sets to select a release rule, the adversary knows the law of this compound randomized mechanism but does not observe the realized calibration samples, the realized branch-specific rule selected from them, or fresh noise draws. The posterior success probability δ_ρ is defined with respect to all remaining randomness in the secret generation and the randomized release process.*

A randomized release rule is viewed as a public algorithm together with private coins. The calibration samples used by Alg, the sampled perturbations, and the adversary's own random choices are all part of the probability space over which the released randomized mechanism and the posterior probability δ_ρ is defined. That is to say, the adversary may know the complete distribution induced by these coins, but not their realized values in a particular execution.

1.3 Privacy Composition

So far, we have focused on a static leakage function $\mathcal{F}(\cdot)$. In practice, leakage may arise in a potentially adversarial interactive environment, and when the secret is reused, we must track its cumulative leakage. To model leakage in such a dynamic setting, we consider a more general function $\mathcal{F}(\mathsf{x}, \mathsf{v})$, where an environment parameter v can be adaptively selected by an adversary to maximize information gained through sequential leakages. For example, given a secret key x, a user may be asked to encrypt multiple messages $\{\mathsf{v}_1, \mathsf{v}_2, \dots, \mathsf{v}_T\}$, which can be adversarially chosen; each encryption produces leakage $\mathcal{F}(\mathsf{x}, \mathsf{v}_t)$. Intuitively, an adversary can adaptively choose the next query v_t based on accumulated information about x, gradually maximizing their posterior success probability.

Existing composition results primarily address Type-(B) secrets under III-style guarantees, and have largely been developed in the DP literature [39–43]. Tighter composition accounting has also motivated new privacy definitions, evolving from pure ϵ-DP to approximate (ϵ, δ)-DP, to concentrated DP [41] and Rényi DP [42], and most recently to f-DP [43]. Typically, these results rely on two key prerequisites: (i) a *global* worst-case III guarantee for *all* possible leakage functions $\mathcal{F}(\cdot, \mathsf{v})$ (independent of all auxiliary information and adversarial adaptivity), and (ii) *independence* of the randomness used at each step of the composed mechanism: the global worst-case guarantee (i) bounds the posterior advantage increment regardless of the adversary's query, while independence in (ii) enables aggregation via concentration or convolution techniques.

For Type-(A) entropic secrets in the *black-box* regime, however, the situation changes dramatically and existing III-style composition analyses do *not* directly apply. First, in a black-box setting, we cannot generally obtain non-trivial input-independent guarantees from finitely many end-to-end simulations. Second, and more importantly, the entropic secret x is generated once and reused: even if fresh randomization is applied at each step, the resulting leakages remain *correlated* through the shared secret. We formalize the composition version of **Problem** 1 as follows.

Problem 2 (Mechanism for tight composition)**.** *Given a prior secret distribution $\mathcal{D}_{\mathsf{x}}$, a fixed privacy budget in terms of posterior success rate δ_ρ for a fixed inference task captured by $\rho : \mathcal{A}_\rho \times \mathcal{X} \to \{0, 1\}$, and a family of (black-box) leakage functions $\mathcal{F}(\cdot, \cdot) : \mathcal{X} \times \mathcal{V} \to \mathcal{O}$, we want to design a joint randomization mechanism* Alg *such that there does* not *exist an adversarial query algorithm $\mathcal{Q}_{adv}$ for which the following experiment between the user and adversary is possible:*

1. *The user samples a secret $\mathsf{x} \sim \mathcal{D}_{\mathsf{x}}$.*

2. *At iteration $t \in [T]$, from the adversary side, possibly based on all previously observed leakages $\{\mathsf{o}_j = \mathcal{F}(\mathsf{x}, \mathsf{v}_j) + \mathsf{e}_j : j \in [t-1]\}$ and past queries $\{\mathsf{v}_j : j \in [t-1]\}$, the adversary chooses a new query $\mathsf{v}_t \leftarrow \mathcal{Q}_{adv}(\{\mathsf{o}_j, \mathsf{v}_j\}_{j<t})$ and sends v_t to the user.*

3. *At iteration $t \in [T]$, from the user side, after receiving the adversarial query v_t, the user performs end-to-end simulations of the leakage functions $\{\mathcal{F}(\cdot, \mathsf{v}_j) : j \in [t]\}$ on a selected input set $\mathsf{X}^{(t)} = \{\bar{\mathsf{x}}_i^{(t)} : i \in [m]\}$, and applies* Alg *to the resulting evaluations*
$$\mathsf{S}^{(t)} = \{(\bar{\mathsf{x}}_i^{(t)}, \mathcal{F}(\bar{\mathsf{x}}_i^{(t)}, \mathsf{v}_j)) : i \in [m],\, j \in [t]\},$$
together with the previously sampled noises $\mathbf{e}^{(t-1)} = (\mathsf{e}_1, \mathsf{e}_2, \ldots, \mathsf{e}_{t-1})$, to determine a noise distribution $\mathcal{D}_e^{(t)}$. The user's utility objective may adapt to the realized transcript, but the inference task and privacy budget remain the user-specified privacy requirement throughout the interaction. The user then samples $\mathsf{e}_t \sim \mathcal{D}_e^{(t)}$ and returns the noisy leakage $\mathsf{o}_t = \mathcal{F}(\mathsf{x}, \mathsf{v}_t) + \mathsf{e}_t$ to the adversary.

4. *After T iterations, the adversary outputs a report $\hat{a} \in \mathcal{A}_\rho$ based on $\{(\mathsf{o}_t, \mathsf{v}_t)\}_{t\in[T]}$ such that $\rho(\hat{a}, \mathsf{x}) = 1$ with probability greater than δ_ρ.*

Fig. 1 illustrates the technical challenge behind **Problem** 2, which we refer to as the *parallel universe challenge*. With an adaptive adversary, the joint distribution of the composed leakages $\{\mathcal{F}(\mathsf{x}, \mathsf{v}_t)\}_{t=1}^T$ is jointly determined by the (unknown) adversarial strategy $\mathcal{Q}_{\text{adv}}$ and the user's actions (secret sampling and noise injection). Under the black-box model, the user cannot infer $\mathcal{Q}_{\text{adv}}$ or the induced global leakage distribution required for a direct Bayesian analysis. The user's only visibility into $\mathcal{Q}_{\text{adv}}$ is through the adversary's reaction *conditional* on the particular randomness realized in the user's own execution (blue line only in Fig. 1). Concretely, even conditioning on a first query v_1 and a first-round noise distribution $\mathcal{D}_e^{(1)}$, the user observes only the adversary's next query v_2 corresponding to the specific sampled noise instance e_1. Different noise realizations $\mathsf{e}_1' \sim \mathcal{D}_e^{(1)}$ could induce different adversarial branches—analogous to "parallel universes" inaccessible to the user (yellow and orange dash lines). **Problem** 2 thus asks the user to infer and mitigate leakage from the *local* universe they observe, while ensuring that aggregating over all (unobserved) branches yields the desired global guarantee.

Motivated by **Problem** 2, a further generalization is to develop composition theory that is agnostic to the specific privacy-preserving mechanism used at each step. A natural application setting is one where multiple parties share a secret but are processing the secret separately, and each party lacks knowledge of the leakage functions and defenses used by others. That is, given T black-box leakage functions $\{\mathcal{F}(\mathsf{x}, \mathsf{v}_t) : t \in [T]\}$ that share the same entropic secret x, and given only that each *marginal* leakage satisfies certain posterior inference hardness with respect to x, how can we upper bound the cumulative Bayesian privacy risk? This motivates the following problem.

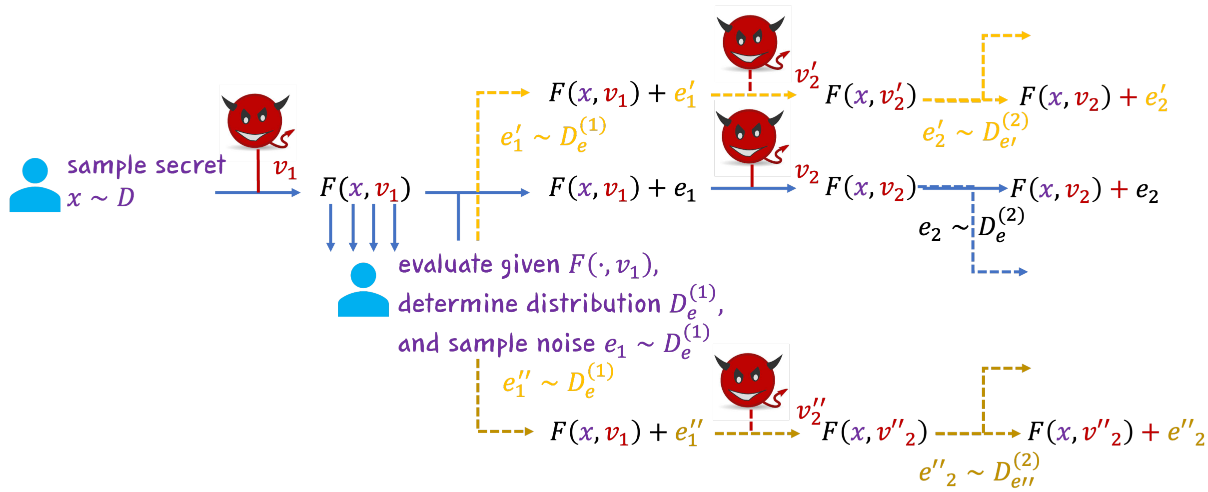

Figure 1: Illustration of the *Parallel Universe Challenge* in adversarially adaptive composition for entropic secrets. The solid blue line corresponds to the specific (local) realization observed by the user, while dashed lines represent alternative (unobserved) realizations induced by randomness in both the secret and the mechanism.

Table 1. Comparison of different approaches including Differential Privacy (DP) and Maximal Leakage (MaxL) [12], representatives of Input-Independent Indistinguishability (III) philosophy; mutual information(MI) based methods [44]; PAC Privacy; and our established framework. As a side note, in MaxL [12] and PAC [1], only static composition (i.e., queries are fixed in advanced) is known, while the adversarially-adaptive setting remained open.

	DP	MaxL	MI	PAC	Ours
Black-box setting	✗	✗	✗	✓	✓
Tight utility-privacy tradeoff	✗	✓	✗	✗	✓
Adversarial composition	✓	✗	✗	✗	✓

Problem 3 (Mechanism-agnostic composition)**.** *Given a random secret* $\mathsf{x} \sim \mathcal{D}_{\mathsf{x}}$ *and an arbitrary sequence of* T *leakage functions* $\{\mathcal{F}(\cdot, \mathsf{v}_t) : t \in [T]\}$*, suppose that, conditioned on the shared secret input* x*, all additional randomness in each* $\mathcal{F}(\cdot, \mathsf{v}_t)$ *is independent across* $t \in [T]$*. Under what types of* marginal *measurements for each* $\mathcal{F}(\cdot, \mathsf{v}_t)$ *can we derive a global bound on the cumulative Bayesian risk?*

Table 1 is meant as an operational comparison. "Black-box setting" means finite end-to-end leakage mitigation without structural information about $\mathcal{F}$. "Tight utility-privacy tradeoff" means that operationally the randomization strategy can be tightly optimized under a given privacy budget. "Adversarial composition" means an accounting for repeated exposure of the same secret under adaptively selected leakages. The checkmarks therefore refer to the availability of a *general* framework in the corresponding regime, not to every specialized instantiation of the listed definitions.

1.4 Techniques Overview

In this section, we provide a roadmap of techniques established to solve **Problems** 1–3.
Learning Privacy: At a high level, our methodology is analogous to *statistical machine learning* [45]. The desired privacy–utility tradeoff can be formulated as a constrained population optimization problem: minimizing utility loss subject to a prescribed privacy-budget constraint. In

the Type-(A) entropic-secret regime, privacy is determined by the secret distribution $\mathcal{D}_{\mathsf{x}}$ and the leakage function $\mathcal{F}$. However, because $\mathcal{F}$ is available only through black-box evaluations, the privacy constraint are thus initially *unseen*. This motivates us to replace the population optimization problem with a *solvable* empirical optimization problem constructed from finitely many samples and leakage evaluations observed. Our results do not require a new general-purpose numerical optimizer: any returned rule that passes the validation certificate is privacy-valid, while the achieved utility depends on how closely the empirical optimizer approaches the feasible privacy–utility frontier.

This learning perspective raises two central challenges. First, we need a privacy-constraint function that is aligned with the Bayesian decision rule: finite-sample calibration must use a conservative certificate, while the optimized population certificate should converge to the exact posterior risk. Second, analogous to controlling *generalization error* in learning theory, we must upper bound the gap between the empirical objective and the unseen population objective caused by finite sampling. Crucially, this bound must still imply rigorous privacy guarantees for the resulting randomized mechanism.

A subtle issue is that the validation loss becomes a higher-order PRW moment as α increases. Without additional structure, the resulting range term can make high-confidence validation expensive. We therefore also identify a finite-output mechanism class that avoids this bottleneck: when the output randomization channel is commonly dominated by a fixed reference distribution, the PRW validation range is controlled at the target posterior-risk scale. This gives a high-confidence correction whose relative size is $O(\sqrt{\log(1/\gamma)/m})$, rather than one with an uncontrolled α-exponential factor.

To illustrate, we begin with a special case of **Problem** 1 where the inference task ρ is fixed to identification (exact reconstruction). In this case, the adversary's optimal inference rule $\mathcal{A}(o)$ given an observation $o = \mathcal{F}(\mathsf{x})$ is the classical *Maximum A Posteriori* (MAP) estimator [46]:

$$\mathcal{A}(o) \in \arg\max_{\hat{\mathsf{x}}} \Pr_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}\big(\mathsf{x} = \hat{\mathsf{x}} \mid \mathcal{F}(\mathsf{x}) = o\big), \tag{1}$$

i.e., the adversary outputs the hypothesis with maximal posterior probability. Write $S(o) := \arg\max_z \Pr_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}\big(\mathsf{x} = z \mid \mathcal{F}(\mathsf{x}) = o\big)$, and accordingly, the optimal identification success probability, denoted by δ_I, can be written as an aggregate of weighted 0–1 losses:

$$\begin{aligned}
\delta_I &= \Pr_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}\big[\mathcal{A}(\mathcal{F}(\mathsf{x})) = \mathsf{x}\big] \\
&= \sum_{\hat{\mathsf{x}}} \Pr_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}(\mathsf{x} = \hat{\mathsf{x}}) \cdot \int_o \mathbf{1}[\hat{\mathsf{x}} \in S(o)] \cdot \Pr(\mathcal{F}(\hat{\mathsf{x}}) = o)\,\mathrm{d}o \\
&= \mathbb{E}_{\hat{\mathsf{x}}\sim\mathcal{D}_{\mathsf{x}}}\bigg[\int_o \mathbf{1}[\hat{\mathsf{x}} \in S(o)] \cdot \Pr(\mathcal{F}(\hat{\mathsf{x}}) = o)\,\mathrm{d}o\bigg].
\end{aligned} \tag{2}$$

Here, $\mathbf{1}(C) = 1$ iff condition C holds. Equation (2) shows that δ_I is the probability mass of observation events for which the *true* secret happens to be MAP-optimal given the observed leakage.

Unfortunately, in the black-box setting, directly computing (2) is typically infeasible because the secret space $\mathcal{X}$ may be exponentially large. However, the expectation form in (2) suggests a route toward ❷: if there exists a *bounded* function $\mathcal{G}(\cdot)$ such that

$$\delta_I = \mathbb{E}_{\bar{\mathsf{x}}\sim\mathcal{D}_{\mathsf{x}}}\big[\mathcal{G}\big(\bar{\mathsf{x}}, \mathcal{F}(\bar{\mathsf{x}})\big)\big], \tag{3}$$

then the empirical average $\frac{1}{m}\sum_{i=1}^{m} \mathcal{G}\big(\bar{\mathsf{x}}_i, \mathcal{F}(\bar{\mathsf{x}}_i)\big)$ over samples $\mathsf{X} = \{\bar{\mathsf{x}}_i\}_{i=1}^{m}$ is an unbiased estimator of δ_I, and concentration inequalities (e.g., Hoeffding's) yield non-asymptotic error bounds.

To incorporate optimal mitigation (❸), we can represent a privacy mechanism by a perturbation term $\mathsf{e} \sim \mathcal{D}_e$ and formulate a constrained optimization problem: under a utility loss measure κ and a privacy requirement $\delta_I \leq \epsilon$,

$$\min_{\mathcal{D}_e} \ \kappa(\mathcal{D}_e) \quad \text{s.t.} \quad \mathbb{E}_{\bar{\mathsf{x}} \sim \mathcal{D}_{\mathsf{x}}}\left[\mathcal{G}\big(\bar{\mathsf{x}}, \mathcal{F}(\bar{\mathsf{x}})\big)\right] \leq \epsilon. \tag{4}$$

We then consider replacing the unseen population expectation constraint in (4) with an empirical proxy:

$$\min_{\mathcal{D}_e} \ \kappa(\mathcal{D}_e) \quad \text{s.t.} \quad \frac{1}{m}\sum_{i=1}^{m} \mathcal{G}\big(\bar{\mathsf{x}}_i, \mathcal{F}(\bar{\mathsf{x}}_i)\big) \leq \epsilon'. \tag{5}$$

To ensure that solutions satisfying (5) also satisfy (4) with high probability (i.e., to control generalization error), a standard approach is to choose a stricter threshold $\epsilon' < \epsilon$ and bound the gap $\epsilon - \epsilon'$ via uniform convergence or complexity measures (e.g., Rademacher complexity) [47]. In black-box settings, one may alternatively adopt a *propose–verify* strategy that gradually increases noise e until an independent test (using fresh samples) is passed.

All of the above depends on constructing such a function $\mathcal{G}$, which is generally challenging. Building on PAC Privacy [1] and α-mutual information [12, 30, 48], we establish a family of prior-reference-weighted (PRW) α-certificates $\{\tilde{\mathcal{G}}_\alpha : \alpha \in \mathbb{R}^+\}$, which represent δ_I in the optimized limit:

$$\delta_I = \mathbb{E}_{\bar{\mathsf{x}} \sim \mathcal{D}_{\mathsf{x}}}\left[\lim_{\alpha \to \infty} \tilde{\mathcal{G}}_\alpha\big(\bar{\mathsf{x}}, \mathcal{F}(\bar{\mathsf{x}})\big)\right]. \tag{6}$$

More importantly, for any fixed α, we have the conservative upper bound

$$\delta_I \leq \mathbb{E}_{\bar{\mathsf{x}} \sim \mathcal{D}_{\mathsf{x}}}\left[\tilde{\mathcal{G}}_\alpha\big(\bar{\mathsf{x}}, \mathcal{F}(\bar{\mathsf{x}})\big)\right],$$

so non-asymptotic calibration based on a fixed $\tilde{\mathcal{G}}_\alpha$ remains safe, while larger α reduces the approximation gap. Building on the identification case, Section 3.2 extends the same PRW construction to arbitrary report spaces and arbitrary inference criteria ρ, which completes the first step toward **Problem** 1.

Composition: For composition, as illustrated in Fig. 1, the central challenge is that intermediate randomness and adversarial adaptivity induce a global leakage distribution: a mixture over potentially infinitely many conditional branches ("parallel universes"), while the user observes only a single realized path (the "local universe"). We fix the privacy semantics, inference task, privacy budget, and PRW order α across the interaction; the adversary adaptively chooses leakage queries from the transcript, and the user may adapt utility objectives and randomization choices subject to the fixed privacy requirement. Rather than performing a direct Bayesian analysis of the global mixture, we impose transcript-dependent local PRW constraints. If all alternate branches satisfy their respective local equalities, the collection of local controls yields a globally tight finite-α PRW accounting identity for the complete adaptive transcript mechanism. Thus finite α gives a conservative global posterior-risk upper bound, and the accounting becomes exact for Bayesian posterior risk in the optimized limit as $\alpha \to \infty$.

To capture this, we consider decomposing the overall privacy risk accounting into a sequence of smaller subproblems, each solvable within its appropriate "universe". Once composed, these local controls provide a global privacy guarantee. We consider two composition setups, distinguished by whether the per-round mechanism may depend on the realized transcript of previous rounds.

In the *history-accessible* setting, the user is allowed to condition each round's mechanism on the full interaction history, including previously selected mechanisms, realized outputs, and adversarial queries. This allows one to track a running "leakage score" for each candidate secret along the current transcript. Consequently, this enables a *branch-dependent* decomposition: per-round subproblems may depend on the current history, allowing the user to adaptively calibrate the randomization at each round.

Conversely, in the *mechanism-agnostic* setting, the decomposition must be determined *independently* of the realized history. In this scenario, the adversary may concentrate on the most vulnerable region of the secret space to maximize inference success, and the user is therefore compelled to enforce a worst-case (uniform) guarantee over all possible, unseen histories with carefully-selected separate measurement of each marginal leakage function.

1.5 Contributions

To summarize, we make five contributions.
(**C1**) We introduce Prior–Reference-Weighted (PRW) α-information (Definition. 6), a posterior-risk certificate valid for every finite $\alpha > 1$ and asymptotically tight as $\alpha \to \infty$. The exact-identification case recovers a known Arimoto–Rényi geometry, but the PRW formulation extends the certificate to *arbitrary* priors and arbitrary Bayesian inference criteria ρ (Theorems 1 and 2). Unlike the f-divergence Fano bound (Lemma 1), which compresses the prior into a single baseline success rate, PRW retains the full prior through the weights and therefore remains tight under non-uniform priors and non-identification tasks (Appendix A.4).
(**C2**) We reduce black-box privatization to empirical utility minimization under a PRW constraint (Algorithm 1 and Theorem 3). For exact identification, the PRW certificate is an expectation of a bounded per-secret loss, so any validation-passing output gives a conservative privacy certificate for the resulting compound randomized channel. General finite report-space tasks use the population PRW certificate of Theorem 2, with finite-sample calibration requiring the corresponding task-score plug-in concentration. As sample size grows and α increases, the PRW certificate approaches the posterior-risk frontier and the mechanism can use the prescribed privacy budget with vanishing slack. The analysis separates optimization error, finite-sample validation error, fallback risk, and finite-α approximation error.
(**C3**) We identify and remove a finite-α sample-complexity bottleneck for an important finite-output mechanism class (Theorem 4). In the most general case, Algorithm 1 needs a range bound for a high-order PRW loss, and this range can be exponentially large in α. We show that if the output randomization channel is commonly dominated by a fixed reference distribution, then the PRW validation range is bounded and the relative confidence correction becomes $O(\sqrt{\log(1/\gamma)/m})$ with no α-exponential factor.
(**C4**) We give a history-accessible adaptive-composition theorem (Section 4) for a fixed inference task, privacy budget, and PRW order α: per-round local conditions on the realized transcript compose into a globally tight finite-α PRW accounting certificate, despite the user never observing the counterfactual interaction tree. There is no additional composition slack beyond the finite-order PRW relaxation, while utility calibration remains local rather than globally optimal in an online setting over the entire adaptive tree.
(**C5**) We give mechanism-agnostic composition bounds from marginal KL and α divergence measurements (Theorems 9 and 10), and show the cost of agnosticism: KL composition requires a worst-case cap, while α-divergence composition requires orders that grow with the round count T, approaching worst-case control.

1.6 Relation to Bayesian Leakage Measures

Our work builds on a long line of Bayesian and quantitative information-flow measures, including Bayes vulnerability and min-entropy leakage [49], gain-function or g-leakage [6], Arimoto–Rényi and Sibson information [29, 48, 50], maximal leakage [12], maximal α-leakage [51], and PAC Privacy [1]. These works are central to the interpretation of posterior inference risk. Our contribution is not to claim any new gain functions or Bayesian decision tasks. Rather, we isolate the properties needed for *tight, black-box* privatization: 1) a conservative certificate, 2) asymptotic tightness for the target posterior success rate, 3) empirical calibration from sampled leakage evaluations with high-confidence validation, 4) finite-output conditions that avoid high-order sample-complexity blowup, and 5) adaptive composition for a reused entropic secret.

Bayes vulnerability [49] and g-leakage [6] already provide the right semantic language for adversarial gain functions under a prior: for a fixed channel, they characterize the adversary's optimal decision value. However, they do not by themselves give a sample-calibratable mechanism-selection algorithm for an arbitrary (black-box) leakage procedure, nor an adaptive composition theorem for reuse of the same secret. Arimoto/Sibson information [48, 50] and maximal α-leakage [51] give sharp operational meanings for important families of inference losses and recover exact-identification geometry as special cases, but they are primarily measurement concepts: they do not address how a user with only finite black-box leakage evaluations should choose a randomized release rule and their tightness is not guaranteed beyond identification tasks.

PAC Privacy [1] takes the crucial operational step of turning Bayesian posterior risk into a black-box auditing and privatization problem. Its f-divergence/Fano certificate is broadly applicable and sample-estimable, but it can be loose because it compresses the posterior decision problem into a Bernoulli divergence around a baseline success rate. PRW is designed to fill this remaining gap: it retains the prior-weighted likelihood geometry of the MAP/gain decision rule, gives a conservative finite-α certificate, becomes tight for arbitrary finite report-space tasks as $\alpha \to \infty$, and supports the local-to-global adaptive accounting developed in Section 4.

2 Preliminaries

2.1 Statistical Divergence

We start with some useful distributional-distance measure.

Definition 1 ((Generalized) f-Divergence)**.** *Let $f : (0, +\infty) \to \mathbb{R}$ be a convex function. Let P and Q be two probability distributions defined on the same measurable space, with P absolutely continuous with respect to Q. The f-divergence between P and Q is defined as*

$$D_f(P \,\|\, Q) \;:=\; \mathbb{E}_{X\sim Q}\left[f\left(\frac{\mathrm{d}P}{\mathrm{d}Q}(X)\right)\right]. \tag{7}$$

Here, $\mathrm{d}P$ (resp., $\mathrm{d}Q$) denotes the probability density function in the continuous case or the probability mass function in the discrete case.

The classical definition of f-divergence [52] further requires $f(1) = 0$ so that identical distributions have divergence zero; this normalization is not needed for our analysis. Many commonly-used statistical divergences are special cases of Definition 1. For example, KL divergence and total variation correspond to $f(z) = z \log z$ and $f(z) = |z - 1|$, respectively. The main variant used in this paper is the α-divergence, obtained from $f(z) = z^{\alpha}$.

Definition 2 (α-Divergence)**.** *Let P and Q be two probability distributions defined on the same measurable space. For any $\alpha > 1$, $D_\alpha(P \,\|\, Q) := \mathbb{E}_{\mathsf{x}\sim Q}\left[\left(\frac{\mathrm{d}P}{\mathrm{d}Q}(\mathsf{x})\right)^{\alpha}\right]$.*

The well-known Rényi divergence [42] is a logarithmic transformation of the α-divergence, $\frac{1}{\alpha-1}\log D_\alpha(P \,\|\, Q)$.

2.2 Privacy Definitions

In the following, we cover the background of privacy definitions. As discussed earlier, depending on the type of secret to protect, there are two main philosophies: *Input-Independent Indistinguishability* (III) for Type-(B) secrets and Bayesian posterior risk for Type-(A) entropic secrets.

Definition 3 (ϵ input-independent indistinguishability [7–9, 11])**.** *Given a leakage function $\mathcal{F}(\cdot) : \mathcal{X} \to \mathcal{O}$ and an arbitrary pair of* adjacent *secret hypotheses $\{\bar{\mathsf{x}}, \bar{\mathsf{x}}'\}$ that differ only in the sensitive information to be protected, $\mathcal{F}(\cdot)$ satisfies ϵ-III if observing the leakage changes the ability to distinguish $\bar{\mathsf{x}}$ from $\bar{\mathsf{x}}'$ by at most ϵ. Equivalently, in an information-theoretic statistical form, for every test $T : \mathcal{O} \to \{0, 1\}$,*

$$\left|\Pr[T(\mathcal{F}(\bar{\mathsf{x}})) = 1] - \Pr[T(\mathcal{F}(\bar{\mathsf{x}}')) = 1]\right| \leq \epsilon.$$

Computational variants restrict the class of tests T, while DP- and Pufferfish-style definitions replace this additive form by their corresponding worst-case indistinguishability inequalities.

For Type-(A) entropic secrets as our main focus, a general and formal definition of Bayesian risk is *PAC Privacy* [1].

Definition 4 (($\mathcal{D}_{\mathsf{x}}, \rho, \delta_\rho$) PAC Privacy [1])**.** *For a leakage function $\mathcal{F} : \mathcal{X} \to \mathcal{O}$, let $\mathcal{D}_{\mathsf{x}}$ be the prior distribution of secret x over $\mathcal{X}$, and let $\mathcal{A}$ be an arbitrary report space. Let $\rho : \mathcal{A} \times \mathcal{X} \to \{0, 1\}$ be an inference criterion: $\rho(\hat{a}, \mathsf{x}) = 1$ iff an adversarial report $\hat{a} \in \mathcal{A}$ is successful. Then $\mathcal{F}$ satisfies $(\mathcal{D}_{\mathsf{x}}, \rho, \delta_\rho)$ PAC Privacy if the following experiment is impossible:*

A user samples $\mathsf{x} \sim \mathcal{D}_{\mathsf{x}}$ and releases $\mathcal{F}(\mathsf{x})$ to an informed adversary who knows $\mathcal{D}_{\mathsf{x}}$, $\mathcal{F}$, and ρ. The adversary returns an estimate $\hat{a} \in \mathcal{A}$ such that with probability greater than δ_ρ, $\rho(\hat{a}, \mathsf{x}) = 1$.

When there is no contextual ambiguity, we refer to $(\mathcal{D}_{\mathsf{x}}, \rho, \delta_\rho)$ PAC Privacy simply as δ_ρ-privacy. Unless stated otherwise, tightness claims below refer to this Bayesian posterior-risk budget. Appendix A.1 gives additional details on the threat model and algorithmic assumptions.

2.3 Bayesian Analysis Tools

In this section, we introduce existing information-theoretic tools for Bayesian posterior-risk analysis. One operationally useful result, which we call the *f-divergence Fano inequality*, is the main theorem of PAC Privacy (Theorem 1 in [1]). It upper bounds the target posterior success rate δ_ρ through an expectation over per-secret/sample leakage divergences.

Lemma 1 (f-divergence Fano Inequality [1])**.** *For any selected f-divergence D_f, an arbitrary leakage function $\mathcal{F} : \mathcal{X} \to \mathcal{O}$ satisfies $(\delta_\rho, \rho, \mathcal{D}_{\mathsf{x}})$* PAC Privacy *where the adversarial posterior success rate δ_ρ can be selected as the minimal number satisfying the following inequality:*

$$\begin{aligned} \Delta_{f,\rho} := D_f\big(\mathbf{1}_{\delta_\rho} \,\big\|\, \mathbf{1}_{\delta_{o,\rho}}\big) \;&\geq\; \inf_{\mathbb{P}_W} D_f\big(\mathbb{P}_{\mathsf{x},\mathcal{F}(\mathsf{x})} \,\big\|\, \mathbb{P}_{\mathsf{x}} \otimes \mathbb{P}_W\big) \\ &= \inf_{\mathbb{P}_W}\; \mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}\big[\, D_f\big(\mathbb{P}_{\mathcal{F}(\mathsf{x})} \,\big\|\, \mathbb{P}_W\big)\big]\,; \end{aligned} \tag{8}$$

Here, $\delta_{o,\rho}$ is the optimal prior success rate of recovering x *before observing the leakage;* $\mathbf{1}_{\delta_\rho}$ *and* $\mathbf{1}_{\delta_{o,\rho}}$ *are two Bernoulli distributions of parameters δ_ρ and $\delta_{o,\rho}$, respectively; $\mathbb{P}_{X,\mathcal{F}(X)}$ and $\mathbb{P}_X$ are the joint distribution and the marginal distribution of* $(\mathsf{x}, \mathcal{F}(\mathsf{x}))$ *and* x*, respectively; and $\mathbb{P}_W$ represents the distribution of an arbitrary random variable W defined over $\mathcal{O}$.*

[1] further shows that the Bernoulli divergence on the left-hand side of (8) is monotone in δ_ρ. Thus, Lemma 1 suggests that, by optimizing a *reference* marginal distribution $\mathbb{P}_W$ over the leakage domain, an expected f-divergence between the conditional leakage likelihood $\mathbb{P}_{\mathcal{F}(\mathsf{x})}$ and $\mathbb{P}_W$ yields an upper bound on δ_ρ. In particular, when D_f is KL divergence, (8) reduces to the well-known Fano inequality [53], connecting Shannon information to Bayesian reconstruction risk. When D_f is the α-divergence in Definition 2, the same expression is related to α-mutual information [48].

Definition 5 (Sibson's α-mutual information [48])**.** *Let $\mathbb{P}_{XY}$ be some joint distribution of random variables (X, Y) whose marginal distribution with respect to X is $\mathbb{P}_X$. For any $\alpha > 1$,* Sibson's α-mutual information *is defined by the Rényi information for an arbitrary marginal distribution $\mathcal{Q}$ of Y,*

$$I_\alpha^{sib}(X;Y) \;:=\; \inf_{Q} \frac{1}{\alpha-1} \cdot \log D_\alpha\big(\mathbb{P}_{XY} \,\big\|\, \mathbb{P}_X \otimes \mathcal{Q}\big). \tag{9}$$

When $\alpha \to \infty$, $I_\infty^{sib}(X;Y)$ forms the foundation of *maximal leakage* [12]. In a *prior-free* setup, taken over the worst case of prior distribution $\mathbb{P}_X$, $\sup_{\mathbb{P}_X} I_\infty^{sib}(X;Y)$ captures the logarithm of the maximal multiplicative posterior gain [12].

3 Tight Privatization for Static Leakage

In this section, we address **Problem** 1 for a static leakage function. Our starting point is the exact Bayesian decision rule. For each observed leakage value, the optimal adversary selects the secret candidate with the largest posterior probability. Consequently, the posterior identification success rate can be written as an integral of a pointwise maximum. We upper-bound this maximum by an unnormalized ℓ_α-norm. The resulting family of bounds is valid for every $\alpha > 1$, and converges to the exact posterior identification success rate as $\alpha \to \infty$. Our construction directly retains the prior-weighted likelihoods and therefore remains asymptotically tight under *arbitrary* priors and *arbitrary* Bayesian inference tasks.

We begin with exact identification because it exposes the main Bayesian geometry most cleanly and connects to known Arimoto–Rényi conditional entropy characterizations. The new role of this subsection is to derive the PRW viewpoint that we then extend to arbitrary priors, report spaces, and inference criteria ρ. Appendix A.4 provides a detailed comparison with the f-divergence Fano inequality. Let $\mathcal{X}$ be a finite secret space and let $\mathsf{x} \sim \mathcal{D}_\mathsf{x}$. We write $\mathcal{M}$ for the final released randomized mechanism. For an output o, let $\mathbb{P}_\mathcal{M}(\mathsf{o} \mid \mathsf{x})$ denote its conditional probability mass or density given the secret x.

3.1 Identification under arbitrary priors via PRW α-Information

For each secret value $\hat{\mathsf{x}} \in \mathcal{X}$ and output o, define the prior-weighted likelihood $q_\mathcal{M}(\hat{\mathsf{x}}, \mathsf{o}) := \Pr_{\mathcal{D}_\mathsf{x}}(\mathsf{x} = \hat{\mathsf{x}})\, \mathbb{P}_\mathcal{M}(\mathsf{o} \mid \hat{\mathsf{x}})$. After observing o, the optimal adversary applies the maximum a posteriori (MAP) rule: $\hat{\mathsf{x}}_{\mathrm{MAP}}(\mathsf{o}) \in \arg\max_{\hat{\mathsf{x}} \in \mathcal{X}} q_\mathcal{M}(\hat{\mathsf{x}}, \mathsf{o})$.

Lemma 2 (Exact maximal posterior identification success [50])**.** *The maximal posterior identification success rate is*

$$\delta_I(\mathcal{M}) = \int_{\mathcal{O}} \max_{\hat{\mathsf{x}} \in \mathcal{X}} q_{\mathcal{M}}(\hat{\mathsf{x}}, \mathsf{o}) \, d\mathsf{o}. \tag{10}$$

When $\mathcal{O}$ is discrete, the integral is interpreted as a sum.

When the secret space $\mathcal{X}$ is too large to enumerate, the exact identification success rate in (10) cannot be computed directly. Our goal is therefore to derive an upper bound that can be expressed as an expectation over $\mathsf{x} \sim \mathcal{D}_{\mathsf{x}}$ and approximated using independent samples. For each output o, the pointwise maximum in (10) is the ℓ_∞-norm of the vector of prior-weighted likelihoods. For any $\alpha > 1$, we upper-bound it using the unnormalized ℓ_α-norm:

$$\left(\sum_{\hat{\mathsf{x}} \in \mathcal{X}} q_{\mathcal{M}}(\hat{\mathsf{x}}, \mathsf{o})^\alpha \right)^{1/\alpha}. \tag{11}$$

This quantity is an upper bound on $\max_{\hat{\mathsf{x}} \in \mathcal{X}} q_{\mathcal{M}}(\hat{\mathsf{x}}, \mathsf{o})$ and converges to it as $\alpha \to \infty$.

Our next task is to express this upper bound as an expectation over the secret distribution. For brevity, write

$$\pi(\hat{\mathsf{x}}) := \Pr_{\mathcal{D}_{\mathsf{x}}}(\mathsf{x} = \hat{\mathsf{x}}). \tag{12}$$

Let W be any reference distribution over $\mathcal{O}$, with probability mass or density $\mathbb{P}_W$. By Hölder's inequality, for any $\alpha > 1$,

$$\begin{aligned}
&\int_{\mathcal{O}} \left(\sum_{\hat{\mathsf{x}} \in \mathcal{X}} q_{\mathcal{M}}(\hat{\mathsf{x}}, \mathsf{o})^\alpha \right)^{\frac{1}{\alpha}} d\mathsf{o} \\
&\leq \left(\int_{\mathcal{O}} \frac{\sum_{\hat{\mathsf{x}} \in \mathcal{X}} q_{\mathcal{M}}(\hat{\mathsf{x}}, \mathsf{o})^\alpha}{\mathbb{P}_W(\mathsf{o})^{\alpha-1}} \, d\mathsf{o} \right)^{\frac{1}{\alpha}} \\
&= \left(\mathbb{E}_{\mathsf{x} \sim \mathcal{D}_{\mathsf{x}}} \left[\pi(\mathsf{x})^{\alpha-1} D_\alpha(\mathbb{P}_{\mathcal{M}(\mathsf{x})} \| \mathbb{P}_W) \right] \right)^{1/\alpha}.
\end{aligned} \tag{13}$$

The final expression in (13) is an expectation over $\mathsf{x} \sim \mathcal{D}_{\mathsf{x}}$, which can be approximated using independent samples from the secret distribution. To ensure tightness by achieving equality in (13), we introduce the following measurement.

Definition 6 (Prior–reference-weighted α-information)**.** *For any reference distribution W and any $\alpha > 1$, define*

$$\mathsf{I}_\alpha^W(\mathcal{M}) := \left(\mathbb{E}_{\mathsf{x} \sim \mathcal{D}_{\mathsf{x}}} \left[\pi(\mathsf{x})^{\alpha-1} D_\alpha\big(\mathbb{P}_{\mathcal{M}(\mathsf{x})} \,\big\|\, \mathbb{P}_W\big) \right] \right)^{1/\alpha}.$$

Since the choice of W is arbitrary, we seek the reference distribution that minimizes the upper bound in (13). It is noted that equality in Hölder's inequality holds when there exists a constant $\lambda > 0$ such that

$$\sum_{\hat{\mathsf{x}} \in \mathcal{X}} q_{\mathcal{M}}(\hat{\mathsf{x}}, \mathsf{o})^\alpha = \lambda \, p_W(\mathsf{o})^\alpha \quad \text{for almost every } \mathsf{o}.$$

Therefore, the optimal reference distribution has probability mass or density

$$\mathbb{P}_{W_\alpha^\star}(\mathsf{o}) \propto \left(\sum_{\hat{\mathsf{x}} \in \mathcal{X}} q_{\mathcal{M}}(\hat{\mathsf{x}}, \mathsf{o})^\alpha \right)^{1/\alpha}.$$

The preceding derivation yields an upper bound for every reference distribution W. Optimizing over W gives the tightest member of this family.

Theorem 1 (Asymptotically tight PRW bound for identification)**.** *Let* $\mathsf{x} \sim \mathcal{D}_{\mathsf{x}}$ *and let* $\mathcal{M}$ *be a randomized mechanism. For any* $\alpha > 1$ *and any reference distribution* W *for which* $\mathsf{I}_\alpha^W(\mathcal{M})$ *is finite,*

$$\delta_I(\mathcal{M}) \le \mathsf{I}_\alpha^W(\mathcal{M}).$$

Moreover, for each fixed α*, the optimal reference distribution* $W_\alpha^\star$ *has probability mass or density*

$$\mathbb{P}_{W_\alpha^\star}(\mathsf{o}) = \frac{\left(\sum_{\hat{\mathsf{x}}\in\mathcal{X}} q_{\mathcal{M}}(\hat{\mathsf{x}}, \mathsf{o})^\alpha\right)^{1/\alpha}}{\displaystyle\int_{\mathcal{O}} \left(\sum_{\hat{\mathsf{x}}\in\mathcal{X}} q_{\mathcal{M}}(\hat{\mathsf{x}}, u)^\alpha\right)^{1/\alpha} \mathrm{d}u}.$$

Also, $\inf_W \mathsf{I}_\alpha^W(\mathcal{M}) = \int_{\mathcal{O}} \left(\sum_{\hat{\mathsf{x}}\in\mathcal{X}} q_{\mathcal{M}}(\hat{\mathsf{x}}, \mathsf{o})^\alpha\right)^{1/\alpha} \mathrm{d}\mathsf{o}$, *and*

$$\lim_{\alpha\to\infty} \inf_W \mathsf{I}_\alpha^W(\mathcal{M}) = \delta_I(\mathcal{M}).$$

For identification task, Theorem 1 provides a sample-estimable upper bound on the posterior rate, and Arimoto–Rényi conditional entropy becomes a special case of PRW [29]. For each finite α, the bound is valid for any reference distribution $\mathbb{P}_W$. With the optimal reference $W_\alpha^\star$, it converges to the exact posterior identification success rate as $\alpha \to \infty$. Thus finite α gives a valid conservative certificate, while increasing α reduces approximation bias at the price of estimating higher-order likelihood-ratio moments.

This bound is compatible with the black-box model. For an additive perturbation e,

$$\mathcal{M}_{\mathcal{D}_e}(\mathsf{x}) = \mathcal{F}(\mathsf{x}) + \mathsf{e}, \qquad \mathsf{e} \sim \mathcal{D}_e,$$

and the conditional output density is

$$\mathbb{P}_{\mathcal{M}_{\mathcal{D}_e}}(\mathsf{o} \mid \mathsf{x}) = \mathbb{P}_{\mathcal{D}_e}\big(\mathsf{o} - \mathcal{F}(\mathsf{x})\big).$$

Therefore, after evaluating $\mathcal{F}(\mathsf{x})$ through black-box simulation, the user can compute the likelihood terms required by the PRW bound without knowing a structural representation of $\mathcal{F}$. Although the exact optimal reference $W_\alpha^\star$ may still be infeasible to compute when $\mathcal{X}$ is large, both the expectation in Definition 6 and an empirical approximation of $W_\alpha^\star$ can be constructed from sampled secrets. We will return to this finite-sample calibration problem in Section 3.3.

3.2 Extension to arbitrary inference task ρ

We then consider the generalization to arbitrary inference ρ. As clarified earlier, there was *no* known operationally-efficient method which can tightly approach the posterior success rate δ_ρ for an arbitrary inference task ρ. All earlier-mentioned methods, including Arimoto–Rényi conditional entropy [29], Fano's inequality [53] and PAC Privacy [1], may only provide loose upper bounds, as demonstrated later in Section 6 and Appendix A.4.

Let $\mathcal{A}$ be a report space (for notation simplicity, we assume it is discrete; otherwise replacing the summation in (14) by an integration) and let $\rho(\hat{a}, \mathsf{x}) \in \{0, 1\}$ be an arbitrary adversarial success criterion, where $\hat{a} \in \mathcal{A}$ is the adversary's report and x is the true secret. Exact identification corresponds to the special case $\mathcal{A} = \mathcal{X}$ and $\rho(\hat{a}, \mathsf{x}) = \mathbf{1}[\hat{a} = \mathsf{x}]$. For a randomized mechanism $\mathcal{M}$, define the task-weighted posterior score

$$H_{\rho,\mathcal{M}}(\hat{a}, \mathsf{o}) := \sum_{\bar{\mathsf{x}}\in\mathcal{X}} \rho(\hat{a}, \bar{\mathsf{x}})\, \pi(\bar{\mathsf{x}})\, \mathbb{P}_{\mathcal{M}}(\mathsf{o} \mid \bar{\mathsf{x}}). \tag{14}$$

Lemma 3 (Exact maximal posterior success for arbitrary tasks). *For a fixed observation* o*, the optimal adversary chooses the report* $\hat{a}$ *that maximizes this score. Therefore the optimal posterior success rate for task* ρ *is*

$$\delta_\rho(\mathcal{M}) = \int_{\mathcal{O}} \max_{\hat{a}\in\mathcal{A}} H_{\rho,\mathcal{M}}(\hat{a},\mathsf{o})\,\mathrm{d}\mathsf{o}. \tag{15}$$

In Lemma 3, exact identification as a special case can be captured by setting $\mathcal{A} = \mathcal{X}$ and $\rho(\hat{a},\mathsf{x}) = \mathbf{1}[\hat{a} = \mathsf{x}]$, in which case $H_{\rho,\mathcal{M}}(\hat{a},\mathsf{o}) = q_{\mathcal{M}}(\hat{a},\mathsf{o})$. For any reference distribution W over $\mathcal{O}$, define the task-aware PRW quantity

$$\mathsf{I}^W_{\alpha,\rho}(\mathcal{M}) := \left(\int_{\mathcal{O}} \frac{\sum_{\hat{a}\in\mathcal{A}} H_{\rho,\mathcal{M}}(\hat{a},\mathsf{o})^\alpha}{\mathbb{P}_W(\mathsf{o})^{\alpha-1}}\,\mathrm{d}\mathsf{o}\right)^{1/\alpha}.$$

The same ℓ_α-relaxation and Hölder argument used for identification in Section 3.1 gives the following theorem.

Theorem 2 (Asymptotically tight PRW bound for arbitrary inference task ρ). *For any finite report space* $\mathcal{A}$*, inference criterion* $\rho : \mathcal{A}\times\mathcal{X} \to \{0,1\}$*, randomized mechanism* $\mathcal{M}$*, reference distribution* W*, and* $\alpha > 1$*,*

$$\delta_\rho(\mathcal{M}) \le \mathsf{I}^W_{\alpha,\rho}(\mathcal{M}).$$

Moreover, the optimal reference distribution satisfies $\mathbb{P}_{W^\star_{\alpha,\rho}}(\mathsf{o}) \propto \left(\sum_{\hat{a}\in\mathcal{A}} H_{\rho,\mathcal{M}}(\hat{a},\mathsf{o})^\alpha\right)^{1/\alpha}$*, and*

$$\lim_{\alpha\to\infty} \inf_W \mathsf{I}^W_{\alpha,\rho}(\mathcal{M}) = \delta_\rho(\mathcal{M}).$$

Theorem 2 is the static privacy measurement used throughout the rest of the paper. As mentioned before, the inference task/success criterion ρ can be freely selected, not tied to exact recovery: approximate reconstruction, predicate recovery, and application-specific unacceptable reports are all handled by changing $\mathcal{A}$ and ρ. From the operational standpoint to apply Theorem 2, the PRW divergence requires computing $H_{\rho,\mathcal{M}}(\hat{a},\mathsf{o})$ in (14) in the first place. Since prior distribution $\mathcal{D}_\mathsf{x}$ is given, $\rho(\hat{a},\bar{\mathsf{x}})\pi(\bar{\mathsf{x}})$, i.e., the probability that adversarial guessing $\hat{a}$ is a successful of the random secret x when $\mathsf{x} = \bar{\mathsf{x}}$, is fully determined by $\mathcal{D}_\mathsf{x}$; the only remaining part $\mathbb{P}_{\mathcal{M}}(\mathsf{o} \mid \bar{\mathsf{x}})$ is the distribution of $\mathcal{F}(\bar{\mathsf{x}}) + \mathsf{e}$. Throughout the main body of this paper, we focus on the scenario where the leakage function $\mathcal{F}(\cdot)$ is deterministic and thus $\mathbb{P}_{\mathcal{M}}(\mathsf{o} \mid \bar{\mathsf{x}})$ corresponds to the distribution $\mathcal{D}_\mathsf{e}$ (conditional on $\bar{\mathsf{x}}$) shifted by $\mathcal{F}(\bar{\mathsf{x}})$, which is also straightforward. In Appendix A.2, we present the generalization to randomized leakage function scenario. Unlike exact identification, however, the task-aware PRW functional is nonlinear in $H_{\rho,\mathcal{M}}$, so finite-sample calibration for a general ρ uses a plug-in or empirical-process concentration argument rather than the single-secret Hoeffding loss used in Algorithm 1; see the appendix for the precise distinction.

3.3 Finite-Sample Algorithm Calibration

We now show how to approximate and calibrate the PRW bound in Definition 6 using finitely many black-box evaluations. The notation below writes the decision variable as an additive perturbation distribution $\mathcal{D}_e$. $\mathcal{D}_e$ here should be read as an index for a candidate randomized mechanism in the admissible class $\mathcal{R}$ and $\mathcal{D}_e$ can be dependent on the secret input x though the most common instantiation is independent perturbation like Gaussian noise. For notation simplicity, let

$$\mathcal{M}_{\mathcal{D}_e}(\mathsf{x}) = \mathcal{F}(\mathsf{x}) + \mathsf{e}, \qquad \mathsf{e} \sim \mathcal{D}_e,$$

and let $W_{\mathcal{D}_e}$ be its associated reference distribution and

$$\mathcal{G}(\mathcal{D}_e, \mathsf{x}) := \pi(\mathsf{x})^{\alpha-1} D_\alpha\left(\mathbb{P}_{\mathcal{M}_{\mathcal{D}_e}(\mathsf{x})} \,\middle\|\, \mathbb{P}_{W_{\mathcal{D}_e}}\right). \tag{16}$$

The α-th power of the PRW bound is then an expectation over the secret distribution:

$$\mathsf{I}_\alpha^{W_{\mathcal{D}_e}}(\mathcal{M}_{\mathcal{D}_e})^\alpha = \mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}\left[\mathcal{G}(\mathcal{D}_e, \mathsf{x})\right].$$

Suppose the target posterior identification success rate is δ_I^{target}. By Theorem 1, it suffices to ensure

$$\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}\left[\mathcal{G}(\mathcal{D}_e, \mathsf{x})\right] \le r, \qquad r := \left(\delta_I^{\text{target}}\right)^\alpha. \tag{17}$$

Given i.i.d. samples $\mathsf{X} = \{\bar{\mathsf{x}}_1, \ldots, \bar{\mathsf{x}}_m\} \sim \mathcal{D}_{\mathsf{x}}^m$, the expectation can be efficiently approximated by the empirical average $1/m \cdot \sum_{i=1}^m \mathcal{G}(\mathcal{D}_e, \bar{\mathsf{x}}_i)$.

To systematically control the "generalization error" from solving the empirical version of the population optimization (17), Algorithm 1 adopts a propose-test method with two independent sample sets. A training set X is used to select an initial perturbation distribution and its associated reference distribution (steps 2-6). Before the validation phase (steps 7-16) starts, this reference rule is fixed. An independent validation set X' is then used to test whether the randomization strategies satisfies the required population bound in (17). This separation is necessary for applying concentration techniques (e.g., Hoeffding's inequality): once a candidate has been selected using X, its empirical mean on the same samples can no longer be treated as an average of independent evaluations of a fixed candidate.

If the initial candidate does not pass validation, the algorithm increases the noise level and validates the updated candidate again while the reference is *not* re-optimized based on the validation samples. Because the validation path is a monotone sequence of post processings of the initial mechanism–reference pair, the same validation set X' can be reused without a union bound over the number of fallback updates. The empirical optimization step may be solved by any optimizer appropriate for the chosen admissible class. If the solver returns an η-suboptimal empirical optimizer, the utility guarantee is weakened by the corresponding η term, while privacy feasibility is still certified by the validation test. We formalize the resulting guarantee below. The returned rule should be viewed as part of a compound randomized channel: the calibration samples $(\mathsf{X}, \mathsf{X}')$ are private mechanism randomness. The adversary knows their distribution and the calibration algorithm, but not their realized values.

Theorem 3 (Finite sample calibration guarantee)**.** *Suppose*

$$B_-(\mathcal{D}_e) \le \mathcal{G}(\mathcal{D}_e, \mathsf{x}) \le B_+(\mathcal{D}_e) \tag{19}$$

for every admissible randomization parameter $\mathcal{D}_e \in \mathcal{Q}$ and every $\mathsf{x} \in \mathcal{X}$. Conditional on the training sample, assume that the validation and fallback candidates lie on a validation-sample-independent common post-processing path $\{\mathcal{D}_e^\lambda : \lambda \ge 0\}$, where the same post-processing channel is applied to the candidate mechanism and to the fixed reference distribution selected before validation. Along this path, assume the population loss is continuous and the pointwise loss and range $B_+(\mathcal{D}_e) - B_-(\mathcal{D}_e)$ are nonincreasing. Assume also that the fallback phase terminates. If $\mathcal{Z} \ge r'$ and

$$(1-\gamma)r' + \gamma\mathcal{Z} \le r,$$

then Algorithm 1, viewed as a compound randomized mechanism over its private calibration samples, satisfies the calibration-sample averaged population bound

$$\mathbb{E}_{\mathsf{X},\mathsf{X}'}\left[\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}\left[\mathcal{G}(\widehat{\mathcal{D}}_e, \mathsf{x})\right]\right] \le r,$$

Algorithm 1 Empirical Randomization Solver

Input: Leakage function $\mathcal{F}$; privacy-loss function $\mathcal{G}(\mathcal{D}_e, \cdot)$; lower- and upper-bound functions $B_-(\mathcal{D}_e)$ and $B_+(\mathcal{D}_e)$; input distribution $\mathcal{D}_{\mathsf{x}}$; admissible randomization class $\mathcal{Q}$; utility loss metric κ; privacy budget r; sample size m; maximum number of validated candidates N; conservative fallback update $\mathsf{Safe}(\cdot, c)$.

Sample a training set $\mathsf{X} = \{\bar{\mathsf{x}}_1, \dots, \bar{\mathsf{x}}_m\}$, i.i.d. from $\mathcal{D}_{\mathsf{x}}$.

Sample once a validation set $\mathsf{X}' = \{\bar{\mathsf{x}}'_1, \dots, \bar{\mathsf{x}}'_m\}$, all i.i.d. from $\mathcal{D}_{\mathsf{x}}$ and independent of X.

▷ Training Phase: solving empirical utility optimization with privacy constraint from training samples X .

Choose a target budget r', a total failure probability $\gamma \in (0,1)$, and a universal bound $\mathcal{Z} \geq r'$ such that

$$(1-\gamma)r' + \gamma \mathcal{Z} \leq r.$$

Solve

$$\min_{\mathcal{D}_e \in \mathcal{Q}} \kappa(\mathcal{D}_e), \qquad \text{such that} \qquad \frac{1}{m}\sum_{i=1}^{m} \mathcal{G}(\mathcal{D}_e, \bar{\mathsf{x}}_i) \leq r' \tag{18}$$

to obtain an initial perturbation distribution $\mathcal{D}_e^{(1)}$.

Fix the reference distribution $\mathbb{P}_W$ associated with $\mathcal{D}_e^{(1)}$ as defined in (16) using the training sample X.

▷ Test Phase: adding additional noises to the randomization solution until it passes population test on X'.

for $k = 1, 2, \dots, N$ **do**

  Based on the fixed validation set X', compute

$$L_k \leftarrow \frac{1}{m}\sum_{i=1}^{m} \mathcal{G}(\mathcal{D}_e^{(k)}, \bar{\mathsf{x}}'_i),$$

$$\beta_k \leftarrow \left(B_+(\mathcal{D}_e^{(k)}) - B_-(\mathcal{D}_e^{(k)})\right)\sqrt{\frac{\log(1/\gamma)}{2m}}.$$

  if $L_k \leq r' - \beta_k$ **and** $B_+(\mathcal{D}_e^{(k)}) \leq \mathcal{Z}$ **then**

    Output: Mechanism $\mathcal{M}_{\mathcal{D}_e^{(k)}}$ and terminate.

  end if

  if $k < N$ **then**

    Construct $\mathcal{D}_e^{(k+1)} \leftarrow \mathsf{Safe}(\mathcal{D}_e^{(k)}, c)$ and update the reference by the same post-processing to $W^{(k+1)}$.

  end if

end for

Let $\mathcal{D}_e \leftarrow \mathcal{D}_e^{(N)}$.

while $B_+(\mathcal{D}_e) > r'$ **do**

  Replace $\mathcal{D}_e \leftarrow \mathsf{Safe}(\mathcal{D}_e, c)$ and apply the same post-processing to the current reference distribution.

end while

Output: Fallback mechanism $\mathcal{M}_{\mathcal{D}_e}$.

where $\widehat{\mathcal{D}}_e$ is the randomization returned by Algorithm 1. More explicitly, except on a validation failure event of probability at most γ, the selected rule satisfies the stronger population loss bound at level r'; on the failure event its loss is controlled by the global bound $\mathcal{Z}$.

A key in the proof of Theorem 3 is to view both the training X and validation X' dataset sampled as inherent randomness in the randomized release $\mathcal{M}(\mathsf{x}) = \mathcal{F}(\mathsf{x}) + \mathsf{e}$. It is worth noting the workflow of Algorithm 1. The final release $\mathcal{M}(\mathsf{x})$ should be viewed as a single compound randomized mechanism: first the user samples $(\mathsf{X}, \mathsf{X}')$, then this sample $(\mathsf{X}, \mathsf{X}')$ determines the randomization rule $\widehat{\mathcal{D}}_e = \mathsf{Alg}(\mathsf{X}, \mathsf{X}')$, and finally a fresh secret $\mathsf{x} \sim \mathcal{D}_{\mathsf{x}}$ is sampled, processed and released through that rule. We thus apply Hoeffding's inequality as a concentration control only to upper bound the fraction of calibration seeds $(\mathsf{X}, \mathsf{X}')$ for which the empirical validation test is misleading. On the good calibration seeds, the selected rule has population PRW loss at most r'. On the remaining seeds, whose probability is at most γ, the algorithm falls back to the global bound $\mathcal{Z}$. Averaging over this private calibration randomness gives $(1-\gamma)r' + \gamma\mathcal{Z} \leq r$, which is the conservative certificate for the compound randomized mechanism. If none of the candidates passes validation, the fallback phase directly enforces $B_+(\mathcal{D}_e) \leq r'$. This creates a tradeoff between approximation and confidence. Increasing α tightens the PRW relaxation, but the corresponding loss function contains higher-order likelihood-ratio terms, so the global range bound used in the validation test can become much larger. The next subsection makes this finite-sample bottleneck explicit and gives a simple structural condition under which the confidence correction no longer suffers from an α-exponential range factor.

3.4 Avoiding the High-Order-α Sample Complexity Curse

In this section, we study the sample complexity m of Algorithm 1. Algorithm 1 is deliberately stated for a very general admissible randomization class. In this level of generality, a candidate rule is validated through a bounded-loss concentration inequality – that is to say, the only requirement for a feasible randomization $\mathcal{D}_{\mathsf{e}}$ is to satisfy a bounded risk loss $\mathcal{G}(\mathcal{D}_e, \mathsf{x})$ as defined in (19). Under an α-PRW divergence, if $0 \leq \mathcal{G}(\mathcal{D}_e, \mathsf{x}) \leq B_\alpha(\mathcal{D}_e)$, then m fresh validation samples give

$$\Pr\left[\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}\mathcal{G}(\mathcal{D}_e, \mathsf{x}) > \frac{1}{m}\sum_{i=1}^{m}\mathcal{G}(\mathcal{D}_e, \bar{\mathsf{x}}'_i) + B_\alpha(\mathcal{D}_e)\sqrt{\frac{\log(1/\gamma)}{2m}}\right] \leq \gamma.$$

For a target posterior success rate $\delta_\rho^{\text{target}}$, the PRW moment budget is $r = (\delta_\rho^{\text{target}})^\alpha$. Therefore, to certify a relative slack ηr, for $\eta \in (0,1)$, using only a global range bound, it is sufficient to take

$$m \geq \frac{B_\alpha(\mathcal{D}_e)^2}{2\eta^2 r^2}\log(1/\gamma). \tag{20}$$

Without additional structure, $B_\alpha(\mathcal{D}_e)$ can grow exponentially in α, because $\mathcal{G}$ contains α-th powers of output likelihood ratios. For example, if the perturbation e is simply an independent Gaussian, i.e., $\mathcal{D}_{\mathsf{e}} = \mathcal{N}(0, \sigma^2)$, then for a l-bit uniform string x with a target $\delta_\rho^{\text{target}}$ posterior success rate and a bounded leakage function $\mathcal{F}(\mathsf{x}) \in [-b/2, b/2]$, (20) then suggests

$$m \geq \frac{1}{2\eta^2}\left(\frac{2^{-l(\alpha-1)}\exp\left(\frac{\alpha(\alpha-1)b^2}{2\sigma^2}\right)}{\left(\delta_\rho^{\text{target}}\right)^\alpha}\right)^2 \cdot \log(1/\gamma), \tag{21}$$

which generally increases exponentially in α.

To circumvent this exponential curse of sample complexity, we introduce a tradeoff by imposing a mild restriction on the randomization rule Q applied to the release. For simplicity, assume the raw leakage output takes values in a discrete alphabet $\mathcal{O}_0$, while the final released value takes values in a finite alphabet $\mathcal{Y}$. A randomized release rule is a channel $Q : \mathcal{O}_0 \to \Delta(\mathcal{Y})$, so that

$$\mathbb{P}_{\mathcal{M}_Q(\mathsf{x})}(y) = Q(y \mid \mathcal{F}(\mathsf{x})).$$

Theorem 4 (Bounded-ratio PRW validation)**.** *Fix $\alpha > 1$, a finite report space $\mathcal{A}_\rho$, and an inference criterion $\rho : \mathcal{A}_\rho \times \mathcal{X} \to \{0,1\}$. Define*

$$p_\rho(\hat{a}) := \Pr_{\mathsf{x}\sim\mathcal{D}_\mathsf{x}}[\rho(\hat{a},\mathsf{x}) = 1], \qquad C_{\rho,\alpha} := \sum_{\hat{a}\in\mathcal{A}_\rho} p_\rho(\hat{a})^\alpha.$$

Suppose there is a reference distribution $W \in \Delta(\mathcal{Y})$ and a constant $L \geq 1$ such that

$$Q(y \mid o) \leq L\,\mathbb{P}_W(y) \qquad \text{for every } o \in \mathcal{O}_0 \text{ and } y \in \mathcal{Y}. \tag{22}$$

Then the task-aware PRW certificate of the randomized release $\mathcal{M}_Q(\mathsf{x}) \sim Q(\cdot \mid \mathcal{F}(\mathsf{x}))$ satisfies

$$\left(\mathsf{I}^W_{\alpha,\rho}(\mathcal{M}_Q)\right)^\alpha \leq L^{\alpha-1} C_{\rho,\alpha}. \tag{23}$$

Consequently, by Theorem 2,

$$\delta_\rho(\mathcal{M}_Q)^\alpha \leq L^{\alpha-1} C_{\rho,\alpha}.$$

In particular, a target posterior success rate $\delta_\rho^{\text{target}}$ is certified whenever

$$L \leq \left(\frac{(\delta_\rho^{\text{target}})^\alpha}{C_{\rho,\alpha}}\right)^{1/(\alpha-1)}.$$

Proof. For every report $\hat{a}$ and output y, define the task-aware joint score

$$H_{\rho,\mathcal{M}_Q}(\hat{a}, y) := \mathbb{E}_{\mathsf{x}\sim\mathcal{D}_\mathsf{x}}[\rho(\hat{a},\mathsf{x})\, Q(y \mid \mathcal{F}(\mathsf{x}))].$$

By (22),

$$H_{\rho,\mathcal{M}_Q}(\hat{a}, y) \leq L\,\mathbb{P}_W(y)\, p_\rho(\hat{a}).$$

Therefore,

$$\sum_{y\in\mathcal{Y}} \mathbb{P}_W(y) \left(\frac{H_{\rho,\mathcal{M}_Q}(\hat{a}, y)}{\mathbb{P}_W(y)}\right)^\alpha \leq L^{\alpha-1} p_\rho(\hat{a})^{\alpha-1} \sum_{y\in\mathcal{Y}} H_{\rho,\mathcal{M}_Q}(\hat{a}, y) = L^{\alpha-1} p_\rho(\hat{a})^\alpha.$$

Summing over $\hat{a} \in \mathcal{A}_\rho$ gives (23). The posterior-risk bound then follows from Theorem 2. □

After imposing the bounded-ratio channel constraint, the privacy certificate is controlled by the task prior masses $p_\rho(\hat{a})$ and the domination constant L, avoiding the α-exponential validation range term in (20). We take exact identification as an example. Suppose $\mathbf{x}$ is an l-bit random string, and $C_{\rho,\alpha} = \sum_{\mathbf{x}} \pi(\mathbf{x})^\alpha = 2^{-(\alpha-1)l}$. In such a setting,

$$L \leq \frac{(\delta_\rho^{\text{target}})^{\alpha/(\alpha-1)}}{2^{-l}} \approx \frac{\delta_\rho^{\text{target}}}{2^{-l}}, \tag{24}$$

which is roughly the multiplicative increase from the prior success rate to the posterior success rate.

As demonstrated later in Fig. 5 and Fig. 6, except in extremely strict privacy regimes, i.e., when the posterior budget $\delta_\rho^{\text{target}}$ is selected too close to the prior $\delta_{0,\rho}$, the bounded-ratio constraint incurs negligible additional utility overhead. As a final note, the finite/discretized-output assumption is not needed for the inequality itself; the core requirement is the common domination condition (22). Finite or discretized outputs simply make the condition operationally clean in the empirical optimization (18), as demonstrated in Appendix B.2.

4 History-Dependent Adaptive Mechanisms

The adaptive setting is more challenging than the static one because the adversary does not commit to all queries in advance. Instead, each query may depend on the previous queries and randomized/noisy outputs. The adversary can repeatedly probe the aspects of the secret that appear most vulnerable from earlier observations, and thereby extract more information from the interaction. Consequently, the distribution of the final transcript is jointly determined by the secret, the user's randomization, and the adversary's adaptive strategy. Even if the user can fully control the noise calibration rule, the induced transcript distribution is not directly visible to the user.

The black-box nature of both the leakage function and the adversarial strategy $\mathcal{A}dv$ creates a "parallel-universe" challenge (Fig. 1) as discussed earlier. In the black-box setting, the user cannot simulate the entire adaptive interaction tree. At each round, the user observes only the realized transcript, while alternative transcripts induced by different secret values, noise realizations, or adversarial responses remain unobserved. Since the adversary's strategy is unknown and may branch on previous observations, the user cannot directly compute the global transcript distribution required for a one-shot Bayesian analysis.

Our key observation is that reconstructing the full interaction tree is unnecessary. Before the interaction begins, the user commits to a single local calibration rule Alg, and the same rule is applied at every reachable branch of the interaction tree. Therefore, if this rule enforces a suitable transcript-dependent *local* condition on *every* realized branch, then these branch-local conditions can be aggregated into a *global* posterior-success certificate. In this sense, the user only needs to certify the local universe they actually observe, while the fact that the same rule would be applied in all alternative branches allows the local guarantees to lift to a global adaptive-composition guarantee.

The terminology "local" and "global" is important. A local universe is a branch conditioned on the transcript that the user actually experiences: the current adversarial query, the previous released outputs, and the user's private coins realized so far. The global universe is the distribution of the entire interactive mechanism after averaging over all secrets, user coins, and adversary coins. Our composition theorem shows that spending the prescribed PRW budget in every local universe induces the desired PRW budget for the global universe. The adversary need not be weak or oblivious: it may know the calibration rule and choose queries adaptively, but the realized private coins are still internal randomness of the randomized mechanism.

As a result, it suffices to prove a *history-dependent* guarantee that holds at each step conditioned on the past; once such a guarantee is established, it automatically lifts to a global guarantee over the entire adaptive interaction. For clarity, the main statements in this section are written for exact identification and use the notation δ_I. The extension to any fixed inference criterion ρ is obtained by replacing the identification PRW score with the task-ρ-aware PRW score from Section 3.2; the local calibration rule is unchanged once the branch-local loss $\mathcal{G}_t$ is instantiated for the chosen task ρ. We spell out this replacement in the appendix. Similarly, we display additive perturbations because they are natural in side-channel applications, but the argument depends only on the conditional law of the released output and therefore applies to general transcript-dependent randomized channels in the admissible class $\mathcal{Q}$.

4.1 Adaptive interaction and global certificate

We now extend the static identification analysis in Section 3 to an adaptive interaction. At each round, the adversary may select its next query based on the previously observed queries and noisy outputs. Our goal is to control the final posterior identification success rate δ_I after all T rounds.

The privacy side of the interaction is fixed throughout: the user fixes the inference task, privacy budget, and a single PRW order α before the composition begins. The adversary adapts only the leakage queries from the observed transcript. The user may adapt utility objectives and randomization rules as part of its defense policy, but these choices remain constrained by the fixed privacy requirement. This distinction is important: adaptive composition here means adversarially adaptive leakage queries against a fixed posterior-risk privacy semantics, not an adversary that chooses a new privacy definition at each round.

We first formalize the adaptive interaction model. We then show that, for any fixed adversarial strategy, the complete interaction can be viewed as a randomized mapping from the secret to the final transcript. This allows us to apply the PRW bound from Theorem 1.

Definition 7 (Adaptive Adversarial Composition)**.**
Spaces and transcripts. *Given a total iteration number T and three spaces: $\mathcal{X}$ is the* secret space*, $\mathcal{V}$ is the* query space *from which the adversary selects the query at each iteration, and $\mathcal{O}$ is the* leakage space*. A secret $\mathsf{x} \sim \mathcal{D}_{\mathsf{x}}$ is sampled once at the beginning and reused throughout all T rounds. For each round t, define the pre-round-t transcript space as*

$$\mathcal{T}_t := \mathcal{V}^{t-1} \times \mathcal{O}^{t-1}.$$

Accordingly, we write the realized pre-round-t transcript as

$$\tau_t := (\mathsf{v}_{<t}, \mathsf{o}_{<t}) \in \mathcal{T}_t,$$

here $\mathsf{v}_{<t} = (\mathsf{v}_1, \ldots, \mathsf{v}_{t-1})$, $\mathsf{o}_{<t} = (\mathsf{o}_1, \ldots, \mathsf{o}_{t-1})$, to record the adversary's past queries and the user's past outputs.
Leakage and randomization. *Given T leakage functions*

$$\mathcal{F}_t : \mathcal{X} \times \mathcal{V} \to \mathcal{O}, \qquad t \in [T],$$

and T transcript-dependent noise distributions $\Sigma_t(\mathsf{v}_t, \tau_t)$, $\quad t \in [T]$, the user's response at round t is generated as follows. Given the current query v_t and transcript τ_t, the user samples $\mathsf{e}_t \sim \Sigma_t(\mathsf{v}_t, \tau_t)$ and releases

$$\mathsf{o}_t = \mathcal{M}_t(\mathsf{x}, \mathsf{v}_t, \tau_t) := \mathcal{F}_t(\mathsf{x}, \mathsf{v}_t) + \mathsf{e}_t.$$

Adaptive interaction experiment. *An adversarial strategy $\mathcal{A}dv = \{\mathcal{A}dv_t\}_{t=1}^{T}$ is a sequence of possibly randomized algorithms. The interaction proceeds for $t \in [T]$ as follows:*

1. *Given the realized transcript τ_t, the adversary selects*

$$\mathsf{v}_t \leftarrow \mathcal{A}dv_t(\tau_t).$$

2. *Given the secret x, the current query v_t, and the transcript τ_t, the user releases*

$$\mathsf{o}_t \leftarrow \mathcal{M}_t(\mathsf{x}, \mathsf{v}_t, \tau_t).$$

The complete interaction induces a randomized transcript mechanism

$$\mathcal{M}^{\mathcal{A}dv} : \mathsf{x} \mapsto (\mathsf{v}_{1:T}, \mathsf{o}_{1:T}).$$

At the end of the interaction, the adversary uses the transcript to form an estimate $\hat{\mathsf{x}} \in \mathcal{X}$ of the secret x.

The additive form in Definition 7 is a presentation choice. For a general admissible channel, one may instead sample o_t from any conditional distribution $Q_t(\cdot \mid \mathcal{F}_t(\mathsf{x}, \mathsf{v}_t), \mathsf{v}_t, \tau_t)$; the proofs below use only this conditional distribution and the corresponding reference distribution $W_t(\mathsf{v}_t, \tau_t)$. When $\Sigma_t(\mathsf{v}_t, \tau_t)$ is Gaussian, the conditional mapping $\mathsf{x} \mapsto \mathcal{M}_t(\mathsf{x}, \mathsf{v}_t, \tau_t)$ is an additive Gaussian mechanism whose noise distribution depends on realized query and transcript.

For any fixed adversarial strategy $\mathcal{A}dv$, the complete interaction induces a randomized transcript mechanism $\mathcal{M}^{\mathcal{A}dv}$ mapping the secret x to the final transcript $(\mathsf{v}_{1:T}, \mathsf{o}_{1:T})$. Although each query may depend on the previous transcript, this induced mapping is still a mechanism in the form considered in Theorem 1. We can therefore apply the static PRW bound to the distribution of the complete transcript.

It thus remains to construct a suitable reference process for the transcript. At each round t, the reference process uses the same conditional query distribution as the adversary and replaces the user's conditional response distribution with an arbitrary reference distribution $W_t(\mathsf{v}_t, \tau_t)$ that is independent of the secret. Since the adversary's query distribution is identical in the original and reference processes, the corresponding query probabilities cancel in the likelihood ratio. This yields the following global certificate.

Theorem 5 (Posterior success bound for adaptive adversary)**.** *Under the setting of Definition 7, for any fixed adversarial strategy $\mathcal{A}dv$ and any fixed order $\alpha > 1$, the α-th power of posterior identification success rate $(\delta_I)^\alpha$ is bounded by*

$$\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_\mathsf{x}} \int_{\mathcal{O}^T \times \mathcal{V}^T} \Pr_{\mathcal{D}_\mathsf{x}}(\mathsf{x})^{\alpha-1} \cdot \prod_{t=1}^{T} \frac{\big(\Pr\big(\mathcal{M}_t(\mathsf{x}, \mathsf{v}_t, \tau_t) = \mathsf{o}_t\big)\big)^\alpha}{\big(\Pr\big(W_t(\mathsf{v}_t, \tau_t) = \mathsf{o}_t\big)\big)^{\alpha-1}} \cdot \prod_{t=1}^{T} \Pr\big(\mathrm{Adv}_t(\tau_t) = \mathsf{v}_t\big)\, \mathrm{d}\mathsf{o}_{[1:T]}\, \mathrm{d}\mathsf{v}_{[1:T]} \tag{25}$$

The reference process uses the adversary's conditional query law and an arbitrary conditional output reference distribution $W_t(\mathsf{v}_t, \tau_t)$ that is independent of the secret.

With full knowledge of the adversarial strategy and unlimited computation, the user could directly control δ_I by increasing the noise scale until the right-hand side of inequality (25) is sufficiently small.

4.2 Local condition

The global certificate in Theorem 5 is not directly computable in the black-box setting. The right-hand side of (25) integrates over the full adaptive interaction tree, including alternative transcripts that the user never observes and query probabilities determined by the unknown adversarial strategy. During an actual interaction, the user can calibrate noise only on the realized branch.

Our goal is therefore to replace the global computation with a local condition that can be enforced at each realized transcript. Because the same calibration rule is applied at every reachable branch, locally enforced conditions can be lifted to a global certificate for the complete adaptive interaction. Conditioned on the current transcript τ_t and query v_t, the user can control the conditional divergence

$$D_t(\mathsf{x}) := \mathcal{D}_\alpha(\mathcal{M}_t(\mathsf{x}, \mathsf{v}_t, \tau_t) \,\|\, W_t(\mathsf{v}_t, \tau_t))\,.$$

However, controlling its unweighted expectation is insufficient because different branches contribute differently to the global certificate. We therefore introduce a prefix coefficient that records the contribution accumulated before round t.

Definition 8 (Prefix coefficient and local condition)**.** *For each round t and each realized transcript τ_t, define $A_{t-1}(\mathsf{x})$*

$$A_{t-1}(\mathsf{x}) := \Pr_{\mathcal{D}_{\mathsf{x}}}(\mathsf{x})^{\alpha-1} \left(\prod_{s=1}^{t-1} \frac{\Pr\big(\mathcal{M}_s(\mathsf{x}, \mathsf{v}_s, \tau_s) = \mathsf{o}_s\big)}{\Pr\big(W_s(\mathsf{v}_s, \tau_s) = \mathsf{o}_s\big)} \right)^{\alpha},$$

as the prefix coefficient, where the empty product gives $A_0(\mathsf{x}) = \Pr_{\mathcal{D}_{\mathsf{x}}}(\mathsf{x})^{\alpha-1}$. The local condition at round t is

$$\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}[A_{t-1}(\mathsf{x})D_t(\mathsf{x})] \le r_t \, \mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}[A_{t-1}(\mathsf{x})]\,. \tag{26}$$

At round t, the prefix coefficient is determined by the realized transcript τ_t before the user calibrates the current noise distribution. The local condition is therefore a branch-local constraint that can be evaluated using the information (τ_t, v_t) available at that round. The left-hand side of (26) is a weighted average of per-secret α-divergences. The conditional divergence $D_t(\mathsf{x})$ measures the leakage associated with the current response, while $A_{t-1}(\mathsf{x})$ records the contribution accumulated along the realized transcript before round t. Secrets with larger prefix coefficients receive greater weight during the current calibration.

4.3 From local condition to global certificate

The following theorem shows that enforcing the local condition at every reachable transcript is sufficient to control the final posterior identification success rate.

Theorem 6 (Adaptive composition of α-divergence)**.** *Under the setting of Definition 7, fix any adversarial strategy $\mathcal{A}dv$ and any conditional reference process $\{W_t(\mathsf{v}_t, \tau_t)\}_{t=1}^{T}$ that may depend on the realized query and transcript but not otherwise on the secret. If, for every round t and every transcript τ_t reachable under $\mathcal{A}dv$, the corresponding branch-local condition* (26) *holds with factor r_t, then the posterior success rate satisfies*

$$\big(\delta_I\big)^{\alpha} \le \mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}\left[\Pr_{\mathcal{D}_{\mathsf{x}}}(\mathsf{x})^{\alpha-1}\right] \prod_{t=1}^{T} r_t. \tag{27}$$

At a high level, the theorem follows by recursively integrating the branch-local condition over the interaction tree. Applying condition (26) successively for $t = 1, \dots, T$ leaves the initial factor $\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}[\Pr_{\mathcal{D}_{\mathsf{x}}}(\mathsf{x})^{\alpha-1}]$ and contributes a multiplicative factor r_t at each round. Combining this recursive bound with the global certificate yields (27).

The theorem makes adaptive randomization implementable in the black-box setting. During an actual execution, the user observes only the realized transcript τ_t and enforces condition (26) on that branch. Because the same calibration rule is invoked at every reachable branch, the corresponding local conditions hold throughout the interaction tree without requiring the user to enumerate alternative transcripts. Under bounded leakage values, the local condition is always feasible with some additive (e.g., Gaussian) perturbations: by choosing an appropriate Gaussian reference distribution and sufficiently large noise variance, $D_t(\mathsf{x})$ approaches 1 uniformly over x, so condition (26) can be satisfied for any $r_t > 1$.

Remark 2 (Multiplicative and additive budget accounting)**.** *The local factors compose multiplicatively: $r_{\mathrm{tot}} = \prod_{t=1}^{T} r_t$. Equivalently, defining $\varepsilon_t := \log r_t$ gives the additive form $\log r_{\mathrm{tot}} = \sum_{t=1}^{T} \varepsilon_t$. Thus, the multiplicative accounting used above is equivalent to the familiar log-additive accounting for Rényi-type privacy losses.*

The PRW bound is asymptotically tight when its reference distribution is optimized. The following theorem shows that, under an equality condition, the globally optimal reference process can be constructed locally along each realized transcript.

Theorem 7 (Local selection realizes the globally optimal reference process)**.** *For every round t and every reachable pair (v_t, τ_t), define $\Pr\big(W_t^\star(\mathsf{v}_t, \tau_t) = \mathsf{o}_t\big)$ proportional to*

$$\propto \left(\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_\mathsf{x}}\left[A_{t-1}(\mathsf{x}) \Pr\big(\mathcal{M}_t(\mathsf{x}, \mathsf{v}_t, \tau_t) = \mathsf{o}_t\big)^\alpha\right]\right)^{1/\alpha}.$$

Suppose for every reachable transcript, the corresponding branch-local condition (26) *holds with equality, using the same factor r_t for all branches at round t. Then the induced reference process $W_{1:T}^\star$ globally minimizes the PRW certificate for the complete transcript mechanism. In particular,*

$$\left(\inf_W \mathsf{I}_\alpha^W\right)^\alpha = \mathbb{E}_{\mathsf{x}\sim\mathcal{D}_\mathsf{x}}\left[\Pr_{\mathcal{D}_\mathsf{x}}(\mathsf{x})^{\alpha-1}\right] \prod_{t=1}^{T} r_t.$$

Consequently, this certificate converges to the posterior identification success rate as $\alpha \to \infty$.

Therefore, when every branch-local condition holds with equality, selecting the best reference distribution at each round incurs no loss compared with optimizing the reference distribution for the complete transcript globally.

This is the sense in which the composition accounting is tight. If the factors r_t are chosen so that the right-hand side of (27) equals the desired total posterior-risk budget, then the branch-local equalities use the entire finite-α PRW accounting budget without additional composition slack. For finite α, this is PRW-tight and remains a conservative upper bound on Bayesian posterior risk; as the optimized PRW certificate converges with $\alpha \to \infty$, the accounting becomes asymptotically exact for the posterior identification risk. This privacy optimality should be distinguished from utility optimality: the user minimizes the utility loss of the current branch-local randomization subject to the current factor r_t. The paper does not claim a global utility optimum over the exponentially large adaptive interaction tree; it claims locally optimal utility calibration together with globally tight finite-order PRW privacy accounting.

4.4 Finite-sample calibration for adaptive leakage

The local condition in (26) depends on an expectation over the secret distribution and cannot generally be evaluated exactly. At each round, we therefore instantiate Algorithm 1 with a privacy-loss function derived from the current prefix coefficient and conditional divergence. In the finite-sample theorem below, these prefix coefficients are those of the compound branch channels obtained after averaging the mechanism's private calibration coins. This yields a finite-sample calibration rule that can be applied along the realized interaction path. For the adaptive application, we use the centered branch-local loss

$$\mathcal{G}_t(\mathcal{D}_e, \mathsf{x}) := A_{t-1}(\mathsf{x})\big(D_t(\mathsf{x}) - r_t\big). \tag{28}$$

Since A_{t-1} is fixed at the current transcript before the fresh calibration seed for round t is sampled, condition (26) is equivalent to

$$\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_\mathsf{x}}[\mathcal{G}_t(\mathcal{D}_e, \mathsf{x})] \le 0.$$

Algorithm 2 Optimizer under Adversarial Composition

Input: secret x; number of rounds T; leakage functions $\{\mathcal{F}_t : \mathcal{X} \times \mathcal{V} \to \mathcal{O}\}_{t=1}^{T}$; input distribution $\mathcal{D}_{\mathsf{x}}$; per-round factors $\{r_t\}_{t=1}^{T}$; parameters required by Algorithm 1.
for $t = 1, 2, \ldots, T$ **do**
  Receive the adversarial query v_t.
  Construct the centered loss $\mathcal{G}_t(\mathcal{D}_e, \cdot)$ in (28) and valid bound functions $B_{t,-}(\mathcal{D}_e)$ and $B_{t,+}(\mathcal{D}_e)$.
  Run Algorithm 1 with fresh private training and validation samples, $\mathcal{F}_t(\cdot, \mathsf{v}_t)$, $\mathcal{G}_t(\mathcal{D}_e, \cdot)$, $B_{t,\pm}(\mathcal{D}_e)$, $\mathcal{D}_{\mathsf{x}}$, and privacy budget 0 to obtain $\Sigma_t(\mathsf{v}_t, \tau_t)$.
  Sample $\mathsf{e}_t \sim \Sigma_t(\mathsf{v}_t, \tau_t)$; release $\mathsf{o}_t = \mathcal{F}_t(\mathsf{x}, \mathsf{v}_t) + \mathsf{e}_t$.
end for

Theorem 8 (Finite-sample calibration guarantee for adaptive leakage)**.** *Suppose that, at each round and for every reachable transcript, Algorithm 1 is instantiated with fresh private calibration samples, the centered loss in* (28)*, privacy budget* 0*, and valid lower and upper bounds* $B_{t,-}(\mathcal{D}_e)$ *and* $B_{t,+}(\mathcal{D}_e)$*. Formally, view the online procedure as a randomized policy that assigns an independent private calibration seed to every reachable transcript node, although an actual execution samples only the seeds on its realized path. If every call of Algorithm 1 satisfies the conditions of Theorem 3, then Algorithm 2, viewed as the compound interactive mechanism obtained by averaging over these private seeds, satisfies the branch-local condition* (26) *for the compound branch channels. Consequently, for any adversarial strategy* $\mathcal{A}dv$*, the corresponding posterior success rate satisfies*

$$\left(\delta_I\right)^{\alpha} \leq \mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}\left[\Pr_{\mathcal{D}_{\mathsf{x}}}(\mathsf{x})^{\alpha-1}\right] \prod_{t=1}^{T} r_t.$$

For bounded leakage functions and additive Gaussian perturbations, valid lower and upper bounds $B_{t,-}(\mathcal{D}_e)$ and $B_{t,+}(\mathcal{D}_e)$ can be constructed from bounds on the prefix coefficient A_{t-1} and the conditional divergence D_t. The explicit construction and the corresponding termination argument are deferred to the full version.

5 Mechanism-Agnostic Adaptive Composition

In this section, we study composition in the *mechanism-agnostic* setup. In this setting, across each round $t \in [T]$, a mechanism $\mathcal{M}_t$ must determine its internal randomization using only the privacy budget allocated to that round and the specifically observed leakage function $\mathcal{F}(\cdot, \mathsf{v}_t)$. The difference from Definition 7 is that τ_t is not available as an input to the mechanism.

A key question we address in this section is which separate measures of marginal leakage can be aggregated into an upper bound on the cumulative privacy risk, independently of the adversarial strategy that selects the leakage functions. We present two such measures, based on KL divergence and α-divergence, and characterize their mechanism-agnostic composition guarantees in the following two theorems.

Theorem 9 (KL agnostic composition)**.** *For each round* $t \in [T]$*, let* $W_t(\mathsf{v}_t)$ *be an arbitrary reference distribution. Suppose that the following worst-case and average-case KL-divergence bounds hold:*

$$\mathcal{D}_{KL}(P_{\mathcal{M}_t(\hat{\mathsf{x}},\mathsf{v}_t)} \| P_{W_t(\mathsf{v}_t)}) \leq R_t, \quad \forall \hat{\mathsf{x}} \in \mathcal{X}, \mathsf{v}_t \in \mathcal{V}_t \tag{29}$$

$$\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}} \mathcal{D}_{KL}\big(P_{\mathcal{M}_t(\mathsf{x},\mathsf{v}_t)} \big\| P_{W_t(\mathsf{v}_t)}\big) \leq r_t, \quad \forall \mathsf{v}_t \in \mathcal{V}_t \tag{30}$$

Then, for any adversarial strategy $\mathcal{A}dv$, the posterior success rate δ_ρ of the resulting interactive mechanism satisfies

$$\mathcal{D}_{KL}(\mathbf{1}_{\delta_\rho} \| \mathbf{1}_{\delta_{o,\rho}}) \leq B_T,$$

where B_t is defined recursively by

$$B_1 = r_1, \tag{31}$$

$$B_t = B_{t-1} + \min\{\sqrt{2B_{t-1}}R_t + r_t, R_t\}, t \geq 2. \tag{32}$$

In Theorem 9, R_t behaves as a universal, input-independent upper bound. From (32), the increment between B_t and B_{t-1} in two consecutive iterations is upper bounded by either $\sqrt{2B_{t-1}}R_t + r_t$, that is, the average increment r_t plus a worst-case increment R_t scaled by a factor $\sqrt{2B_{t-1}}$ depending on the current accumulated risk, or by the worst-case increment R_t. Before the worst-case cap becomes active, the recursive bound may exhibit quadratic growth in T. We also show the result through α-divergence in the following theorem:

Theorem 10 (α-divergence agnostic composition)**.** *Fix any $\alpha > 1$. Let $p_1, \ldots, p_T > 1$ satisfy $\sum_{t=1}^{T} \frac{1}{p_t} = 1$. For each round $t \in [T]$, let $W_t(\mathsf{v}_t)$ be an arbitrary reference distribution. Suppose that, for all $\mathsf{v}_t \in \mathcal{V}_t$,*

$$\mathbb{E}_{\mathsf{x} \sim \mathcal{D}_\mathsf{x}}\left[\Pr_{\mathcal{D}_\mathsf{x}}(\mathsf{x})^{\alpha-1} D_{\alpha p_t}\left(P_{\mathcal{M}_t(\mathsf{x},\mathsf{v}_t)} \,\|\, P_{W_t(\mathsf{v}_t)}\right)\right] \leq r_t.$$

Then, for any adversarial strategy $\mathcal{A}dv$, the posterior identification success rate of the final interactive mechanism satisfies $\delta_I^\alpha \leq \prod_{t=1}^{T} r_t^{1/p_t}$.

Compared with KL-agnostic composition in Theorem 9, the Hölder-style α-divergence composition in Theorem 10 does not explicitly require a worst-case guarantee. Instead, it controls marginal divergences of orders αp_t, where $\sum_{t=1}^{T} 1/p_t = 1$. As T grows, at least one p_t must be at least T. Consequently, the required divergence orders increase with the number of rounds. High-order divergences increasingly emphasize the largest likelihood ratios and approach worst-case likelihood-ratio control in the limit. Thus, as the number of rounds grows, mechanism-agnostic composition gradually requires a worst-case-like guarantee. This reflects the cost of mechanism agnosticism: unlike the history-accessible setting in Section 4, the user cannot adapt the mechanism to the realized transcript.

6 Experiments

Experiment setting. We use the following experiments as illustrative validations of the proposed black-box privatization framework on two side-channel models: (i) power leakage from AES secret-key generation and (ii) timing leakage from RSA modular exponentiation. In (i), we model the power trace as aggregate Hamming-weight leakage, a standard first-order abstraction for power side channels [16, 54, 55], capturing a bytewise AES secret-key generator after discretization. For an l-bit secret key x and a plaintext v, the raw leakage $\mathcal{F}(\mathsf{x}, \mathsf{v}) = \mathsf{HW}(\mathsf{x} \oplus \mathsf{v})$ is the Hamming weight of $\mathsf{x} \oplus \mathsf{v}$. In (ii), we use a black-box square-and-multiply timing simulator for non-constant-time exponentiation [56]: $\mathcal{F}(\mathsf{x}, \mathsf{v})$ is the running time, rounded to a 1 ns bin, of encrypting a (possibly adversarially chosen) plaintext v under the key x.

6.1 Illustrative Experiments without Confidence Validation

In the first set of experiments, we ignore the confidence validation step in the main Algorithm 1 and only report the results from solving the empirical optimization (18). We use this set of experiments to illustrate the tightness of the PRW α-divergence with comparison to other Bayesian analysis tools. In Section 6.2, we show the full implementation of Algorithm 1 with confidence validation assisted with bounded ratio trick in Section 3.4. We highlight several observations below.

Tightness. In panels (a)–(d) of Fig. 2 and Fig. 3, we consider noisy leakage of the form $\mathcal{F}(\mathsf{x})+\mathsf{e}$ with independent Gaussian noise $\mathsf{e}\sim\mathcal{N}(0,\sigma^2)$, and compare the tightness of the posterior-success-rate bound obtained from the classical mutual-information (Fano) bound [53], PAC Privacy with the standard α-divergence [1], and our proposed PRW divergence for various choices of α. Comparing (a) against (b), (c), and (d) isolates the effects of three factors: entropy, the prior distribution $\mathcal{D}_{\mathsf{x}}$, and the inference task ρ. Globally, as α increases the PRW bound converges to the ground truth, as established in Theorems 1 and 2; in these controlled accounting curves, $\alpha\geq 64$ already yields a tight bound (within 3 bits of the ground truth). Moreover, in the high-noise regime the Fano mutual-information bound is exponentially looser than PAC Privacy with an optimized α-divergence, and PRW with sufficiently large α further remarkably improves upon PAC Privacy. The gap widens further when the entropy increases ($l=256$ in (b) versus $l=128$ in (a)), when the prior is non-uniform ((c) uses a secret whose Hamming weight follows a truncated Poisson law), or when the task departs from exact identification ((d) requires recovering at least 240 of 256 bits): in all three cases mutual information and PAC Privacy become even looser relative to PRW.

Entropy. Panel (e) studies the effect of entropy on inference hardness. For Gaussian-perturbed leakage $\mathcal{F}(\mathsf{x})+\mathsf{e}$ under the full-identification task ρ and a fixed target posterior success rate (varied from 2^{-248} to 2^{-252}), we report the minimal Gaussian noise required across secret lengths (l from 253 to 256). Secrets with higher entropy require less noise to attain the same posterior hardness. This matches the intuition that the posterior success rate is governed by both input randomness (the secret's entropy) and mechanism randomness (the injected noise): in a fixed setting, a lower prior success rate also induces a lower posterior success rate.

Optimal constrained randomization. Although Gaussian noise is a common instantiation of randomization, it is neither universally optimal nor always compatible with physical constraints. For example, in panel (f), under the utility loss $\kappa(\mathsf{e})=\mathbb{E}[|\mathsf{e}|]$ in terms of expected absolute value, the optimal zero-mean noise (purple) attains a considerably smaller expected absolute perturbation than Gaussian (blue). In addition, many side channels may admit only one-sided perturbation: power can be obfuscated by running dummy programs and timing by adding delay, whereas reducing power or accelerating computation is potentially infeasible or performance-degrading. This translates into a non-negativity constraint on the noise [34]. Leveraging our ability to handle arbitrary constrained utility losses, the red line in (f) reports the optimal non-negative noise with minimal expected absolute perturbation; under this constraint the noise must be biased rather than zero-mean, and its scale exceeds that of the unconstrained zero-mean optimum.

Composition. Finally, panels (g) and (h) demonstrate history-dependent adversarial composition (Section 4) under total privacy budgets of 2^{-240} and 2^{-220} posterior identification rate, respectively; as expected, a weaker privacy target requires less randomization. Asymptotically, the per-round noise scale grows as $\tilde{O}(\sqrt{T})$ over T composition rounds, matching differential-privacy composition bounds [43]. Non-asymptotically, however, history-dependent composition is tighter than its mechanism-agnostic counterpart, as shown in Fig. 4.

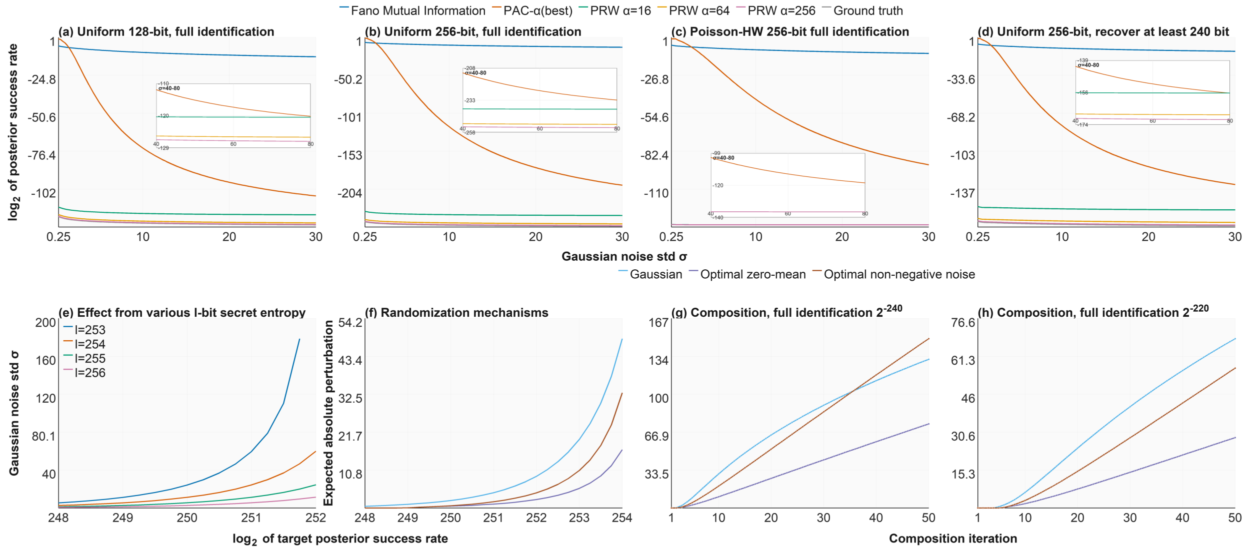

Figure 2: Power leakage experiments. The upper row compares posterior-success accounting under Gaussian perturbation under different entropy, prior distribution and inference task. The lower row studies the effect of secret entropy, (constrained) randomization families, and history-dependent adaptive composition with two different privacy budgets.

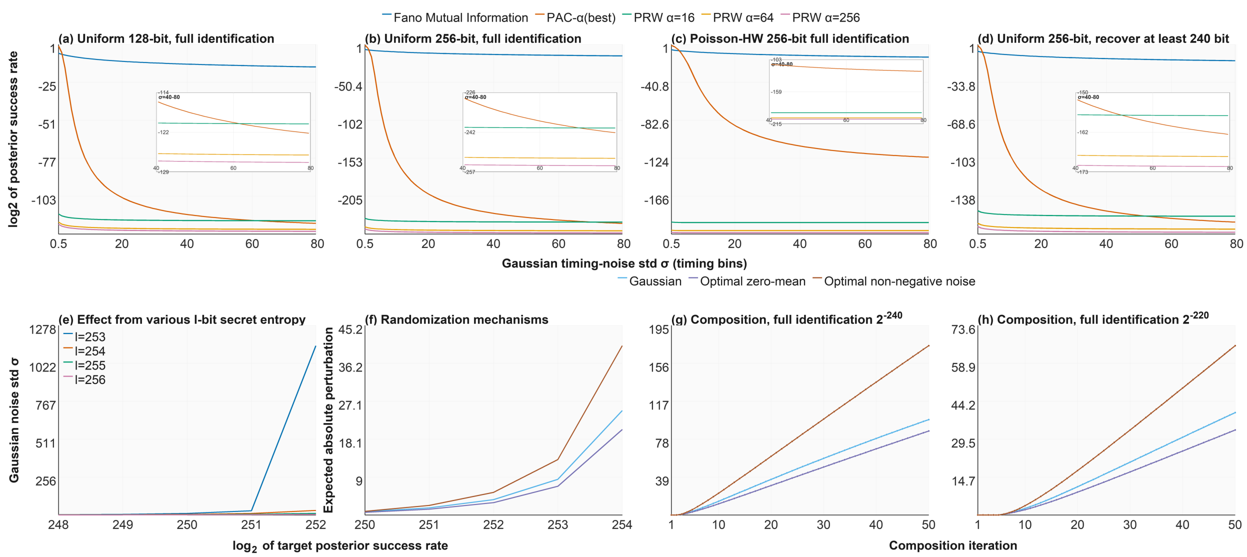

Figure 3: RSA timing leakage experiments. The upper row compares posterior-success accounting under Gaussian perturbation under different entropy, prior distribution and inference task. The lower row studies the effect of secret entropy, (constrained) randomization families, and history-dependent adaptive composition with two different privacy budgets.

6.2 Practical Implementation with Confidence Validation

In this section, we present the evaluation results for a full implementation of Algorithm 1 incorporating high-confidence validation, and we analyze the additional utility overhead incurred by leveraging the bounded-ratio technique (Section 3.4). While the experimental setup remains identical to the

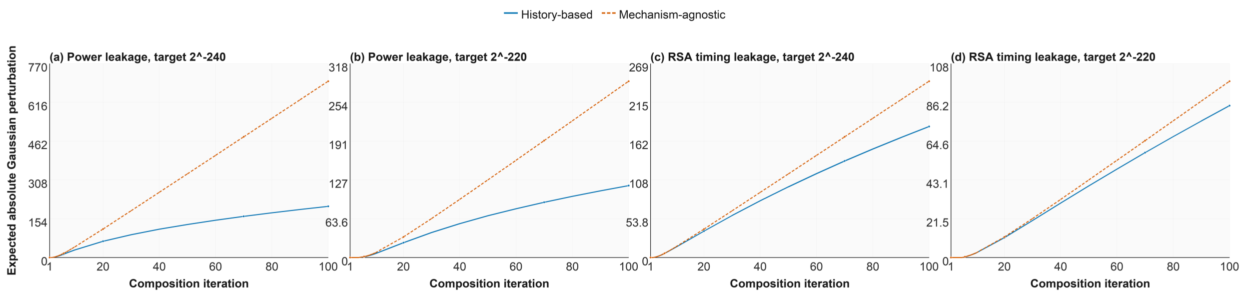

Figure 4: Comparison between the scale of Gaussian noise required through history-dependent adversarial composition and the mechanism-agnostic adversarial composition with different privacy budgets in terms of posterior identification success rate of a 256-bit uniform secret string.

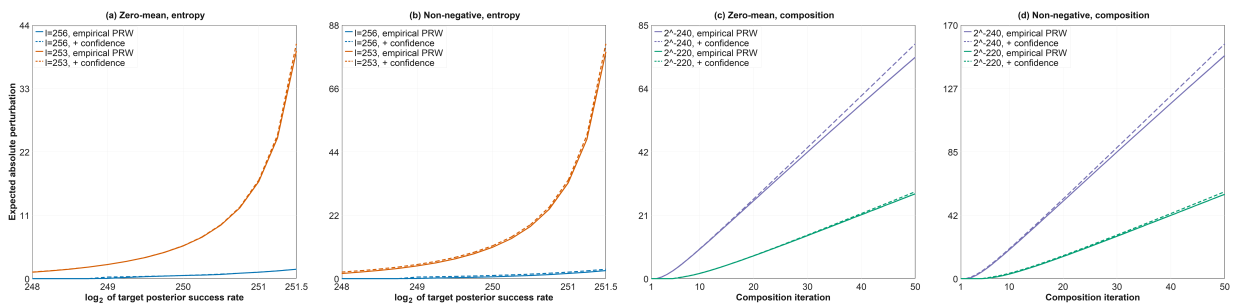

Figure 5: Bounded-ratio PRW validation with sample size $m = 100,000$, confidence $\gamma = 2^{-256}$ for AES power leakage privatization in both static and adversarially-compositional settings under different entropy and randomization constraints.

previous configuration, the results displayed in Fig. 5 and Fig. 6 are evaluated against those in configurations (e) and (g) of Fig. 2 and Fig. 3, respectively.

For the evaluations in Fig. 5 and Fig. 6, we set the sample size m for both the training dataset X and the validation dataset X' to 100,000, with a high-confidence level of $(1-\gamma) = 1-2^{-256}$. When combined with the bounded-ratio technique, this sample complexity m ensures that the validation test is passed in a single round with overwhelming probability.

In Fig. 2 and Fig. 3, the dashed lines represent the randomization strategies under varying entropy and non-negativity constraints when the bounded-ratio constraint is applied. In contrast, the solid lines depict the optimal randomization strategies derived in Section 6.1 via the empirical objective (18), which omits both the bounded-ratio constraint and confidence validation. As expected, a slight utility overhead from the bounded-ratio restriction emerges only in scenarios featuring stricter privacy budgets—namely, those requiring a higher posterior success rate or involving a longer composition sequence.

7 Conclusion and Future Prospects

In this paper, we focused on entropic secrets and establish an asymptotically-tight Bayesian analysis framework to optimize privacy-preserving solutions for both static and adversarially-adaptive composition settings given black-box leakage functions. Such an improvement through our tight Bayesian inference hardness measurement can be exponentially significant compared to classic Fano's

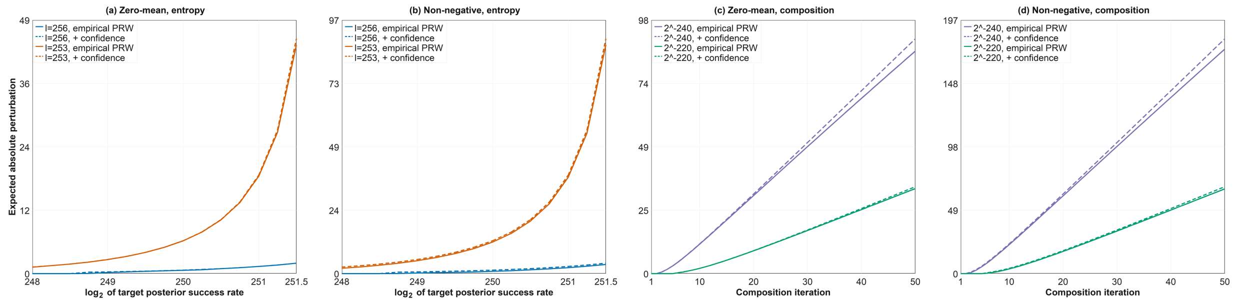

Figure 6: Bounded-ratio PRW validation with sample size $m = 100,000$, confidence $\gamma = 2^{-256}$ for RSA timing leakage privatization in both static and adversarially-compositional settings under different entropy and randomization constraints.

inequality based methods [1,53], especially through mutual information, as elaborated in Appendix A.4.

On promising future directions, one is software/hardware co-design for side-channel leakage mitigation. Given a specific leakage channel (e.g., power or timing consumption from a hardware environment) and an objective task (e.g., encryption), tight black-box privatization enables the free exploration of different processing functions (e.g., crypto protocols) to identify the sharpest utility-privacy tradeoff.

Several technical questions also remain open within our framework. First, there is a tradeoff between approximation and confidence in the finite-sample regime. Larger α makes the PRW relaxation closer to the exact posterior success rate, but it also raises higher-order output-likelihood terms. In the most general admissible mechanism class, the global bound $\mathscr{Z}$ in Theorem 3 can increase rapidly with α, and high-confidence validation may therefore require many more calibration samples. Section 3.4 gives one positive resolution for finite or discretized output domains: a common bounded-ratio constraint on the output randomization channel controls the PRW range at the target posterior-risk scale and removes the α-exponential factor from the relative confidence correction. Extending such stable finite-sample validation to richer continuous mechanism classes, including unbounded Gaussian perturbations, remains an important direction for future work. Developing specialized solvers for rich admissible randomization classes is another. Finally, regarding mechanism-agnostic composition, we note that computing tight composition for III-style worst-case guarantees is known to be #P-hard [57]. A natural open problem is whether this complexity result also holds for entropic secrets and how to generalize it to other divergence-based marginal privacy risk measurements beyond the KL- and α-divergence.

References

[1] H. Xiao and S. Devadas, "Pac privacy: Automatic privacy measurement and control of data processing," in *Annual International Cryptology Conference*. Springer, 2023, pp. 611–644.

[2] Y. Dodis and A. Smith, "Entropic security and the encryption of high entropy messages," in *Theory of Cryptography Conference*. Springer, 2005, pp. 556–577.

[3] A. Russell and H. Wang, "How to fool an unbounded adversary with a short key," *IEEE Transactions on Information Theory*, vol. 52, no. 3, pp. 1130–1140, 2006.

[4] B. Köpf and D. Basin, "Automatically deriving information-theoretic bounds for adaptive side-channel attacks," *Journal of Computer Security*, vol. 19, no. 1, pp. 1–31, 2011.

[5] J. L. Massey, "Guessing and entropy," in *Proceedings of 1994 IEEE International Symposium on Information Theory*. IEEE, 1994, p. 204.

[6] S. A. M'rio, K. Chatzikokolakis, C. Palamidessi, and G. Smith, "Measuring information leakage using generalized gain functions," in *2012 IEEE 25th Computer Security Foundations Symposium*. IEEE, 2012, pp. 265–279.
[7] C. E. Shannon, "Communication theory of secrecy systems," *The Bell system technical journal*, vol. 28, no. 4, pp. 656–715, 1949.
[8] S. Goldwasser and S. Micali, "Probabilistic encryption," *Journal of computer and system sciences*, vol. 28, no. 2, pp. 270–299, 1984.
[9] C. Dwork, F. McSherry, K. Nissim, and A. Smith, "Calibrating noise to sensitivity in private data analysis," in *Theory of Cryptography: Third Theory of Cryptography Conference, TCC 2006, New York, NY, USA, March 4-7, 2006. Proceedings 3*. Springer, 2006, pp. 265–284.
[10] C. Dwork, K. Kenthapadi, F. McSherry, I. Mironov, and M. Naor, "Our data, ourselves: Privacy via distributed noise generation," in *Advances in Cryptology-EUROCRYPT 2006: 24th Annual International Conference on the Theory and Applications of Cryptographic Techniques, St. Petersburg, Russia, May 28-June 1, 2006. Proceedings 25*. Springer, 2006, pp. 486–503.
[11] D. Kifer and A. Machanavajjhala, "Pufferfish: A framework for mathematical privacy definitions," *ACM Transactions on Database Systems (TODS)*, vol. 39, no. 1, pp. 1–36, 2014.
[12] I. Issa, A. B. Wagner, and S. Kamath, "An operational approach to information leakage," *IEEE Transactions on Information Theory*, vol. 66, no. 3, pp. 1625–1657, 2019.
[13] K. Abouelmehdi, A. Beni-Hessane, and H. Khaloufi, "Big healthcare data: preserving security and privacy," *Journal of big data*, vol. 5, no. 1, pp. 1–18, 2018.
[14] B. Coppens, I. Verbauwhede, K. De Bosschere, and B. De Sutter, "Practical mitigations for timing-based side-channel attacks on modern x86 processors," in *2009 30th IEEE symposium on security and privacy*. IEEE, 2009, pp. 45–60.
[15] J. Zhang, C. Chen, J. Cui, and K. Li, "Timing side-channel attacks and countermeasures in cpu microarchitectures," *ACM Computing Surveys*, vol. 56, no. 7, pp. 1–40, 2024.
[16] M. Randolph and W. Diehl, "Power side-channel attack analysis: A review of 20 years of study for the layman," *Cryptography*, vol. 4, no. 2, p. 15, 2020.
[17] S. Aga and S. Narayanasamy, "Invisimem: Smart memory defenses for memory bus side channel," *ACM SIGARCH Computer Architecture News*, vol. 45, no. 2, pp. 94–106, 2017.
[18] J. Van Den Hooff, D. Lazar, M. Zaharia, and N. Zeldovich, "Vuvuzela: Scalable private messaging resistant to traffic analysis," in *Proceedings of the 25th Symposium on Operating Systems Principles*, 2015, pp. 137–152.
[19] X. Xiao and Y. Tao, "Output perturbation with query relaxation," *Proceedings of the VLDB Endowment*, vol. 1, no. 1, pp. 857–869, 2008.
[20] K. Nissim, S. Raskhodnikova, and A. Smith, "Smooth sensitivity and sampling in private data analysis," in *Proceedings of the thirty-ninth annual ACM symposium on Theory of computing*, 2007, pp. 75–84.
[21] M. Lyu, D. Su, and N. Li, "Understanding the sparse vector technique for differential privacy," *Proceedings of the VLDB Endowment*, vol. 10, no. 6, pp. 637–648, 2017.
[22] M. Abadi, A. Chu, I. Goodfellow, H. B. McMahan, I. Mironov, K. Talwar, and L. Zhang, "Deep learning with differential privacy," in *Proceedings of the 2016 ACM SIGSAC conference on computer and communications security*, 2016, pp. 308–318.
[23] H. Xiao, Z. Xiang, D. Wang, and S. Devadas, "A theory to instruct differentially-private learning via clipping bias reduction," in *2023 IEEE Symposium on Security and Privacy (SP)*. IEEE, 2023, pp. 2170–2189.
[24] H. Xiao, J. Wan, and S. Devadas, "Geometry of sensitivity: Twice sampling and hybrid clipping in differential privacy with optimal gaussian noise and application to deep learning," in *Proceedings of the 2023 ACM SIGSAC Conference on Computer and Communications Security*, 2023, pp. 2636–2650.
[25] N. Papernot, S. Song, I. Mironov, A. Raghunathan, K. Talwar, and U. Erlingsson, "Scalable private learning with PATE," in *International Conference on Learning Representations*, 2018. [Online]. Available: https://openreview.net/forum?id=rkZB1XbRZ
[26] Z. Lin, S. Gopi, J. Kulkarni, H. Nori, and S. Yekhanin, "Differentially private synthetic data via foundation model APIs 1: Images," in *The Twelfth International Conference on Learning Representations*, 2024. [Online]. Available: https://openreview.net/forum?id=YEhQs8POIo
[27] B. Ghazi, R. Kumar, and P. Manurangsi, "Differentially private clustering: Tight approximation ratios," *Advances in Neural Information Processing Systems*, vol. 33, pp. 4040–4054, 2020.
[28] N. Ponomareva, H. Hazimeh, A. Kurakin, Z. Xu, C. Denison, H. B. McMahan, S. Vassilvitskii, S. Chien, and A. G. Thakurta, "How to dp-fy ml: A practical guide to machine learning with differential privacy," *Journal of Artificial Intelligence Research*, vol. 77, pp. 1113–1201, 2023.
[29] I. Sason and S. Verdú, "Arimoto–rényi conditional entropy and Bayesian M-ary hypothesis testing," *IEEE Transactions on Information Theory*, vol. 64, no. 1, pp. 4–25, Jan. 2018.
[30] S. Verdú, "α-mutual information," in *2015 Information Theory and Applications Workshop (ITA)*. IEEE, 2015, pp. 1–6.

[31] H. Xiao, "Automated and provable privatization for black-box processing," Ph.D. dissertation, Massachusetts Institute of Technology, 2024.
[32] M. Sridhar, H. Xiao, and S. Devadas, "Pac-private algorithms," in *2025 IEEE Symposium on Security and Privacy (SP)*. IEEE, 2025, pp. 3839–3857.
[33] H. Xiao, G. E. Suh, and S. Devadas, "Formal privacy proof of data encoding: The possibility and impossibility of learnable encryption," in *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security*, 2024, pp. 1834–1848.
[34] H. Xiao, J. Wan, E. Shi, and S. Devadas, "One-sided bounded noise: Theory, optimization algorithms and applications," in *Proceedings of the 2025 ACM SIGSAC Conference on Computer and Communications Security*, 2025, pp. 4214–4228.
[35] H. Xiao and S. Devadas, "Pac privacy and black-box privatization," *IEEE Security & Privacy*, vol. 23, no. 4, pp. 92–97, 2025.
[36] M. Sridhar, M. A. Noguera, C. Jain, K. Kristensen, S. Devadas, H. Xiao, and X. Yu, "Pac-private databases," *Cryptology ePrint Archive*, 2026.
[37] S. P. Kasiviswanathan and A. Smith, "On the'semantics' of differential privacy: A bayesian formulation," *Journal of Privacy and Confidentiality*, vol. 6, no. 1, 2014.
[38] A. Kerckhoffs, "La cryptographie militaire," *Journal des sciences militaires*, vol. IX, pp. 5–38, 161–191, 1883.
[39] C. Dwork, G. N. Rothblum, and S. Vadhan, "Boosting and differential privacy," in *2010 IEEE 51st annual symposium on foundations of computer science*. IEEE, 2010, pp. 51–60.
[40] P. Kairouz, S. Oh, and P. Viswanath, "The composition theorem for differential privacy," in *International conference on machine learning*. PMLR, 2015, pp. 1376–1385.
[41] M. Bun and T. Steinke, "Concentrated differential privacy: Simplifications, extensions, and lower bounds," in *Theory of cryptography conference*. Springer, 2016, pp. 635–658.
[42] I. Mironov, "Rényi differential privacy," in *2017 IEEE 30th computer security foundations symposium (CSF)*. IEEE, 2017, pp. 263–275.
[43] J. Dong, A. Roth, and W. J. Su, "Gaussian differential privacy," *Journal of the Royal Statistical Society Series B: Statistical Methodology*, vol. 84, no. 1, pp. 3–37, 2022.
[44] P. Cuff and L. Yu, "Differential privacy as a mutual information constraint," in *Proceedings of the 2016 ACM SIGSAC Conference on Computer and Communications Security*, 2016, pp. 43–54.
[45] M. Sugiyama, *Introduction to statistical machine learning*. Morgan Kaufmann, 2015.
[46] G. A. Young, R. L. Smith, and R. L. Smith, *Essentials of statistical inference*. Cambridge University Press, 2005, vol. 16.
[47] M. J. Wainwright, *High-dimensional statistics: A non-asymptotic viewpoint*. Cambridge university press, 2019, vol. 48.
[48] R. Sibson, "Information radius," *Zeitschrift für Wahrscheinlichkeitstheorie und verwandte Gebiete*, vol. 14, no. 2, pp. 149–160, 1969.
[49] G. Smith, "On the foundations of quantitative information flow," in *International Conference on Foundations of Software Science and Computational Structures*. Springer, 2009, pp. 288–302.
[50] S. Arimoto, "Information measures and capacity of order α of discrete memoryless channels," in *Topics in Information Theory*, ser. Colloquia Mathematica Societatis János Bolyai, 1977, vol. 16, pp. 41–52.
[51] J. Liao, O. Kosut, L. Sankar, and F. P. Calmon, "Tunable measures for information leakage and applications to privacy–utility tradeoffs," *IEEE Transactions on Information Theory*, vol. 65, no. 12, pp. 8043–8066, 2019.
[52] I. Sason and S. Verdú, "f-divergence inequalities," *IEEE Transactions on Information Theory*, vol. 62, no. 11, pp. 5973–6006, 2016.
[53] R. M. Fano, "Fano inequality," *Scholarpedia*, vol. 3, no. 10, p. 6648, 2008.
[54] P. Kocher, J. Jaffe, and B. Jun, "Differential power analysis," in *Advances in Cryptology – CRYPTO '99*, ser. Lecture Notes in Computer Science, vol. 1666. Springer, 1999, pp. 388–397.
[55] F.-X. Standaert, T. G. Malkin, and M. Yung, "A unified framework for the analysis of side-channel key recovery attacks," in *Annual international conference on the theory and applications of cryptographic techniques*. Springer, 2009, pp. 443–461.
[56] P. C. Kocher, "Timing attacks on implementations of diffie-hellman, rsa, dss, and other systems," in *Advances in Cryptology – CRYPTO '96*, ser. Lecture Notes in Computer Science, vol. 1109. Springer, 1996, pp. 104–113.
[57] J. Murtagh and S. Vadhan, "The complexity of computing the optimal composition of differential privacy," in *Theory of Cryptography Conference*. Springer, 2015, pp. 157–175.
[58] M. S. Alvim, M. E. Andrés, K. Chatzikokolakis, P. Degano, and C. Palamidessi, "Differential privacy: on the trade-off between utility and information leakage," in *International workshop on formal aspects in security and trust*. Springer, 2011, pp. 39–54.
[59] K. Pietrzak, "A leakage-resilient mode of operation," in *Annual International Conference on the Theory and Applications of Cryptographic Techniques*. Springer, 2009, pp. 462–482.

[60] Y. Ishai, A. Sahai, and D. Wagner, "Private circuits: Securing hardware against probing attacks," in *Annual International Cryptology Conference*. Springer, 2003, pp. 463–481.
[61] S. Dziembowski and K. Pietrzak, "Leakage-resilient cryptography," in *2008 49th Annual IEEE Symposium on Foundations of Computer Science*. IEEE, 2008, pp. 293–302.
[62] J. Liu, T. Jager, S. A. Kakvi, and B. Warinschi, "How to build time-lock encryption," *Designs, Codes and Cryptography*, vol. 86, no. 11, pp. 2549–2586, 2018.
[63] X. Zhu, M. Sridhar, and S. Devadas, "Pac-private responses with adversarial composition," *arXiv preprint arXiv:2601.14033*, 2026.
[64] D. Brumley and D. Boneh, "Remote timing attacks are practical," in *Proceedings of the 12th USENIX Security Symposium*. USENIX Association, 2003.

A Settings and Comments

A.1 Summary of Settings

For clarity, we restate the terminology, threat model and assumptions we have used in the paper.

Specified entropic prior. The secret x is sampled from a specified prior distribution $\mathcal{D}_{\mathsf{x}}$. The user can draw independent samples $\bar{\mathsf{x}} \sim \mathcal{D}_{\mathsf{x}}$, and can evaluate the prior mass or density $\pi(\bar{\mathsf{x}})$ for sampled secrets. This separates the privatization problem from the separate problem of learning the prior itself.[7]

Black-box leakage access. The black-box assumption is a restriction on the user, not on the adversary. In the static setting, the user can sample $\bar{\mathsf{x}} \sim \mathcal{D}_{\mathsf{x}}$ and evaluate the selected leakage $\mathcal{F}(\bar{\mathsf{x}})$. If the leakage depends on a public or environmental query v, the user can evaluate the selected one-round leakage $\mathcal{F}(\bar{\mathsf{x}}, \mathsf{v})$. In the adaptive setting, the adversary selects the query v_t from the realized transcript, and the user can simulate only the realized branch $\mathcal{F}_t(\bar{\mathsf{x}}, \mathsf{v}_t, \tau_t)$ on sampled secrets; the user is not assumed to enumerate the full counterfactual interaction tree. In all cases, the user is not assumed to know any structural information (e.g., source code, sensitivity, gradients, or an analytical likelihood model) for $\mathcal{F}$.

White-box adversary and arbitrary task. The adversary is computationally unbounded and white-box: it knows $\mathcal{D}_{\mathsf{x}}$, $\mathcal{F}$, the inference criterion ρ, and the user's randomized mechanism and calibration rule. The only unknowns are the realized secret and the mechanism's private randomness. The relation $\rho(\hat{a}, \mathsf{x})$ is arbitrary and specifies the adversarial task; exact identification is only the special case in which the report space is $\mathcal{X}$ and $\rho(\hat{a}, \mathsf{x}) = \mathbf{1}[\hat{a} = \mathsf{x}]$.

Admissible randomization and utility. The user specifies an admissible class $\mathcal{R}$ of randomized release mechanisms and a utility loss $\kappa(R)$. A mechanism $R \in \mathcal{R}$ may be a general randomized channel from raw leakage values to released outputs, a finite transition matrix, a transcript-dependent channel, or an additive perturbation rule. Our theory is not restricted to input-independent additive noise. Once $\mathcal{F}(\bar{\mathsf{x}}, \mathsf{v})$ has been obtained by black-box evaluation, the conditional law of the released output is induced by the chosen R, so the likelihood terms in the PRW bounds can be evaluated without an analytical model of $\mathcal{F}$.

Empirical optimization and validation. The algorithms construct an empirical PRW constrained optimization problem over the admissible class $\mathcal{R}$. We assume that the resulting empirical optimization can be solved, or solved up to an optimization error η. An η-approximate optimizer contributes the corresponding optimization slack in addition to the statistical validation error. The finite-sample guarantees use independent validation samples to control the gap between empirical feasibility and population posterior-risk feasibility; asymptotic statements are taken as the number of sampled secrets grows.

[7] If the prior mass or density can be consistently estimated from samples, the same empirical-learning viewpoint can be combined with that prior-estimation step but we do not pursue this extension here.

A.2 Generalization to Randomized $\mathcal{F}(\cdot, \theta)$

For simplicity, the main paper assumes the leakage function $\mathcal{F}$ to be deterministic. The same PRW certification viewpoint extends to a randomized leakage function $\mathcal{F}(\mathsf{x}, \theta)$, where θ denotes a random seed, provided the randomized leakage channel and the reference channel are averaged over the same private seed source as described below. Without loss of generality, we take θ to be a uniform g-bit string; conditional on a fixed θ, $\mathcal{F}(\cdot, \theta)$ reduces to a deterministic function.

In the randomized case, the empirical channel used inside Algorithms 1 and 2 must include sampled leakage seeds. Previously we sampled a training set $\mathsf{X} = \{\bar{\mathsf{x}}_1, \ldots, \bar{\mathsf{x}}_m\}$ and an independent validation set $\mathsf{X}' = \{\bar{\mathsf{x}}'_1, \ldots, \bar{\mathsf{x}}'_m\}$; we now additionally sample a training random-seed set $\Theta = \{\bar{\theta}_1, \ldots, \bar{\theta}_m\}$ and an independent validation random-seed set $\Theta' = \{\bar{\theta}'_1, \ldots, \bar{\theta}'_m\}$. We then define the distribution of the noisy mechanism applied to a particular instance $\bar{\mathsf{x}}$ and evaluated over a given random-seed set Θ as

$$\mathbb{P}_{\tilde{\mathcal{M}}_{\mathcal{D}_{\mathsf{e}},\Theta}(\bar{\mathsf{x}})} = \frac{1}{m} \sum_{i=1}^{m} \mathbb{P}_{\mathcal{F}(\bar{\mathsf{x}}, \bar{\theta}_i) + \mathsf{e}}, \tag{33}$$

that is, a mixture of the noisy deterministic mechanisms conditioned on the sampled seeds in Θ. The PRW certificate then follows by applying the same analysis to the compound channel and compound reference defined below.

To see why, we invoke another key property of f-divergence, namely *joint convexity*, stated below.

Lemma 4 (Joint convexity of f-Divergence). *Let $f : (0, +\infty) \to \mathbb{R}$ be a convex function. Then the function $(z, v) \mapsto v\, f\left(\frac{z}{v}\right)$ is jointly convex in (z, v) over $z \geq 0$ and $v > 0$. Consequently, f-divergence is jointly convex with respect to the pair of mixture distributions*

$$\big(c \cdot P_1 + (1-c)P_2, cQ_1 + (1-c)Q_2\big)$$

for any $c \in [0, 1]$, such that

$$\begin{aligned} &D_f(c \cdot P_1 + (1-c) \cdot P_2 \,\|\, c \cdot Q_1 + (1-c) \cdot Q_2) \\ &\qquad \leq cD_f(P_1 \,\|\, Q_1) + (1-c)D_f(P_2 \,\|\, Q_2). \end{aligned}$$

The ground-truth distribution of the noisy output $\mathcal{M}_{\mathcal{D}_{\mathsf{e}}}(\mathsf{x}) = \mathcal{F}(\mathsf{x}, \theta) + \mathsf{e}$, with $\mathsf{x} \sim \mathcal{D}_{\mathsf{x}}$, $\theta \sim \mathsf{U}\{0, 1\}^g$, and $\mathsf{e} \sim \mathcal{D}_{\mathsf{e}}$, is

$$\mathbb{P}_{\mathcal{M}_{\mathcal{D}_{\mathsf{e}}}(\mathsf{x})} = \mathbb{E}_{\theta \sim \{0,1\}^g} \mathbb{P}_{\mathcal{F}(\mathsf{x}, \theta) + \mathsf{e}}, \tag{34}$$

and our generalized PRW analysis should likewise be based on $\mathbb{P}_{\mathcal{M}_{\mathcal{D}_{\mathsf{e}}}(\mathsf{x})}$. The reference distribution must be averaged over the same random-seed source. To make this precise, view the sampled seed set Θ as part of the mechanism's private randomness, write

$$P^{\mathsf{x}}_{\Theta} := \mathbb{P}_{\tilde{\mathcal{M}}_{\mathcal{D}_{\mathsf{e}},\Theta}(\mathsf{x})},$$

and let Q_Θ be the reference distribution used for that realized seed set. Define the compound mechanism and compound reference by

$$P^{\mathsf{x}} := \mathbb{E}_{\Theta} P^{\mathsf{x}}_{\Theta}, \qquad Q := \mathbb{E}_{\Theta} Q_{\Theta}.$$

By joint convexity of f-divergences,

$$D_f(P^{\mathsf{x}} \| Q) \le \mathbb{E}_\Theta D_f(P^{\mathsf{x}}_\Theta \| Q_\Theta). \tag{35}$$

The special case $Q_\Theta \equiv W$ gives a common-reference certificate $D_f(P^{\mathsf{x}} \| W) \le \mathbb{E}_\Theta D_f(P^{\mathsf{x}}_\Theta \| W)$. If the empirical PRW reference is recomputed for each seed set, then the left-hand side should be read with the mixture reference $Q = \mathbb{E}_\Theta Q_\Theta$, not with a separately optimized ground-truth reference. This is sufficient for privacy certification, since Theorems 1 and 2 hold for arbitrary references. As the number of sampled seeds grows, the empirical mixture channel converges to the true randomized leakage channel; under the same consistency assumptions for the empirical reference construction, the resulting certificate approaches the certificate for the true randomized leakage mechanism.

A.3 Additional Related Work

Looseness of KL-divergence and Mutual Information. Mutual Information (MI) is often the de facto metric for quantifying the leakage of entropic secrets. However, independent of our findings regarding the looseness of posterior inference bounds, existing literature also identifies other limitations of MI. For instance, MI generally captures average leakage rather than a high-probability sense [49] or the worst case [58]. Furthermore, it may not accurately reflect the empirical success rate of practical attackers [55].

Leakage-Resilient Cryptography. In contrast to our approach, which applies a universal randomization mechanism to any (potentially black-box) leakage, Leakage-Resilient Cryptography [59] focuses on redesigning cryptographic primitives to mitigate leakage directly. These methods typically rely on specific assumptions about the leakage channel, such as the t-probing model (where an adversary observes at most t wires) [60] or the bounded-leakage model [61]. Common techniques include masking and periodic key updates. Hardware- and software-level countermeasures like constant-time encryption [62] also exist, but they are often computationally expensive or platform-specific.

Adversarial PAC Composition: During the preparation of this paper, we became aware that Zhu, Sridhar, and Devadas were independently studying adversarial composition for PAC Privacy. Since their methodology and application focus differ from ours, the two works are being presented separately. Their concurrent work [63] develops adversarial composition for PAC Privacy through mutual information and applies it to privatizing inference outputs of machine-learning models. Technically, however, mutual information does not in general provide a tight characterization of posterior inference success, as illustrated in Figs. 2 and 3(a)–(d). Moreover, the history-dependent composition accounting in [63] does not directly yield a tight posterior-success accounting that exhausts the available Bayesian privacy budget. Their analysis is also specialized to a finite input domain with 128 elements. In contrast, our framework addresses the asymptotic tightness of both static and adversarial composition accounting, and applies to general black-box leakage under arbitrary specified priors $\mathcal{D}_{\mathsf{x}}$.

A.4 Comparison with f-Divergence Fano Bound

We compare Theorem 1 with the classic f-divergence Fano bound in Lemma 1. Consider identification under a uniform prior over a secret space of size $N = |\mathcal{X}|$. The optimal prior success rate before observing the leakage is $\delta_{o,I} = \frac{1}{N}$. Here, the subscript o denotes the optimal success rate based only on the prior, before observing any leakage.

For $f(z) = z^\alpha$, define

$$R^W_\alpha(\mathcal{M}) := \mathbb{E}_{\mathsf{x} \sim \mathcal{D}_{\mathsf{x}}} \left[D_\alpha\big(\mathbb{P}_{\mathcal{M}(\mathsf{x})} \,\big\|\, \mathbb{P}_W\big) \right]. \tag{36}$$

Under the uniform prior, our PRW bound reduces to

$$\mathsf{I}_\alpha^W(\mathcal{M}) = \left(\delta_{o,I}^{\alpha-1} R_\alpha^W(\mathcal{M})\right)^{1/\alpha}.$$

The f-divergence Fano bound uses the complete Bernoulli divergence

$$D_\alpha\big(\mathbf{1}_{\delta_I} \,\big\|\, \mathbf{1}_{\delta_{o,I}}\big) = (1-\delta_{o,I})\left(\frac{1-\delta_I}{1-\delta_{o,I}}\right)^\alpha + \delta_{o,I}\left(\frac{\delta_I}{\delta_{o,I}}\right)^\alpha.$$

For finite α, retaining both terms may yield a slightly sharper certificate than the explicit PRW expression above. As $\alpha \to \infty$, however, both approaches become tight for identification under a uniform prior.

The difference appears under nonuniform priors. The f-divergence Fano route summarizes the prior through the single baseline success rate

$$\delta_{o,I} = \max_{\hat{\mathsf{x}} \in \mathcal{X}} \Pr_{\mathcal{D}_\mathsf{x}}(\mathsf{x} = \hat{\mathsf{x}}).$$

In contrast, the PRW bound retains the complete prior through the weights $\pi(\mathsf{x})^{\alpha-1}$. Consequently, Theorem 1 remains asymptotically tight for identification under arbitrary priors.

We next explain why the classical mutual-information-based bound can be loose for posterior identification. Selecting $f(t) = t\log t$ in Lemma 1 recovers the KL-divergence form of Fano's inequality. Optimizing over the reference distribution W yields mutual information:

$$I(\mathsf{x}; \mathcal{M}(\mathsf{x})) = \inf_W \mathbb{E}_{\mathsf{x}\sim\mathcal{D}_\mathsf{x}}\left[D_{\mathrm{KL}}\big(\mathbb{P}_{\mathcal{M}(\mathsf{x})} \,\big\|\, \mathbb{P}_W\big)\right].$$

The infimum is attained when $\mathbb{P}_W$ is the marginal output distribution of $\mathcal{M}(\mathsf{x})$.

The difference is structural. Posterior identification success is governed by a pointwise maximum: for each observed output o, the MAP adversary selects the secret candidate with the largest prior-weighted likelihood $q_\mathcal{M}(\hat{\mathsf{x}}, \mathsf{o})$. Mutual information instead averages KL-divergence contributions across secret values and output events. Translating this average information quantity into an upper bound on MAP success through Fano's inequality can therefore introduce substantial slack, especially in high-entropy settings.

The PRW construction is more directly aligned with the MAP objective. Its unnormalized ℓ_α-norm increasingly emphasizes the largest prior-weighted likelihoods as α grows and converges to the pointwise maximum as $\alpha \to \infty$. Having established this asymptotically tight characterization, we next show how to approximate the resulting expectation using finitely many black-box evaluations.

B Experimental Results

B.1 Common Evaluation Pipeline

We use scalar side-channel leakage in all experiments. The raw output is discretized into integer bins before accounting: the AES power leakage is already an integer Hamming weight, while RSA timing is rounded to a 1 ns bin. After discretization, the black-box simulation induces a finite leakage law over atoms $b \in \mathcal{O}$. For the power model this law is available in a closed form. For RSA timing, the law is empirical: unless otherwise stated, we draw 100,000 secret samples from $\mathcal{D}_\mathsf{x}$, evaluate the timing simulator on the selected public input, and use the empirical distribution over the resulting bins as the leakage law.

For panels (a)–(d), we fix a Gaussian perturbation $\mathsf{e} \sim \mathcal{N}(0, \sigma^2)$ and evaluate each privacy certificate on the same discretized leakage distribution. The Fano curve is obtained by converting mutual information into a posterior-success bound. The PAC-α curve uses the standard PAC Privacy α-divergence bound and optimizes the order over the numerical grid used by the implementation. The PRW curves use $\alpha \in \{16, 64, 256\}$. In the closed-form power experiment we also plot the MAP posterior success rate as ground truth. In RSA timing we do not plot a separate analytic ground-truth curve, because the timing law is supplied only by black-box simulation; the empirical MAP value is used only as an internal diagnostic.

Panel (c) uses a non-uniform prior in which the Hamming weight $H = \mathrm{HW}(\mathsf{x})$ follows a truncated Poisson law on $\{0, \ldots, 256\}$ with mean 96, and the secret is uniform conditional on its Hamming weight. Thus

$$\Pr_{\mathcal{D}_{\mathsf{x}}}(\mathsf{x}) = \frac{\Pr[H = \mathrm{HW}(\mathsf{x})]}{\binom{256}{\mathrm{HW}(\mathsf{x})}}.$$

For panels (e)-(h), we uniformly select $\alpha = 256$ for the evaluation.

B.2 Linear Programming (LP) for Optimal Discrete Perturbation

We now spell out the finite-dimensional optimization used for the optimized discrete-noise curves. Let the perturbation support be

$$\mathcal{U} = \{u_1, \ldots, u_K\},$$

where $\mathcal{U} = [-2048, 2048] \cap \mathbb{Z}$ for the zero-mean mechanism and $\mathcal{U} = [0, 2048] \cap \mathbb{Z}$ for the non-negative mechanism in the reported figures. The decision variable is the probability vector $q \in \mathbb{R}_+^K$, where $q_j = \Pr[\mathsf{e} = u_j]$. We minimize expected absolute perturbation,

$$\kappa(q) = \mathbb{E}[|\mathsf{e}|] = \sum_{j=1}^{K} |u_j| q_j.$$

If a second-moment objective is desired instead, the same LP below is obtained by replacing the cost $|u_j|$ by u_j^2.

Consider first exact identification. Group the secret space by leakage atom: $\mathcal{X}_b = \{\mathsf{x} : \mathcal{F}(\mathsf{x}) = b\}$. Define

$$m_b := \max_{\mathsf{x} \in \mathcal{X}_b} \Pr_{\mathcal{D}_{\mathsf{x}}}(\mathsf{x}).$$

For a candidate discrete perturbation q, the released output is $Y = b + \mathsf{e}$, and the MAP identification success rate is

$$\delta_I(q) = \sum_{y \in \mathcal{O} + \mathcal{U}} \max_{b \in \mathcal{O}:\, y - b \in \mathcal{U}} m_b \, q_{j(y,b)},$$

where $u_{j(y,b)} = y - b$. This expression is convex in q, but it becomes a linear program by introducing epigraph variables t_y for the pointwise maxima:

$$\begin{aligned}\min_{q,t}\quad & \sum_{j=1}^{K} |u_j| q_j \\ \text{s.t.}\quad & q_j \ge 0, \qquad \sum_{j=1}^{K} q_j = 1, \\ & \sum_{j=1}^{K} u_j q_j = 0 \quad \text{for the zero-mean mechanism,} \\ & t_y \ge m_b\, q_{j(y,b)} \quad \forall y,\ \forall b \text{ with } y-b \in \mathcal{U}, \\ & \sum_{y\in\mathcal{O}+\mathcal{U}} t_y \le \delta_{\text{tar}}.\end{aligned} \tag{37}$$

For the non-negative mechanism, the zero-mean equality is omitted and the support $\mathcal{U}$ itself enforces $\mathsf{e} \ge 0$. Thus the objective, probability simplex, mean constraint, MAP epigraph constraints, and posterior-risk budget are all linear.

The same construction handles an arbitrary inference task ρ. Let $\mathcal{A}$ be the finite set of candidate adversarial reports considered in the discretized experiment and define

$$\lambda_{a,b} := \sum_{\mathsf{x}\in\mathcal{X}_b} \Pr_{\mathcal{D}_{\mathsf{x}}}(\mathsf{x})\rho(a,\mathsf{x}).$$

Then

$$\delta_\rho(q) = \sum_y \max_{a\in\mathcal{A}} \sum_{b:\, y-b\in\mathcal{U}} \lambda_{a,b} q_{j(y,b)}.$$

Replacing the third line of (37) by

$$t_y \ge \sum_{b:\, y-b\in\mathcal{U}} \lambda_{a,b} q_{j(y,b)} \qquad \forall y,\ \forall a \in \mathcal{A}$$

gives the LP for task-ρ posterior success. This is the empirical MAP, equivalently the $\alpha \to \infty$ limit of the PRW certificate.

For the high-confidence finite-output variant of Algorithm 1, we use the bounded-ratio specialization from Section 3.4. This confidence-control step is separate from the main empirical PRW plots: the main panels report the finite-α empirical calibration without the validation correction, while the bounded-ratio construction quantifies the extra utility cost needed to turn that empirical calibration into a high-confidence certificate. It is important that the required condition is not literally $Q(y \mid o)/W(y) < 1$ for every y: this would be impossible for a nontrivial channel, since $\sum_y W(y)\, Q(y \mid o)/W(y) = 1$ for each o. The implementable condition is instead a common domination bound

$$Q(y \mid o) \le L W(y) \qquad \forall o, y,$$

with $L \ge 1$. The PRW validation range is then controlled by

$$B_\alpha(L) = (\pi_{\max} L)^{\alpha-1}, \qquad \pi_{\max} = \sup_{\mathsf{x}} \Pr_{\mathcal{D}_{\mathsf{x}}}(\mathsf{x}).$$

Thus, for target posterior success δ_{tar}, moment budget $r = \delta_{\text{tar}}^\alpha$, and a desired range-to-target ratio C, we choose

$$L \le C^{1/(\alpha-1)} \frac{\delta_{\text{tar}}^{\alpha/(\alpha-1)}}{\pi_{\max}}.$$

This is the sense in which the likelihood ratio is kept below the relevant target scale: the product $\pi_{\max}L$, not Q/W alone, is what enters the PRW loss.

For additive discrete perturbation $Q(y \mid b) = q_{y-b}$, the smallest domination constant over all references W is

$$L(q) = \sum_{y \in \mathcal{O}+\mathcal{U}} \max_{b:\, y-b \in \mathcal{U}} q_{j(y,b)}.$$

Indeed, the optimal reference is proportional to $\max_b Q(y \mid b)$. Therefore the finite-output PRW confidence constraint is again linear after introducing epigraph variables s_y:

$$\begin{aligned} \min_{q,s} \quad & \sum_{j=1}^{K} |u_j| q_j \\ \text{s.t.} \quad & q_j \geq 0, \qquad \sum_{j=1}^{K} q_j = 1, \\ & \sum_{j=1}^{K} u_j q_j = 0 \quad \text{for the zero-mean mechanism}, \\ & s_y \geq q_{j(y,b)} \quad \forall y,\ \forall b \text{ with } y-b \in \mathcal{U}, \\ & \sum_{y \in \mathcal{O}+\mathcal{U}} s_y \leq L_{\mathrm{tar}}. \end{aligned} \tag{38}$$

For non-negative perturbation, we use the support $\mathcal{U} = [0, 2048] \cap \mathbb{Z}$ and omit the zero-mean equality. In the confidence-control comparison, L_{tar} is chosen from the displayed formula with $\alpha = 256$, $\gamma = 2^{-256}$, $m = 100{,}000$, and $C = 1$. The empirical PRW target is tightened by the corresponding relative Hoeffding correction $\sqrt{\log(1/\gamma)/(2m)}$. Gaussian perturbations are evaluated by finite-α PRW but are not certified by this bounded-ratio condition because their support is unbounded.

B.3 Power Leakage

We model power leakage from AES secret-key generation by aggregate Hamming weight, a standard first-order abstraction for power side channels [16, 54, 55]. For an l-bit secret x and a public mask or plaintext v,

$$\mathcal{F}(\mathsf{x}, \mathsf{v}) = \mathrm{HW}(\mathsf{x} \oplus \mathsf{v}) \in \{0, \dots, l\}.$$

For a uniform secret and any fixed v, this leakage has the binomial law $\Pr[\mathcal{F} = h] = \binom{l}{h} 2^{-l}$. Since the AES S-box is a permutation, applying an S-box before the Hamming-weight measurement preserves the uniform byte law; the aggregate Hamming-weight model therefore captures the one-dimensional discretized power signal used in the experiment.

Panel (c) uses a non-uniform prior in which the Hamming weight $H = \mathrm{HW}(\mathsf{x})$ follows a truncated Poisson law on $\{0, \dots, 256\}$ with mean 96, and the secret is uniform conditional on its Hamming weight. Thus

$$\Pr_{\mathcal{D}_{\mathsf{x}}}(\mathsf{x}) = \frac{\Pr[H = \mathrm{HW}(\mathsf{x})]}{\binom{256}{\mathrm{HW}(\mathsf{x})}}.$$

Panel (d) changes the task from exact identification to approximate recovery: the adversary succeeds if the reported 256-bit string differs from the true secret in at most 16 bit positions, i.e., if at least 240 bits are recovered. The corresponding task coefficients $\lambda_{a,b}$ in the LP above are computed using Hamming-ball counts.

For panels (e)–(h), the reported vertical axis is either the Gaussian standard deviation σ or the expected absolute perturbation $\mathbb{E}[|\mathsf{e}|]$. In panel (f), the optimized zero-mean and non-negative mechanisms freely optimize the discrete perturbation distribution over the admissible support, without imposing the bounded-ratio L-constraint. The implemented LP uses the weighted max-norm form induced by the finite-α PRW objective with $\alpha = 256$, together with the standard ℓ_α-to-ℓ_∞ correction factor; this avoids introducing the high-confidence validation constraint into the main utility comparison. In panels (g) and (h), we simulate history-dependent composition with total full-identification targets 2^{-240} and 2^{-220}. For each trial, a sequence of public masks $\mathsf{v}_1, \ldots, \mathsf{v}_T$ is generated and the leakage in round t is $\mathrm{HW}(\mathsf{x} \oplus \mathsf{v}_t)$. Under the uniform prior, the marginal law is binomial for every fixed mask, but generating the masks explicitly matches the adaptive-query interpretation. We report the prefix-average perturbation

$$\frac{1}{t}\sum_{i=1}^{t}\mathbb{E}[|N_i|].$$

All three curves in panels (e)–(h) use finite-α PRW accounting with $\alpha = 256$. For composition, the total finite-α PRW budget is split equally across rounds. The optimized discrete curves use the same empirical PRW calibration at each local budget, while the Gaussian curve uses finite-α PRW evaluation of the Gaussian perturbation. The separate confidence-control comparison applies the bounded-ratio condition to the same empirical PRW calibration and reports the additional utility cost.

B.4 RSA Timing Leakage

For RSA timing, the secret x is the private exponent bit string. The public input v represents the plaintext/base and public operand-dependent context. We use a black-box square-and-multiply timing simulator for non-constant-time modular exponentiation [56,64]: every exponent bit contributes a squaring cost, one-bits contribute an additional multiplication cost, and a small public-input-dependent operand term models timing variation from modular arithmetic. The resulting scalar time is rounded to a 1 ns bin. This model is not meant to claim a particular CPU measurement; it is a standard black-box-style timing channel that exposes the dependence of running time on the secret exponent and the chosen public input.

For the upper-row RSA panels we draw 100,000 exponent samples for each case and evaluate one public query. The cases match the power experiment: uniform 128-bit full identification, uniform 256-bit full identification, Poisson-Hamming-weight 256-bit full identification, and uniform 256-bit recovery of at least 240 bits. For the Poisson-Hamming-weight case, the Hamming weight is sampled from the same truncated Poisson law and then the exponent is sampled uniformly among strings with that weight. Since the RSA timing distribution is available only through these black-box samples, every accounting curve is an empirical finite-population curve.

Panels (e)–(h) repeat the entropy, mechanism-constraint, and composition experiments for timing leakage. In this application, $\mathbb{E}[|\mathsf{e}|]$ has a direct timing interpretation: it is the expected absolute timing perturbation per release, and for the non-negative mechanism it is the expected added delay. For composition, each trial generates a sequence of public timing queries and the plotted value is averaged across five trials. The history-dependent versus mechanism-agnostic comparison in Fig. 4 uses Gaussian perturbations, $\alpha = 256$, and composition lengths up to 100; the mechanism-agnostic curve applies the balanced Holder allocation from Theorem 10.

Overall, the experiments show three consistent trends. Higher entropy and looser posterior targets require less perturbation; one-sided constraints increase overhead; and PRW approaches the closed-form ground truth in the power model while avoiding the systematic looseness of KL/Fano accounting, especially for non-uniform priors and non-identification tasks.

C Proof of asymptotic tightness of the α-Fano bound for uniform-prior identification

This comparison concerns exact identification under a uniform prior. In this case $\delta_{o,I} = 1/|\mathcal{X}|$, and taking the $1/\alpha$ power in Lemma 1 with $f(u) = u^\alpha$, then sending $\alpha \to \infty$, gives

$$\frac{\delta_I}{\delta_{o,I}} \leq \exp\{I_\infty^{sib}(\mathsf{x}; \mathcal{F}(\mathsf{x}))\}.$$

For a uniform prior, Sibson information of order infinity satisfies

$$\exp\{I_\infty^{sib}(\mathsf{x}; \mathcal{F}(\mathsf{x}))\} = \int_{\mathcal{O}} \max_{\hat{\mathsf{x}} \in \mathcal{X}} \Pr\big(\mathcal{F}(\hat{\mathsf{x}}) = \mathsf{o}\big)\, d\mathsf{o}.$$

Multiplying by $\delta_{o,I} = 1/|\mathcal{X}|$ yields

$$\begin{aligned}
\delta_{o,I} \exp\{I_\infty^{sib}(\mathsf{x}; \mathcal{F}(\mathsf{x}))\} &= \int_{\mathcal{O}} \max_{\hat{\mathsf{x}} \in \mathcal{X}} \Pr(\mathsf{x} = \hat{\mathsf{x}}) \Pr\big(\mathcal{F}(\hat{\mathsf{x}}) = \mathsf{o}\big)\, d\mathsf{o} &\quad (39)\\
&= \int_{\mathcal{O}} \max_{\hat{\mathsf{x}} \in \mathcal{X}} \Pr\big(\mathsf{x} = \hat{\mathsf{x}}, \mathcal{F}(\hat{\mathsf{x}}) = \mathsf{o}\big)\, d\mathsf{o} &\quad (40)\\
&= \delta_I. &\quad (41)
\end{aligned}$$

Thus the α-Fano/PAC route is asymptotically tight in this special uniform-prior exact-identification case. This argument does not establish tightness for arbitrary priors or arbitrary inference criteria ρ, which is precisely the role of the PRW construction in Theorem 2.

D Proof of Lemma 3

Let $g : \mathcal{O} \to \mathcal{A}$ be any deterministic decision rule, where $g(\mathsf{o})$ is the report when the observation is o. For this proof write

$$H(\hat{a}, \mathsf{o}) := \sum_{\bar{\mathsf{x}} \in \mathcal{X}} \mathbf{1}_{\{\rho(\hat{a}, \bar{\mathsf{x}}) = 1\}} \Pr(\mathsf{x} = \bar{\mathsf{x}}) \Pr(\mathsf{o} \mid \mathsf{x} = \bar{\mathsf{x}}).$$

The success event is $\{\rho(g(\mathsf{o}), \mathsf{x}) = 1\}$, so

$$\begin{aligned}
\Pr\big(\rho(g(\mathsf{o}), \mathsf{x}) = 1\big) &= \int_{\mathcal{O}} \sum_{\bar{\mathsf{x}} \in \mathcal{X}} \mathbf{1}_{\{\rho(g(\mathsf{o}), \bar{\mathsf{x}}) = 1\}} \Pr(\mathsf{x} = \bar{\mathsf{x}}, \mathsf{o})\, d\mathsf{o}\\
&= \int_{\mathcal{O}} \sum_{\bar{\mathsf{x}} \in \mathcal{X}} \mathbf{1}_{\{\rho(g(\mathsf{o}), \bar{\mathsf{x}}) = 1\}} \Pr(\mathsf{x} = \bar{\mathsf{x}}) \Pr(\mathsf{o} \mid \mathsf{x} = \bar{\mathsf{x}})\, d\mathsf{o}.
\end{aligned}$$

For each fixed o, the inner sum is exactly $H(g(\mathsf{o}), \mathsf{o})$ by the definition of H, hence

$$\Pr\big(\rho(g(\mathsf{o}), \mathsf{x}) = 1\big) = \int_{\mathcal{O}} H(g(\mathsf{o}), \mathsf{o})\, d\mathsf{o} \;\leq\; \int_{\mathcal{O}} \max_{\hat{a} \in \mathcal{A}} H(\hat{a}, \mathsf{o})\, d\mathsf{o},$$

since $H(g(\mathsf{o}), \mathsf{o}) \leq \max_{\hat{a}} H(\hat{a}, \mathsf{o})$ for every o.

Conversely, for each $\mathsf{o} \in \mathcal{O}$ we can choose $g^\star(\mathsf{o}) \in \arg\max_{\hat{a}\in\mathcal{A}} H(\hat{a}, \mathsf{o})$ (choosing any maximizer when there are ties). For this $g^\star$ we have $H(g^\star(\mathsf{o}), \mathsf{o}) = \max_{\hat{a}} H(\hat{a}, \mathsf{o})$ for all o, and thus

$$\Pr\big(\rho(g^\star(\mathsf{o}), \mathsf{x}) = 1\big) = \int_{\mathcal{O}} \max_{\hat{a}\in\mathcal{A}} H(\hat{a}, \mathsf{o})\, \mathrm{d}\mathsf{o}.$$

Combining the upper bound with the equality achieved by $g^\star$ proves the claim.

E Proof of (11)

Notice that

$$\left(\sum_{\mathsf{x}\in\mathcal{X}} H(\mathsf{x}, \mathsf{o})^\alpha\right)^{1/\alpha} \;\geq\; \left(\max_{\mathsf{x}\in\mathcal{X}} H(\mathsf{x}, \mathsf{o})^\alpha\right)^{1/\alpha} \;=\; \max_{\mathsf{x}\in\mathcal{X}} H(\mathsf{x}, \mathsf{o}), \tag{42}$$

take integral over $\mathcal{O}$ we directly get

$$\int_{\mathcal{O}} \left(\sum_{\mathsf{x}\in\mathcal{X}} H(\mathsf{x}, \mathsf{o})^\alpha\right)^{1/\alpha} \mathrm{d}\mathsf{o} \geq \int_{\mathcal{O}} \max_{\mathsf{x}\in\mathcal{X}} H(\mathsf{x}, \mathsf{o})\, \mathrm{d}\mathsf{o}.$$

F Proof of the PRW α-norm upper bound

Fix $\alpha > 1$ and define, for each $\mathsf{o} \in \mathcal{O}$,

$$G(\mathsf{o}) \;:=\; \left(\sum_{\mathsf{x}\in\mathcal{X}} H(\mathsf{x}, \mathsf{o})^\alpha\right)^{1/\alpha}.$$

Then the left-hand side of the inequality is simply $\int_{\mathcal{O}} G(\mathsf{o})\, \mathrm{d}\mathsf{o}$. We insert the reference density W and apply Hölder's inequality with exponents α and $\frac{\alpha}{\alpha-1}$:

$$\begin{aligned}
\int_{\mathcal{O}} G(\mathsf{o})\, \mathrm{d}\mathsf{o} &= \int_{\mathcal{O}} G(\mathsf{o})\ \Pr(W = \mathsf{o})^{-\frac{\alpha-1}{\alpha}}\ \Pr(W = \mathsf{o})^{\frac{\alpha-1}{\alpha}}\ \mathrm{d}\mathsf{o} \\
&\leq \left(\int_{\mathcal{O}} \left(G(\mathsf{o})\ \Pr(W = \mathsf{o})^{-\frac{\alpha-1}{\alpha}}\right)^{\alpha} \mathrm{d}\mathsf{o}\right)^{1/\alpha} \left(\int_{\mathcal{O}} \left(\Pr(W = \mathsf{o})^{\frac{\alpha-1}{\alpha}}\right)^{\frac{\alpha}{\alpha-1}} \mathrm{d}\mathsf{o}\right)^{\frac{\alpha-1}{\alpha}} \\
&= \left(\int_{\mathcal{O}} \frac{G(\mathsf{o})^\alpha}{\Pr(W = \mathsf{o})^{\alpha-1}} \mathrm{d}\mathsf{o}\right)^{1/\alpha} \left(\int_{\mathcal{O}} \Pr(W = \mathsf{o})\, \mathrm{d}\mathsf{o}\right)^{\frac{\alpha-1}{\alpha}}.
\end{aligned}$$

Since W is a probability distribution, $\int_{\mathcal{O}} \Pr(W = \mathsf{o})\, \mathrm{d}\mathsf{o} = 1$, so we obtain

$$\int_{\mathcal{O}} G(\mathsf{o})\, \mathrm{d}\mathsf{o} \;\leq\; \left(\int_{\mathcal{O}} \frac{G(\mathsf{o})^\alpha}{\Pr(W = \mathsf{o})^{\alpha-1}} \mathrm{d}\mathsf{o}\right)^{1/\alpha} = \left(\int_{\mathcal{O}} \frac{\sum_{\mathsf{x}\in\mathcal{X}} H(\mathsf{x}, \mathsf{o})^\alpha}{\Pr(W = \mathsf{o})^{\alpha-1}} \mathrm{d}\mathsf{o}\right)^{1/\alpha},$$

which is the first inequality.

For the expectation form, note that

$$\sum_{\mathsf{x}\in\mathcal{X}} H(\mathsf{x}, \mathsf{o})^\alpha = \sum_{\mathsf{x}\in\mathcal{X}} \Pr(\mathsf{x})\, \frac{H(\mathsf{x}, \mathsf{o})^\alpha}{\Pr(\mathsf{x})},$$

so

$$\int_{\mathcal{O}} \frac{\sum_{\mathsf{x}\in\mathcal{X}} H(\mathsf{x},\mathsf{o})^{\alpha}}{\Pr(W=\mathsf{o})^{\alpha-1}}\,\mathrm{d}\mathsf{o} = \mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}} \int_{\mathcal{O}} \frac{H(\mathsf{x},\mathsf{o})^{\alpha}/\Pr(\mathsf{x})}{\Pr(W=\mathsf{o})^{\alpha-1}}\,\mathrm{d}\mathsf{o},$$

which yields the second equality.

Finally, Hölder's inequality is tight if and only if there exists a constant $\lambda > 0$ such that

$$G(\mathsf{o})^{\alpha} \;=\; \lambda \Pr(W=\mathsf{o})^{\alpha} \quad \text{for almost every } \mathsf{o},$$

which is equivalent to

$$\sum_{\mathsf{x}\in\mathcal{X}} H(\mathsf{x},\mathsf{o})^{\alpha} \;=\; \lambda \Pr(W=\mathsf{o})^{\alpha} \quad \text{a.e. } \mathsf{o}.$$

Rewriting this gives the stated condition that the bound is tight when W is chosen proportional to the α-norm of $H(\cdot,\mathsf{o})$:

$$\Pr(W=\mathsf{o}) \;\propto\; \left(\sum_{\mathsf{x}\in\mathcal{X}} H(\mathsf{x},\mathsf{o})^{\alpha}\right)^{1/\alpha}.$$

G Proof of Theorem 3

Let $\mathsf{S}_{\mathrm{cal}} := (\mathsf{X},\mathsf{X}')$ denote the private calibration randomness used by Algorithm 1. For each realized $\mathsf{S}_{\mathrm{cal}}$, the algorithm outputs a rule $\widehat{\mathcal{D}}_e(\mathsf{S}_{\mathrm{cal}})$ and a corresponding reference distribution $W_{\widehat{\mathcal{D}}_e(\mathsf{S}_{\mathrm{cal}})}$. Define the population loss of any candidate by

$$\mu(\mathcal{D}_e) := \mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}\big[\mathcal{G}(\mathcal{D}_e,\mathsf{x})\big], \qquad \Delta(\mathcal{D}_e) := B_{+}(\mathcal{D}_e) - B_{-}(\mathcal{D}_e).$$

The theorem is a guarantee over the internal random seed $\mathsf{S}_{\mathrm{cal}}$. It does not claim that every realized pair (X,X') produces a rule whose population loss is at most r'. Instead, we show that the set of misleading calibration seeds has probability at most γ, and that even on those seeds the fallback bound controls the loss by $\mathcal{Z}$.

Condition on the training sample X. The empirical optimizer and the subsequent fallback path are now fixed before the validation sample X' is drawn; X' only determines the stopping point. For the additive-noise instantiation used in the paper, write this monotone path as $\{\mathcal{D}_e^{\lambda} : \lambda \geq 0\}$, where increasing λ corresponds to applying additional common post-processing noise to the candidate mechanism and to the reference distribution. The tested candidates in Algorithm 1 form a subsequence of this path. By assumption, for $\lambda_2 \geq \lambda_1$,

$$\mathcal{G}(\mathcal{D}_e^{\lambda_2},\mathsf{x}) \leq \mathcal{G}(\mathcal{D}_e^{\lambda_1},\mathsf{x}) \quad \text{for all } \mathsf{x}, \qquad \Delta(\mathcal{D}_e^{\lambda_2}) \leq \Delta(\mathcal{D}_e^{\lambda_1}).$$

Hence $\mu(\mathcal{D}_e^{\lambda})$ is nonincreasing in λ, while the validation threshold $r' - \Delta(\mathcal{D}_e^{\lambda})\sqrt{\log(1/\gamma)/(2m)}$ is nondecreasing.

If the initial candidate already satisfies $\mu(\mathcal{D}_e^{0}) \leq r'$, then every later candidate and every fallback candidate also has population loss at most r', and the claim follows. Otherwise, by monotonicity and the continuity of the common post-processing path, let $\lambda^{\star}$ be the first point at which

$$\mu(\mathcal{D}_e^{\lambda^{\star}}) = r'.$$

This benchmark depends on the fixed training sample X, but it is independent of the validation sample X'. Let

$$L^{\star} := \frac{1}{m}\sum_{i=1}^{m} \mathcal{G}(\mathcal{D}_e^{\lambda^{\star}}, \bar{\mathsf{x}}_i'), \qquad \beta^{\star} := \Delta(\mathcal{D}_e^{\lambda^{\star}})\sqrt{\frac{\log(1/\gamma)}{2m}}.$$

Conditioned on X, the summands in $L^\star$ are i.i.d., bounded in an interval of length $\Delta(\mathcal{D}_e^{\lambda^\star})$, and have mean r'. Hoeffding's inequality gives

$$\Pr_{\mathsf{X}'}\big[L^\star \le r' - \beta^\star \,\big|\, \mathsf{X}\big] \le \gamma.$$

Define the good calibration event

$$\mathsf{E} := \{L^\star > r' - \beta^\star\}.$$

On E, every tested candidate before $\lambda^\star$ fails validation. Indeed, for any $\lambda < \lambda^\star$, pointwise monotonicity gives $L(\mathcal{D}_e^\lambda) \ge L^\star$, while $\beta(\mathcal{D}_e^\lambda) \ge \beta^\star$, and therefore

$$L(\mathcal{D}_e^\lambda) > r' - \beta^\star \ge r' - \beta(\mathcal{D}_e^\lambda).$$

Thus the algorithm cannot stop before the population loss has fallen to at most r'. If it later accepts a validated candidate, then $\mu(\widehat{\mathcal{D}}_e) \le r'$. If the validation loop exhausts all candidates, the fallback phase applies only further common post-processing until $B_+(\widehat{\mathcal{D}}_e) \le r'$, which also implies $\mu(\widehat{\mathcal{D}}_e) \le r'$.

On the complement event E^c, we only use the global fallback bound. If the algorithm accepts during validation, its stopping rule enforces $B_+(\widehat{\mathcal{D}}_e) \le \mathcal{Z}$. If it reaches the fallback phase, then $B_+(\widehat{\mathcal{D}}_e) \le r' \le \mathcal{Z}$. In either case,

$$\mu(\widehat{\mathcal{D}}_e) \le B_+(\widehat{\mathcal{D}}_e) \le \mathcal{Z} \qquad \text{on } \mathsf{E}^c.$$

Taking expectation over X', still conditioned on X, yields

$$\mathbb{E}_{\mathsf{X}'}\left[\mu(\widehat{\mathcal{D}}_e) \,\middle|\, \mathsf{X}\right] \le (1-\gamma)r' + \gamma\mathcal{Z}.$$

Taking expectation over X and using $(1-\gamma)r' + \gamma\mathcal{Z} \le r$ proves

$$\mathbb{E}_{\mathsf{X},\mathsf{X}'}\left[\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_\mathsf{x}}\big[\mathcal{G}(\widehat{\mathcal{D}}_e, \mathsf{x})\big]\right] \le r.$$

Finally, when $\mathcal{G}$ is instantiated as the PRW loss, this averaged bound is exactly the certificate for the compound randomized mechanism. Let

$$P_{\overline{\mathcal{M}}(\mathsf{x})} := \mathbb{E}_{\mathsf{S}_{\mathrm{cal}}} P_{\mathcal{M}_{\widehat{\mathcal{D}}_e(\mathsf{S}_{\mathrm{cal}})}(\mathsf{x})}, \qquad P_{\overline{W}} := \mathbb{E}_{\mathsf{S}_{\mathrm{cal}}} P_{W_{\widehat{\mathcal{D}}_e(\mathsf{S}_{\mathrm{cal}})}}.$$

The adversary knows these mixture laws but not the realized $\mathsf{S}_{\mathrm{cal}}$. By joint convexity of f-divergences, including α-divergence,

$$\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_\mathsf{x}}\left[\Pr_{\mathcal{D}_\mathsf{x}}(\mathsf{x})^{\alpha-1}\mathcal{D}_\alpha\big(P_{\overline{\mathcal{M}}(\mathsf{x})} \| P_{\overline{W}}\big)\right] \le \mathbb{E}_{\mathsf{X},\mathsf{X}'}\left[\mu(\widehat{\mathcal{D}}_e)\right] \le r.$$

Thus the private calibration samples are correctly accounted for as internal randomness of the released mechanism, and the theorem follows.

H Proof of Theorem 2

Apply Lemma 3, the ℓ_α-relaxation in (11), and the PRW α-norm upper bound to obtain

$$\delta_\rho = \int_{\mathcal{O}} \max_{\hat{a}\in\mathcal{A}} H_{\rho,\mathcal{M}}(\hat{a},\mathsf{o})\,\mathrm{d}\mathsf{o} \tag{43}$$

$$\leq \int_{\mathcal{O}} \left(\sum_{\hat{a}\in\mathcal{A}} H_{\rho,\mathcal{M}}(\hat{a},\mathsf{o})^\alpha\right)^{1/\alpha} \mathrm{d}\mathsf{o} \tag{44}$$

$$\leq \left(\int_{\mathcal{O}} \frac{\sum_{\hat{a}\in\mathcal{A}} H_{\rho,\mathcal{M}}(\hat{a},\mathsf{o})^\alpha}{\Pr(W=\mathsf{o})^{\alpha-1}}\mathrm{d}\mathsf{o}\right)^{1/\alpha} \tag{45}$$

$$= \mathsf{I}^{W}_{\alpha,\rho}\big(\mathsf{x};\mathcal{F}(\mathsf{x})\big) \tag{46}$$

Since this holds for arbitrary W, the next inequality also holds:

$$\delta_\rho \;\leq\; \mathsf{I}_{\alpha,\rho}\big(\mathsf{x};\mathcal{F}(\mathsf{x})\big).$$

To show the closed-form optimal PRW reference, applying the PRW α-norm upper bound, we see that choosing

$$\Pr(W=\mathsf{o}) \;\propto\; \left(\sum_{\hat{a}\in\mathcal{A}} H_{\rho,\mathcal{M}}(\hat{a},\mathsf{o})^\alpha\right)^{1/\alpha}$$

is exactly the condition under which Hölder's inequality becomes tight.

As for the asymptotic tightness of the PRW bound,

$$\lim_{\alpha\to\infty} \mathsf{I}_{\alpha,\rho}\big(\mathsf{x};\mathcal{F}(\mathsf{x})\big) \tag{47}$$

$$= \int_{\mathcal{O}} \left(\sum_{\hat{a}\in\mathcal{A}} H_{\rho,\mathcal{M}}(\hat{a},\mathsf{o})^\alpha\right)^{1/\alpha} \mathrm{d}\mathsf{o}, \tag{48}$$

$$= \int_{\mathcal{O}} \max_{\hat{a}\in\mathcal{A}} H_{\rho,\mathcal{M}}(\hat{a},\mathsf{o})\,\mathrm{d}\mathsf{o} \tag{49}$$

$$= \;\delta_\rho \tag{50}$$

I Proof of Theorem 6

Let Q denote the reference transcript process constructed in the previous proof. For a realized transcript, define

$$A_t(\mathsf{x}) := \Pr_{\mathcal{D}_\mathsf{x}}(\mathsf{x})^{\alpha-1}\left(\prod_{s=1}^{t}\frac{\Pr\big(\mathcal{M}_s(\mathsf{x},\mathsf{v}_s,\tau_s)=\mathsf{o}_s\big)}{\Pr\big(W_s(\mathsf{v}_s,\tau_s)=\mathsf{o}_s\big)}\right)^{\alpha},$$

with $A_0(\mathsf{x}) = \Pr_{\mathcal{D}_\mathsf{x}}(\mathsf{x})^{\alpha-1}$. Theorem 5 can be written compactly as

$$\delta_I^\alpha \leq \mathbb{E}_{\tau_{T+1}\sim Q}\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_\mathsf{x}}[A_T(\mathsf{x})]\,.$$

Now condition on any reachable pre-round transcript τ_t and query v_t. Averaging only over the reference output $\mathsf{o}_t \sim W_t(\mathsf{v}_t,\tau_t)$, we have

$$\mathbb{E}_{\mathsf{o}_t\sim W_t}\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_\mathsf{x}}[A_t(\mathsf{x})] = \mathbb{E}_{\mathsf{x}\sim\mathcal{D}_\mathsf{x}}[A_{t-1}(\mathsf{x})D_t(\mathsf{x})]\,.$$

By the branch-local condition (26), this is at most

$$r_t\,\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}[A_{t-1}(\mathsf{x})]\,.$$

The condition holds for every reachable branch and query, so averaging over the adversary's conditional query law and then applying the tower property gives

$$\mathbb{E}_Q\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}[A_t(\mathsf{x})] \le r_t\,\mathbb{E}_Q\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}[A_{t-1}(\mathsf{x})]\,.$$

Iterating this inequality from $t = T$ down to $t = 1$ yields

$$\delta_I^\alpha \le \left(\prod_{t=1}^{T} r_t\right)\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}[A_0(\mathsf{x})] = \mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}\left[\Pr_{\mathcal{D}_{\mathsf{x}}}(\mathsf{x})^{\alpha-1}\right]\prod_{t=1}^{T} r_t,$$

as claimed.

J Proof of Theorem 7

We suppress the adversarial query probabilities in this proof because the reference process uses the same query kernels as the real process; these factors cancel exactly as in the proof of Theorem 5. For a fixed prefix, write

$$P_t^{\mathsf{x}}(\mathsf{o}_t) := \Pr\big(\mathcal{M}_t(\mathsf{x}, \mathsf{v}_t, \tau_t) = \mathsf{o}_t\big), \qquad W_t(\mathsf{o}_t) := \Pr\big(W_t(\mathsf{v}_t, \tau_t) = \mathsf{o}_t\big).$$

Define the prefix likelihood weight

$$L_{t-1}(\mathsf{x}) := \Pr_{\mathcal{D}_{\mathsf{x}}}(\mathsf{x})\prod_{s=1}^{t-1}\frac{P_s^{\mathsf{x}}(\mathsf{o}_s)}{W_s(\mathsf{o}_s)}$$

and its α-mass

$$C_{t-1}(\tau_t) := \sum_{\mathsf{x}\in\mathcal{X}} L_{t-1}(\mathsf{x})^\alpha = \mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}[A_{t-1}(\mathsf{x})]\,.$$

The empty product gives $C_0 = \mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}[\Pr_{\mathcal{D}_{\mathsf{x}}}(\mathsf{x})^{\alpha-1}]$.

At round t, the locally optimal reference density has the form stated in the theorem:

$$W_t^\star(\mathsf{o}_t) = \frac{H_t(\mathsf{o}_t)}{Z_t(\tau_t, \mathsf{v}_t)}, \qquad H_t(\mathsf{o}_t) := \left(\sum_{\mathsf{x}\in\mathcal{X}} L_{t-1}(\mathsf{x})^\alpha P_t^{\mathsf{x}}(\mathsf{o}_t)^\alpha\right)^{1/\alpha}.$$

Its normalizing constant satisfies

$$Z_t(\tau_t, \mathsf{v}_t)^\alpha = \int \frac{H_t(\mathsf{o}_t)^\alpha}{W_t^\star(\mathsf{o}_t)^{\alpha-1}}\,\mathrm{d}\mathsf{o}_t = \mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}[A_{t-1}(\mathsf{x})D_t(\mathsf{x})]\,.$$

If the branch-local condition holds with equality and factor r_t, then

$$Z_t(\tau_t, \mathsf{v}_t)^\alpha = r_t\,C_{t-1}(\tau_t).$$

Moreover, after observing o_t, the next prefix mass is

$$C_t(\tau_{t+1}) = \sum_{\mathsf{x}\in\mathcal{X}}\left(L_{t-1}(\mathsf{x})\frac{P_t^{\mathsf{x}}(\mathsf{o}_t)}{W_t^\star(\mathsf{o}_t)}\right)^\alpha = Z_t(\tau_t, \mathsf{v}_t)^\alpha.$$

Thus, if C_{t-1} is constant over all reachable prefixes at round t, then $C_t = r_t C_{t-1}$ is also constant over all reachable prefixes at round $t+1$. Since C_0 is constant, induction gives

$$C_T = C_0 \prod_{t=1}^{T} r_t = \mathbb{E}_{\mathsf{x}\sim\mathcal{D}_\mathsf{x}}\left[\Pr_{\mathcal{D}_\mathsf{x}}(\mathsf{x})^{\alpha-1}\right] \prod_{t=1}^{T} r_t$$

on every complete transcript.

It remains to see that the resulting product reference is the global Hölder optimizer. Let

$$G_t(\mathsf{o}_{1:t}, \mathsf{v}_{1:t}) := \left(\sum_{\mathsf{x}\in\mathcal{X}} \Pr_{\mathcal{D}_\mathsf{x}}(\mathsf{x})^\alpha \prod_{s=1}^{t} P_s^\mathsf{x}(\mathsf{o}_s)^\alpha \right)^{1/\alpha}.$$

By the definition of L_{t-1},

$$H_t(\mathsf{o}_t) = \frac{G_t(\mathsf{o}_{1:t}, \mathsf{v}_{1:t})}{\prod_{s=1}^{t-1} W_s^\star(\mathsf{o}_s)}.$$

Because the normalizers Z_t are branch-independent under the equality condition, these identities telescope:

$$\prod_{t=1}^{T} W_t^\star(\mathsf{o}_t) = \frac{G_T(\mathsf{o}_{1:T}, \mathsf{v}_{1:T})}{\prod_{t=1}^{T} Z_t}.$$

Therefore the joint reference transcript law is proportional to the ℓ_α-norm of the prior-weighted transcript likelihoods, which is exactly the equality condition for the global Hölder/PRW bound. Hence the locally selected reference process globally minimizes the PRW certificate, and the minimum value is

$$\left(\inf_W \mathsf{I}_\alpha^W\right)^\alpha = \mathbb{E}_{\mathsf{x}\sim\mathcal{D}_\mathsf{x}}\left[\Pr_{\mathcal{D}_\mathsf{x}}(\mathsf{x})^{\alpha-1}\right] \prod_{t=1}^{T} r_t.$$

K Proof of Theorem 8

Formally expose the private calibration randomness as follows. For every reachable node $(t, \tau_t, \mathsf{v}_t)$, let $\mathsf{S}_t(\tau_t, \mathsf{v}_t)$ be an independent private seed used by Algorithm 1. An online execution samples only the seeds on its realized branch, but for analysis this defines a randomized policy on the entire interaction tree.

For a fixed node, write

$$P_{t,s}^\mathsf{x}(\mathsf{o}) := \Pr\big(\mathcal{M}_{t,s}(\mathsf{x}, \mathsf{v}_t, \tau_t) = \mathsf{o}\big), \qquad W_{t,s}(\mathsf{o}) := \Pr\big(W_{t,s}(\mathsf{v}_t, \tau_t) = \mathsf{o}\big)$$

for the channel and reference selected when the current calibration seed is s. The compound branch channel and compound branch reference are

$$\bar{P}_t^\mathsf{x} := \mathbb{E}_{\mathsf{S}_t} P_{t,\mathsf{S}_t}^\mathsf{x}, \qquad \bar{W}_t := \mathbb{E}_{\mathsf{S}_t} W_{t,\mathsf{S}_t}.$$

Let $\bar{A}_{t-1}(\mathsf{x})$ be the prefix coefficient for the compound prefix channel along the current transcript. Equivalently, it is built from the previously averaged pairs $(\bar{P}_s, \bar{W}_s)$, $s < t$. Once the node (τ_t, v_t) is fixed, $\bar{A}_{t-1}$ is fixed before the fresh seed S_t is drawn.

Applying Theorem 3 at this node to the centered loss

$$\mathcal{G}_t(\mathcal{D}_e, \hat{\mathsf{x}}) := \bar{A}_{t-1}(\hat{\mathsf{x}})\,(D_t^{\mathsf{S}_t}(\hat{\mathsf{x}}) - r_t)$$

gives

$$\mathbb{E}_{\mathsf{S}_t}\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}\big[\bar{A}_{t-1}(\mathsf{x})\,(D_t^{\mathsf{S}_t}(\mathsf{x})-r_t)\big] \;\le\; 0, \tag{51}$$

where

$$D_t^s(\mathsf{x}) := D_\alpha(P_{t,s}^{\mathsf{x}}\|W_{t,s}).$$

By joint convexity of the α-divergence,

$$D_\alpha(\bar{P}_t^{\mathsf{x}}\|\bar{W}_t) \le \mathbb{E}_{\mathsf{S}_t} D_t^{\mathsf{S}_t}(\mathsf{x}).$$

Multiplying by the nonnegative fixed prefix coefficient $\bar{A}_{t-1}(\mathsf{x})$ and averaging over $\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}$ yields

$$\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}\big[\bar{A}_{t-1}(\mathsf{x})\,D_\alpha(\bar{P}_t^{\mathsf{x}}\|\bar{W}_t)\big] \le r_t\,\mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}\big[\bar{A}_{t-1}(\mathsf{x})\big]. \tag{52}$$

This is exactly the branch-local condition (26) for the compound branch channels $(\bar{P}_t,\bar{W}_t)$. The argument applies to every reachable node of the randomized policy, and the seed families are independent across nodes. Applying Theorem 6 to the compound interactive mechanism then gives

$$\delta_I^\alpha \le \mathbb{E}_{\mathsf{x}\sim\mathcal{D}_{\mathsf{x}}}\left[\Pr_{\mathcal{D}_{\mathsf{x}}}(\mathsf{x})^{\alpha-1}\right]\prod_{t=1}^{T} r_t. \tag{53}$$

L Extension of the adaptive proof to arbitrary inference criteria

The adaptive-composition theorems in the main body are stated for exact identification in order to keep the notation readable. The same proof extends to any fixed inference criterion $\rho:\mathcal{A}_\rho\times\mathcal{X}\to\{0,1\}$ by replacing the identification PRW score with the task-aware score from Section 3.2. We outline the substitution here.

For a complete adaptive transcript $\tau_{T+1}=(\mathsf{v}_{1:T},\mathsf{o}_{1:T})$, define

$$H_{\rho,\mathcal{M}^{\mathrm{Adv}}}(\hat{a},\tau_{T+1}) := \sum_{\mathsf{x}\in\mathcal{X}}\rho(\hat{a},\mathsf{x})\Pr_{\mathcal{D}_{\mathsf{x}}}(\mathsf{x})\Pr\!\big(\mathcal{M}^{\mathrm{Adv}}(\mathsf{x})=\tau_{T+1}\big).$$

The exact posterior task-success rate is the integral of $\max_{\hat{a}} H_{\rho,\mathcal{M}^{\mathrm{Adv}}}(\hat{a},\tau_{T+1})$, and Theorem 2 upper bounds it by the corresponding task-aware PRW norm. The adversary-query probabilities are treated exactly as in the proof above: the reference process uses the same conditional query law, so those likelihood-ratio terms cancel and only the user-output reference kernels $W_t(\mathsf{v}_t,\tau_t)$ remain.

The branch-local state is then indexed by reports rather than by a single secret. For each partial transcript τ_t, set

$$B_{t-1}^\rho(\hat{a};\tau_t) := \sum_{\mathsf{x}\in\mathcal{X}}\rho(\hat{a},\mathsf{x})\Pr_{\mathcal{D}_{\mathsf{x}}}(\mathsf{x})\prod_{s=1}^{t-1}\frac{\Pr\!\big(\mathcal{M}_s(\mathsf{x},\mathsf{v}_s,\tau_s)=\mathsf{o}_s\big)}{\Pr\!\big(W_s(\mathsf{v}_s,\tau_s)=\mathsf{o}_s\big)}$$

and define

$$\Phi_{t-1}^\rho(\tau_t) := \sum_{\hat{a}\in\mathcal{A}_\rho}\big(B_{t-1}^\rho(\hat{a};\tau_t)\big)^\alpha.$$

The task-aware local calibration condition is that, after choosing the round-t randomized channel and reference,

$$\mathbb{E}_{\mathsf{o}_t\sim W_t(\mathsf{v}_t,\tau_t)}\left[\sum_{\hat{a}\in\mathcal{A}_\rho}\left(\sum_{\mathsf{x}\in\mathcal{X}}\rho(\hat{a},\mathsf{x})\Pr_{\mathcal{D}_{\mathsf{x}}}(\mathsf{x})\prod_{s=1}^{t}\frac{\Pr\!\big(\mathcal{M}_s(\mathsf{x},\mathsf{v}_s,\tau_s)=\mathsf{o}_s\big)}{\Pr\!\big(W_s(\mathsf{v}_s,\tau_s)=\mathsf{o}_s\big)}\right)^\alpha\right] \le r_t\,\Phi_{t-1}^\rho(\tau_t).$$

Integrating this inequality over the reference process and applying it recursively over $t = 1, \ldots, T$ gives

$$\delta_\rho^\alpha \le \left(\sum_{\hat{a} \in \mathcal{A}_\rho} \Pr_{\mathsf{x} \sim \mathcal{D}_\mathsf{x}} [\rho(\hat{a}, \mathsf{x}) = 1]^\alpha \right) \prod_{t=1}^{T} r_t.$$

For exact identification, $\mathcal{A}_\rho = \mathcal{X}$ and $\rho(\hat{a}, \mathsf{x}) = \mathbf{1}[\hat{a} = \mathsf{x}]$, this reduces to the δ_I accounting used in the main composition theorem.

M Proof of Theorem 9

Let $\Gamma_t = (\mathsf{v}_{1:t}, \mathsf{o}_{1:t})$ denote the transcript after round t. It is enough to upper bound $I(\mathsf{x}; \Gamma_T)$, because Lemma 1 with $f(u) = u \log u$ and the transcript marginal as the reference gives

$$D_{\mathrm{KL}}(\mathbf{1}_{\delta_\rho} \| \mathbf{1}_{\delta_{o,\rho}}) \le I(\mathsf{x}; \Gamma_T).$$

We prove by induction that $I(\mathsf{x}; \Gamma_t) \le B_t$.

For $t = 1$, the query v_1 is independent of x. By the chain rule for mutual information and the optimality of the marginal reference for KL,

$$I(\mathsf{x}; \Gamma_1) = I(\mathsf{x}; \mathsf{o}_1 \mid \mathsf{v}_1) \le \mathbb{E}_{\mathsf{v}_1} \mathbb{E}_{\mathsf{x} \sim \mathcal{D}_\mathsf{x}} D_{\mathrm{KL}}\big(P_{\mathcal{M}_1(\mathsf{x}, \mathsf{v}_1)} \big\| P_{W_1(\mathsf{v}_1)}\big) \le r_1 = B_1.$$

Assume $I(\mathsf{x}; \Gamma_{t-1}) \le B_{t-1}$. Since v_t is sampled from the adversary's conditional law given Γ_{t-1}, it is conditionally independent of x once Γ_{t-1} is fixed. Hence

$$I(\mathsf{x}; \Gamma_t) = I(\mathsf{x}; \Gamma_{t-1}) + I(\mathsf{x}; \mathsf{o}_t \mid \Gamma_{t-1}, \mathsf{v}_t).$$

For the second term, define

$$h_t(\mathsf{x}, \mathsf{v}) := D_{\mathrm{KL}}\big(P_{\mathcal{M}_t(\mathsf{x}, \mathsf{v})} \big\| P_{W_t(\mathsf{v})}\big), \qquad 0 \le h_t(\mathsf{x}, \mathsf{v}) \le R_t.$$

Again using the KL-optimality of the conditional output marginal,

$$I(\mathsf{x}; \mathsf{o}_t \mid \Gamma_{t-1}, \mathsf{v}_t) \le \mathbb{E}[h_t(\mathsf{x}, \mathsf{v}_t)],$$

where the expectation is under the real joint law of $(\mathsf{x}, \Gamma_{t-1}, \mathsf{v}_t)$. Let $P_{\mathsf{x},\mathsf{v}_t}$ be the induced joint law of $(\mathsf{x}, \mathsf{v}_t)$ and let $\mathcal{D}_\mathsf{x} P_{\mathsf{v}_t}$ be the product of its marginals. The average-case assumption gives

$$\mathbb{E}_{(\mathsf{x}, \mathsf{v}_t) \sim \mathcal{D}_\mathsf{x} P_{\mathsf{v}_t}} h_t(\mathsf{x}, \mathsf{v}_t) \le r_t.$$

The worst-case bound gives

$$\mathbb{E}_{P_{\mathsf{x},\mathsf{v}_t}} h_t \le r_t + R_t \, \mathrm{TV}(P_{\mathsf{x},\mathsf{v}_t}, \mathcal{D}_\mathsf{x} P_{\mathsf{v}_t}).$$

By data processing,

$$D_{\mathrm{KL}}(P_{\mathsf{x},\mathsf{v}_t} \| \mathcal{D}_\mathsf{x} P_{\mathsf{v}_t}) = I(\mathsf{x}; \mathsf{v}_t) \le I(\mathsf{x}; \Gamma_{t-1}) \le B_{t-1},$$

and Pinsker's inequality therefore gives

$$\mathrm{TV}(P_{\mathsf{x},\mathsf{v}_t}, \mathcal{D}_\mathsf{x} P_{\mathsf{v}_t}) \le \sqrt{2B_{t-1}}.$$

Thus the incremental leakage is at most

$$r_t + R_t \sqrt{2B_{t-1}}.$$

It is also at most R_t, because $h_t \le R_t$ pointwise. Therefore

$$I(\mathsf{x}; \Gamma_t) \le B_{t-1} + \min\{r_t + R_t \sqrt{2B_{t-1}}, R_t\} = B_t.$$

Induction gives $I(\mathsf{x}; \Gamma_T) \le B_T$, proving the theorem.

N Proof of Theorem 10

Fix an adversarial strategy and define a reference transcript process that uses the same adversarial query kernels as the real interaction, but replaces the round-t output law with $W_t(\mathsf{v}_t)$. Applying Theorem 1 to the complete transcript mechanism, and canceling the identical query kernels as in Theorem 5, gives

$$\delta_I^\alpha \leq \sum_{\mathsf{x}\in\mathcal{X}} \Pr_{\mathcal{D}_\mathsf{x}}(\mathsf{x})^\alpha \mathbb{E}_Q \prod_{t=1}^{T} \left(\frac{\Pr(\mathcal{M}_t(\mathsf{x},\mathsf{v}_t)=\mathsf{o}_t)}{\Pr(W_t(\mathsf{v}_t)=\mathsf{o}_t)} \right)^\alpha,$$

where Q denotes the reference transcript law. For fixed x, apply Hölder's inequality over Q with exponents $p_1,\ldots,p_T$:

$$\mathbb{E}_Q \prod_{t=1}^{T} \left(\frac{\Pr(\mathcal{M}_t(\mathsf{x},\mathsf{v}_t)=\mathsf{o}_t)}{\Pr(W_t(\mathsf{v}_t)=\mathsf{o}_t)} \right)^\alpha \leq \prod_{t=1}^{T} \left(\mathbb{E}_Q \left[\left(\frac{\Pr(\mathcal{M}_t(\mathsf{x},\mathsf{v}_t)=\mathsf{o}_t)}{\Pr(W_t(\mathsf{v}_t)=\mathsf{o}_t)} \right)^{\alpha p_t} \right] \right)^{1/p_t}.$$

Under Q, the query v_t may depend on the previous reference transcript but is independent of x, and conditional on v_t the output o_t is sampled from $W_t(\mathsf{v}_t)$. Hence

$$\mathbb{E}_Q \left[\left(\frac{\Pr(\mathcal{M}_t(\mathsf{x},\mathsf{v}_t)=\mathsf{o}_t)}{\Pr(W_t(\mathsf{v}_t)=\mathsf{o}_t)} \right)^{\alpha p_t} \right] = \mathbb{E}_{\mathsf{v}_t\sim Q} D_{\alpha p_t}\left(P_{\mathcal{M}_t(\mathsf{x},\mathsf{v}_t)} \middle\| P_{W_t(\mathsf{v}_t)}\right).$$

A second application of Hölder, now over the finite secret space, yields

$$\delta_I^\alpha \leq \prod_{t=1}^{T} \left(\sum_{\mathsf{x}\in\mathcal{X}} \Pr_{\mathcal{D}_\mathsf{x}}(\mathsf{x})^\alpha \mathbb{E}_{\mathsf{v}_t\sim Q} D_{\alpha p_t}\left(P_{\mathcal{M}_t(\mathsf{x},\mathsf{v}_t)} \middle\| P_{W_t(\mathsf{v}_t)}\right) \right)^{1/p_t}.$$

The assumption of the theorem holds for every query value v_t, so it also holds after averaging over the query distribution induced by Q. Each factor is therefore at most r_t^{1/p_t}, and

$$\delta_I^\alpha \leq \prod_{t=1}^{T} r_t^{1/p_t}.$$